

From Roots to Categories: Deriving Nouns, Verbs, and Adjectives in Santali

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DEPARTMENT OF HUMANITIES AND SOCIAL SCIENCES

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by

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6. All sources of information, including data, ideas, and text, have been duly acknowledged and appropriately cited in the thesis.
7. The use of artificial intelligence (AI) tools (if any), including large language models, has been appropriately disclosed. All analysis interpretations, and conclusions presented in this thesis are my own. Alternatively, I declare that no AI tools, including large language models, have been used in the preparation of this thesis.

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Abstract

Formation of lexical categories in the syntactic workspace has been a part of generative investigation since last few decades. This dissertation investigates the syntactic formation of lexical categories like nouns, verbs, and adjectives in Santali, an understudied Austroasiatic Language spoken in India. The central concern of the current research is the extent to which Syntax affects the internal structure of words/categories. This study employs Distributed Morphology as the theoretical framework to analyze the formation of nouns, verbs and adjectives in Santali because it provides an ideal empirical domain due to its fluid categorial system.

This dissertation addresses three specific questions, each forming the focus of a dedicated chapter. The first question examines the structural position of Santali nominalization. It specifically asks how morphologically zero nominals, traditionally analyzed as root-derived in the literature, can exhibit complex internal structure by showing stem-derivation. By using diagnostics like the presence of argument structure, adverbial modification, aspectual modification, and productivity, this dissertation shows that Santali zero nominals possess verbal characteristics and lack root-level idiosyncrasies. By using these tests, the dissertation also presents that Santali zero nominals are formed at higher derivational points, like *vP*, *VoiceP*, and *AspectP* levels. Connecting it to the nouniness squish (Ross, 1973), this dissertation shows that Santali nominalization happens at different structural heights: prototypical and affixal nominals are root-derived, while zero nominals, despite being morphologically simplex, exhibit syntactic complexity and access extended verbal functional structures.

The second research question addresses how verbalization, widely claimed to be an idiosyncratic operation cross-linguistically, demonstrates systematic syntactic compositionality yielding productive change-of-state semantics in Santali. Santali exhibits both root-derived verbs and stem-derived verbs formed from already categorized nouns or adjectives. In the case

of stem-derivation, Santali verbalizes any categorial or phrasal stem productively by assigning a change-of-state meaning. This dissertation proposes a feature-based account to analyze this phenomenon where the verbalizer carries a [FIENT] feature that encodes the transition event assigning the change-of-state interpretation to its complement and yielding inchoative and causative interpretation.

The third question addresses the internal structure of adjectives, a category whose presence has been reported to be uncertain in the typological literature. Using specific diagnostics the dissertation first establishes that Santali possesses adjectives as a composite category formed through adjectivization of both roots and stems. Root-derived adjectives are idiosyncratic, lack overt morphological adjectivizers, and are formed low in the derivation. Stem-derived adjectives, in contrast, are regular and compositional, marked by the overt adjectivizer *-ih(aa)*. This chapter also demonstrates the structural asymmetry in the stem-derived adjectivization by showing how Santali adjectivizers can merge with extended functional projections like NumP and DP in the nominals domain, while it is restricted to the *v*P-level in the verbal domain. This asymmetry shows how categorization is input sensitive and the selectional restriction of categorizers to merge at different structural points may depend on input categories.

The findings of this dissertation unveils several theoretical implications. Firstly, morphologically simplex units can also carry derivational complexity as categories can contain complex internal structures to be formed at various derivational heights. Secondly, regular semantics on productive categorization can be a result of the feature present on the categorizer head. Thirdly, categorization is also input sensitive selecting different structural heights as the complement depending on the input category, as we learn from Santali adjectivization. This also means that even when the categorization system of a language shows extreme fluidity, it maintains certain constraints in the derivational operation. Since words are uniformly formed using the merge of specific categorizers, merge can be taken as the single fundamental operation sufficient for category formation in syntax cross-linguistically.

सारांश

पिछले कुछ दशकों से सिंटैक्टिक वर्कस्पेस में लेक्सिकल कैटेगरी बनाना जेनरेटिव इन्वेस्टिगेशन का हिस्सा रहा है। यह डिसर्टेशन भारत में बोली जानेवाली एक कम रिसर्च की गई ऑस्ट्रोएशियाई भाषा संताली में नाउन, वर्ब और एडजेक्टिव जैसी लेक्सिकल कैटेगरी के सिंटैक्टिक बनाने की जांच करती है। इस रिसर्च का मुख्य उद्देश्य यह है कि सिंटैक्स शब्दों/कैटेगरी के इंटरनल स्ट्रक्चर को किस हद तक प्रभावित करता है। यह स्टडी संताली में नाउन, वर्ब और एडजेक्टिव के बनने का एनालिसिस करने के लिए डिस्ट्रिब्यूटेड मॉर्फोलॉजी को थ्योरेटिकल फ्रेमवर्क के तौर पर इस्तेमाल करती है क्योंकि यह अपने फ्लूइड कैटेगरी सिस्टम के कारण एक आइडियल एंपिरिकल डोमेन देता है।

यह डिसर्टेशन तीन महत्वपूर्ण सवालों पर बात करता है, जिनमें से हर एक, एक डेडिकेटेड चैप्टर का फोकस है। पहला सवाल संताली नॉमिनलाइजेशन की स्ट्रक्चरल पोজीशन की जांच करता है। यह खास तौर पर पूछता है कि मॉर्फोलॉजिकली जीरो नॉमिनल, जिन्हें ट्रेडिशनली लिटरेचर में रूट-डेरिव्ड के रूप में एनालाइज किया जाता है, स्टेम-डेरिवेशन दिखाकर कॉम्प्लेक्स इंटरनल स्ट्रक्चर कैसे दिखा सकते हैं। आर्गुमेंट स्ट्रक्चर की मौजूदगी, एडवर्बियल मॉडिफिकेशन, एस्पेक्टुअल मॉडिफिकेशन, और प्रोडक्टिविटी जैसे डायग्नोस्टिक्स का इस्तेमाल करके, यह डिसर्टेशन दिखाता है कि संताली जीरो नॉमिनल्स में वर्बल खासियतें होती हैं और उनमें रूट-लेवल की खासियतें नहीं होती हैं। इन टेस्ट का इस्तेमाल करके, यह डिसर्टेशन यह भी दिखाता है कि संताली जीरो नॉमिनल्स हायर डेरिवेशनल पॉइंट्स, जैसे *vP*, *VoiceP*, और *AspectP* लेवल पर बनते हैं। इसे नाउननेस स्क्वश (रॉस, 1973) से जोड़ते हुए, यह डिसर्टेशन दिखाता है कि संताली नॉमिनलाइजेशन अलग-अलग स्ट्रक्चरल हाइट्स पर होता है: प्रोटोटाइपिकल और एफिक्सल नॉमिनल्स रूट-डेरिव्ड होते हैं, जबकि जीरो नॉमिनल्स, मॉर्फोलॉजिकली सिंप्लेक्स होने के बावजूद, सिंटैक्टिक कॉम्प्लेक्सिटी दिखाते हैं और एक्सटेंडेड वर्बल फंक्शनल स्ट्रक्चर तक पहुँचते हैं।

दूसरा रिसर्च सवाल यह बताता है कि वर्बलाइजेशन, जिसे आमतौर पर क्रॉस-लिंग्विस्टिकली एक खास ऑपरेशन माना जाता है, कैसे सिस्टमैटिक सिंटैक्टिक कंपोजिशनैलिटी दिखाता है, जिससे संताली में प्रोडक्टिव चेंज-ऑफ-स्टेट सिमेंटिक्स मिलता है। संताली में रूट-डेरिव्ड वर्ब और स्टेम-डेरिव्ड वर्ब दोनों होते हैं, जो पहले से कैटेगराइज़ किए गए नाउन या एडजेक्टिव से बने होते हैं। स्टेम-डेरिवेशन के मामले में, संताली किसी भी कैटेगराइज़ या फ्रेज़ल स्टेम को प्रोडक्टिव तरीके से वर्बलाइज़ करता है, जिसमें चेंज-ऑफ-स्टेट का मतलब बताया जाता है। यह डिसर्टेशन इस फेनोमेना का एनालिसिस करने के लिए एक फीचर-बेस्ड अकाउंट प्रपोज़ करती है, जहाँ वर्बलाइज़र में एक [FIENT] फीचर होता है जो ट्रांज़िशन इवेंट को एनकोड करता है, चेंज-ऑफ-स्टेट इंटरप्रीटेशन को उसके कॉम्प्लिमेंट को असाइन करता है और इनकोहेटिव और कॉज़ेटिव इंटरप्रीटेशन देता है।

तीसरा सवाल एडजेक्टिव के इंटरनल स्ट्रक्चर के बारे में है, एक ऐसी कैटेगरी जिसकी मौजूदगी टाइपोलॉजिकल लिटरेचर में अनिश्चित बताई गई है। खास डायग्नोस्टिक्स का इस्तेमाल करके, डिसर्टेशन सबसे पहले यह बताती है कि संताली में एडजेक्टिव एक कम्पोजिट कैटेगरी के रूप में होते हैं जो रूट और स्टेम दोनों के एडजेक्टिवाइजेशन से बनते हैं। रूट-डेरिव्ड एडजेक्टिव इडियोसिंक्रेटिक होते हैं, उनमें साफ़ मॉर्फोलॉजिकल एडजेक्टिव नहीं होते हैं, और वे डेरिवेशन में नीचे बनते हैं। इसके विपरीत, स्टेम से निकले एडजेक्टिव रेगुलर और कम्पोजिशनल होते हैं, जिन्हें ओवरट एडजेक्टिवाइज़र *-ih(aa)* से पहचाना जाता है। यह चैप्टर स्टेम-डेरिव्ड एडजेक्टिवाइजेशन में स्ट्रक्चरल एसिमेट्री को भी दिखाता है, यह दिखाकर कि कैसे संताली एडजेक्टिवाइज़र नॉमिनल्स डोमेन में NumP और DP जैसे एक्सटेंडेड फंक्शनल प्रोजेक्शन के साथ मर्ज हो सकते हैं, जबकि यह वर्बल डोमेन में *vP*-लेवल तक ही सीमित है। यह एसिमेट्री दिखाती है कि कैसे कैटेगराइजेशन इनपुट सेंसिटिव होता है और अलग-अलग स्ट्रक्चरल पॉइंट पर मर्ज करने के लिए कैटेगराइज़र का सिलेक्शनल रिस्ट्रिक्शन इनपुट कैटेगरीज़ पर निर्भर हो सकता है।

इस डिसर्टेशन के निष्कर्ष से कई थ्योरेटिकल निहितार्थ सामने आते हैं। सबसे पहले, मॉर्फोलॉजिकली सिंप्लेक्स यूनिट्स में डेरिवेशनल कॉम्प्लेक्सिटी भी हो सकती है क्योंकि कैटेगरीज़ में अलग-अलग डेरिवेशनल हाइट्स पर बनने वाले कॉम्प्लेक्स इंटरनल स्ट्रक्चर हो सकते हैं। दूसरा, प्रोडक्टिव कैटेगराइजेशन पर रेगुलर सिमेंटिक्स, कैटेगराइज़र हेड पर मौजूद फीचर का नतीजा हो सकता है। तीसरा, कैटेगराइजेशन भी इनपुट सेंसिटिव होता है, जो इनपुट कैटेगरी के आधार पर कॉम्प्लिमेंट के तौर पर अलग-अलग स्ट्रक्चरल हाइट्स चुनता है, जैसा कि हम संताली एडजेक्टिवाइजेशन से सीखते हैं। इसका यह भी मतलब है कि जब किसी भाषा का कैटेगराइजेशन सिस्टम बहुत ज्यादा फ्लूइडिटी दिखाता है, तब भी यह डेरिवेशनल ऑपरेशन में कुछ रुकावटें बनाए रखता है। क्योंकि शब्द खास कैटेगराइज़र के मर्ज का इस्तेमाल करके एक जैसे बनते हैं, इसलिए मर्ज को सिंटैक्स में क्रॉस-लिंग्विस्टिकली कैटेगरी बनाने के लिए पर्याप्त एकमात्र फ़ंडामेंटल ऑपरेशन माना जा सकता है।

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List of Abbreviations

1	1st Person	GEN	Genitive
2	2nd Person	HON	Honorificity
3	3rd Person	IA	Indo Aryan
A	Adjective	IE	Indo European
AA	Austro-Asiatic	LC	Lexical Category
ACC	Accusative	LCIR	Locality Constraint on the Interpretation of Roots
ACT	Active Voice		
ADJZ	Adjectivizer	LF	Logical Form
ANIM	Animate	MID	Middle Voice
AP	Adjective Phrase	MS	Morphological Structure
C	Complementizer	N	Noun
CID	Closed Interpretation Domain	NP	Noun Phrase
COP	Copula	NMLZ	Nominalizer
CUC	Canonical Use Constraint	NOM	Nominative
D	Determiner	OBJ	Object marker
DM	Distributed Morphology	P	Preposition
DP	Determiner Phrase	PF	Phonetic Form
FC	Functional Category	Phi	Person-Number-Gender
FIN	Finite	PL	Plural

POS	Part(s) of Speech	T	Tense
PP	Preposition Phrase	TAM	Tense-Aspect-Mood
PST	Past Tense	TP	Tense Phrase
SC	Small Clause	V	Verb
SG	Singular	VI	Vocabulary Item
SUBJ	Subject marker	VP	Verb Phrase

Chapter 1

What are Lexical Categories

1.1 Introduction

Categorization is the most rudimentary trait of human cognition (Harnad, 2005). Humans conceptualize not only physical objects but also abstract concepts, like words, based on specific shared properties. In syntactic theory, two major sub-parts of categorization are lexical categories like nouns, verbs, and adjectives and functional categories like determiners, tense, complementizers, etc. Lexical categories such as nouns, verbs, and adjectives are content words that denote referential, predicative, and modification functions, respectively. The current study focuses on lexical categories, which are, in Baker's (2003) words, the fundamental concepts that humans learn.

Formation of lexical categories in the syntactic workspace has been a part of generative investigation since Halle and Marantz (1993). Non-lexicalists posit that syntax, the derivational engine associated with sentence formation, also forms complex words. A fundamental question addressed in this dissertation concerns the extent to which syntax affects the internal structure of words. More specifically, I investigate how much phrasal structural and semantic information categories encode when they are formed in the syntax. This dissertation addresses the above question by examining the categorial fluidity of nouns, verbs, and adjectives in Santali, an Austroasiatic language spoken in India.

This chapter discusses the nature and treatment of lexical categories in various traditional and theoretical traditions. Beginning with a historical overview of the parts of speech system

and the content vs. functional distinction in Section 1.2, the chapter moves to examine how categorization has been approached within generative grammar in Section 1.3. Further, the discussion moves to a fluid categorization system focusing particularly on Santali that challenges traditional categorial distinction in the lexicon and calls for a syntactocentric analysis. Following this, the chapter moves to outline the research questions and the methodology employed in the current work.

1.2 Historical Prelude to Categorization

1.2.1 Parts of Speech

Traditionally, prior to the lexical-functional division, the most rudimentary categorization was the distinction among the parts of speech. The most notable work on the classification of words is the Greek grammar *Téchnē Grammatikē* by Dionysius Thrax, which clearly categorized the eight parts of speech: noun, verb, participle, article, pronoun, preposition, adverb, and conjunction. However, the Paninian and Platonic grammars were the first to clearly categorize word classes (Robins, 1997). Paninian grammar notes four word classes, namely noun, verb, preposition, and particle. Since nouns are inflected for case and verbs are inflected for person, number, and tense, both nouns and verbs fall under the inflected subclass. Prepositions and particles were grouped as the uninflected subclass in Paninian grammar. Since the inflectional properties of a verb could allow it to stand alone in a sentence in Sanskrit, the verb was regarded as the head of the sentence, carrying its core semantic and grammatical properties.

On the other hand, the categorization of words in Europe begins with Plato's dialogue, *Cratylus*, where a primary distinction between *ónoma* (nouns) and *rhēma* (verbs) is observed. Plato's noun-verb distinction wasn't very different from the Paninian one since both saw the classification in terms of the subject vs. predicate distinction (Robins, 1997; Vinokurova, 2005).

According to Robins (ibid., p. 33), Aristotle further defines nouns and verbs in terms of their ability to indicate temporal reference. Nouns are defined to be entities carrying a meaning but no temporal reference. Verbs, on the contrary, are defined to carry temporal characteristics. Aristotle also observed that nouns have gender marking, adjectives have degrees, and verbs

have tense inflections other than present. Both Plato and Aristotle view parts of speech (POS) as syntactic elements, treating nouns and verbs as NP and VP.

Subsequently, Latin grammar suggests four inflectionally different classes of words according to their ability to mark tenses and cases (Table 1.1).

Class of Words	Inflectional Property
Nouns	Case inflection
Verbs	Tense inflection
Participles	Case and tense inflection
Adverbs	Neither

Table 1.1: Inflectionally Different Classes of Words in Latin Grammar

Due to the lack of definite articles in Latin, other Latin grammarians substituted the Greek article with a separate POS, *interjection*.

Traditionally, these POS are partly language-specific. While these traditional classifications of POS provided the foundation for grammatical analysis, a more fundamental distinction existed to accommodate these categories. This led to a broader dichotomy in modern word class classification: the distinction between content and function words.

1.2.2 The Content vs. Functional Distinction

The POS were categorized according to their grammatical and semantic meanings as they evolved into a more finely tuned system. A more specific fundamental dichotomy of categories having a contentive and grammatical/functional description of lexical items is the distinction between lexical (LC) and functional categories (FC). The traditional morphological variety of content words, or open class of words, corresponds to the lexical category (LC) in the generative paradigm (see Rauh, 2010).

Content words are words with individual core meanings of a lexical item that also contribute to the overall meaning of a sentence. The major content words in the literature are nouns, specifying objects, places, and abstract concepts; verbs, specifying actions and states; and adjectives, denoting modification properties. Adjectives were developed as separate POSs, which were a subclass of nouns, by Greek and Latin grammarians. Content words are also known as the open class since they do not have any limitations on adding new words to the language. Processes such as compounding, coining, lexical borrowing, and derivation con-

tribute to the content words because of the openness of the class. There are pre-theoretical inferential diagnostics of the LCs like nouns, verbs, and adjectives that decide if a word belongs to any of the lexical categories. The empirical classification of the three major LCs is summarized in Table 1.2.

Noun	Verb	Adjective
Determiner	Tense, Aspect, Mood	Modifying properties
Adjectives	Phi features	Degree items
Quantifier/Numeral (Classifier)	Honorificity	Measurement items
Case-assigning properties	Transitivity	
Plurality		
Possessor/Possessum		

Table 1.2: Empirical Classification of Lexical Categories

However, according to Gil (2005), such traditional distinctions of lexical categories are not cross-linguistically attainable. The presence of a determiner (Gil, *ibid.*) with a noun and tense on a verb (Ritter and Wiltschko, 2014) are language-specific. For Gil, nouns have the semantic category of things prototypically, and verbs have a prototypical association of the semantic category of activity.

1.2.3 Distinction in the Prototype Theory

According to the prototype theory (Givón, 1986, ch. 3; Croft, 1991), categories are thought of as prototypes occupying fuzzy areas across the spectrum of time stability. Categories like nouns, verbs and adjectives are taken to be distinguished from one another depending on their prototypical nature. Nouns are, for example, understood to be the most time-stable, and verbs are the least time-stable LCs, putting adjectives somewhere in between in the spectrum of temporal stability.

The theory is criticized as it is taken to be syntax external (Newmeyer, 1998, ch. 4). According to Newmeyer, a prototypical noun like ‘rock’ and a less prototypical noun like ‘theory’ behave the same way. Similarly, there is no categorial difference between a prototypical verb like ‘buy’ and a less prototypical verb like ‘instantiate’. In the same line of thought, Baker (2003) discusses how ‘eat’ is a prototypical verb, whereas ‘hunger’ as a verb is less prototypical. ‘Hunger’, in fact, is more prototypically closer to the adjective ‘hungry’. Even if ‘hunger’

is prototypically close to ‘hungry’, the syntax of both have differences. ‘Hungry’ requires a copula (1a), can modify a noun (1c), can be compatible with degree expressions (1e), and can be a resultative secondary predicate, whereas ‘hunger’ does not (1a, 1d, 1f). Also, ‘hunger’ can bear tense morphology (1g), but ‘hungry’ cannot (1h).

- (1) a. Chris **(is)* hungry.
 b. Chris hungers.
 c. a hungry person
 d. **a* hunger person
 e. Chris is as hungry as Pat.
 f. **Chris* is as hunger as Pat.
 g. Chris hungered.
 h. **Chris* (was) hungried.

Panagiotidis (2015) argues against the prototype theoretic approach based on its limited predictive power. He discusses the example of the non-prototypical noun ‘tachyon’ because it is not time-stable, and the non-prototypical verb ‘persist’ as a time-stable verb. Having traced the distribution of POS in the pre-generative tradition historically, we now turn our attention to the treatment of categories in the generative linguistic tradition. While traditional grammars primarily focused on the distribution of the POS in sentences, the generative tradition introduced a more formalized approach, emphasizing the formal nature of the categories. In the next section we explore how categorization is studied in the generative literature.

1.3 Categorization in the Generative Tradition

In the generative paradigm, languages are considered to differentiate between lexical and functional categories (Abney, 1987; Chomsky, 1995). In the generative literature, the major questions are (i) where the lexical categories were formed and (ii) what constitutes an LC. Zooming out, words have been questioned in the generative tradition for more than last fifty years. There are two opposing ways of answering the above questions. The first is the lexicalist answer that posits that words are primitive to grammar. They are the minimal units and are listed in a non-generative repository, named the lexicon. LCs have their categorial status and the insertion

frames (subcategorization, selections restriction, etc.) in the lexicon. Their categorial status is dependent on the binary feature bundle ($\pm N$, $\pm V$, etc.). The second way of answering the above questions is from a non-lexicalist point of view. It assumes that roots (not words) are primitive to grammar. Words are complex units, and they are formed using similar kind of combinatorial rules used for sentence formation.

The next subsections deal with the development of these models in the last five decades and how these models explain word formation, more specifically, category formation.

1.3.1 The Lexicalist Position

The lexicalist view, a theoretical standpoint associated with Chomsky (1970), suggests a system for word formation distinct from the system of sentence formation. The fundamental understanding of the model is that words are formed in a pre-syntactic lexicon by the application of lexical rules, different from the syntactic transformations. The lexicalist tradition emerges with Chomsky's (1965) introduction of the 'lexicon', conceptualized as a non-generative module containing an unordered set of lexical entries, each equipped with feature bundles and associated redundant rules. Before the introduction of the lexicon, categories were part of the rewrite system. Even the difference between verb types was also a part of the transformational system. Borer (2017), using verbs like 'head', 'read', and 'think', shows how the rewrite rule, before *Aspects*, dealt with verbs (2) by dividing it as V_T (transitive), V_I (intransitive), V_D (directional), and V_P (propositional).

(2) $VP \rightarrow V_T NP$

$VP \rightarrow V_I$

$VP \rightarrow V_D PP$

$VP \rightarrow V_P S$

$V_T \rightarrow \text{head, think, read}$

$V_I \rightarrow \text{read, think}$

$V_P \rightarrow \text{think, read}$

$V_D \rightarrow \text{head}$

(Chomsky, 1965, p. 77, 115; Borer, 2017, p. 112)

In "*Aspects of the Theory of Syntax*" (*Aspects*) Chomsky (1965) puts the unpredictable and idiosyncratic part of words in the lexicon. In *Aspects*, the redundant rewrite rules of different

types of V got changed to the general rewrite rule like (3).

- (3) $VP \rightarrow V (NP) (S)$
 $V \rightarrow \text{'think'}, \text{'head'}, \text{'read'}$

This elimination of lexical terms from the syntactic workspace required a formal component: the lexicon. After the introduction of the lexicon, the insertion frame of a verb like 'read' would look like (4), where the verb carries its transitivity, semantics, and pronunciation.

- (4) 'read': [+V, +[____ {NP, $\emptyset$ }], +*READ*, +/rɪd/]

(Borer, 2017, p. 113)

Borer shows how after the lexicon was introduced, the different meanings of the identical PF of 'head' in (5), became a part of the listed elements of the lexicon.

- (5) a. Kim **headed** the team. (head: *LEAD*)
b. Kim **headed** towards the team. (head: *ADVANCE(D)*)
c. Kim's **head** is big. (head: *BODYPART*)

The idiosyncratic meanings and argument structure of the categories ('head' in 5) cannot be predicted from their insertion frames. There is no regularity or semantic compositionality between 'head': *BODYPART* and 'head': *LEAD*. Also, syntax cannot predict the presence of the obligatory direct object or the absence of the sentential argument of 'head': *LEAD*. This semantic idiosyncrasy and the syntactic unpredictability, like the argument structure information, led Chomsky to posit that the information, like in (6), is a part of the lexicon.

- (6) Borer (2017)
a. *head1*: [+V, +[____ NP], *LEAD*, +/hɛd/]
b. *head2*: [+V, +[____ DIR], *ADVANCE*, +/hɛd/]
c. *head3*: [+N, +count, *BODY PART*, +/hɛd/]

In "Remarks on Nominalization" (*Remarks*), Chomsky (1970) continues the lexicalist agenda to strengthen the lexicon. Chomsky makes the pre-syntactic lexicon so strong that it is not a mere repository of the idiosyncrasies of a language. A lexicon contains pronunciation, semantic properties, morphological properties like the categorial features, phi features, etc., and syntactic properties like the argument structure of its entries (Spencer, 1991, p. 47). According to

Borer (ibid., p. 114), *Remarks* focused on the comparison between the syntactically predictable and the syntactically unpredictable elements, considering the former transformational and the latter lexical. The ultimate motivation *Remarks* carried was to remove all relations from the transformational component that required a lexically specified rule. According to Chomsky, the sentences in (8) can be transformationally formed from the sentences in (7). The nominal counterparts (8) of the verbs in (7) are compositional and predictable from the verbal structures in (7). Chomsky calls the nominals in (8), gerundive nominals¹ ('verbal gerunds' according to Abney (1987)).

(7) Chomsky (1970), Borer (2017)

- a. The enemy has **destroyed** the city.
- b. The scientist **knew** the solution.
- c. The builder **enhanced** the foundations.

(8) Chomsky (1970), Borer (2017)

- a. (The enemy) **having** destroyed the city.
- b. (The scientist) **knowing** the solution.
- c. (The builder) **enhancing** the foundations.

Following Chomsky, Borer postulates the transformational rules in (9) that maps the verbs in (7) to the gerundive nominals in (8).

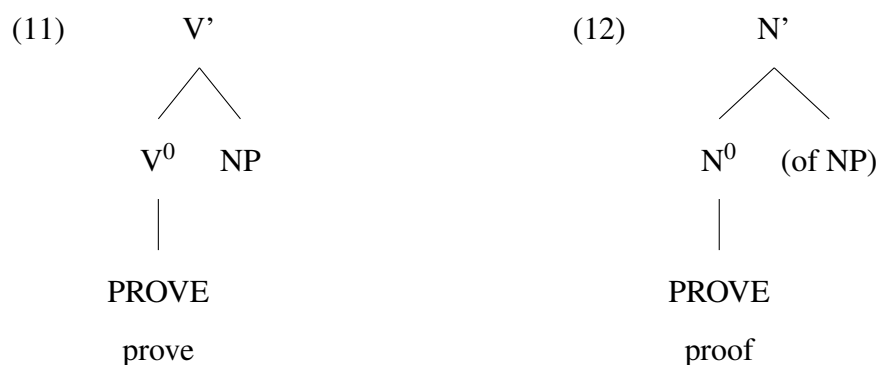
- (9)
- a. Eliminate Tense
 - b. make the subject optional
 - c. Add *-ing* to the verb or the highest auxiliary

However, there is no compositional or predictable relation between the sentences in (7) and the nominals in (10).

- (10)
- a. (The enemy's) **destruction** (of the city)
 - b. (The scientist's) **knowledge** (of the solution)
 - c. (The builder's) **enhancement** (of the foundations)

¹ See Section 2.4 for a discussion of derived and gerundive nominals.

According to Borer (ibid.), even if the gerundive nominals in (10) are related to the verbs ‘destroy’, ‘know’, and ‘enhance’, respectively, they require the item-specific properties like phonological unpredictability (choice of nominalizer), and semantic unpredictability (meaning difference from the source verb). However, Chomsky also notes in *Remarks* that some derived nominals are related to their verbal counterparts. To capture this, Chomsky posits the ‘X bar’ structure, a cross-categorial syntactic insertion frame. The hierarchical structure is the same across categories. In the case of the derived nominals, which are related to their source verbs (proof, destruction, civilization, etc.), there is a single lexical entry with the semantics. This entry lacks the PF. Chomsky shows that the verbal PF representation happens if the insertion happens under a V^0 , (11) and the nominal PF representation is a result of the insertion under the N^0 (12).



In these specific derived nominals, which are connected to their verbal bases, the categorial label emerges from the syntax. However, this is not a general rule in the lexicalist tradition because derived nominals, which are not compositional, must be listed in the lexicon.

According to the lexicalist model, the interaction of words with sentences is unidirectional, i.e., the output of the system for word formation is the input for the system of phrase/sentence building (Bruening, 2018). This can be better understood through Williams (2007), where ‘construction’, a noun formed as ‘construct_v+ -ion’ in the lexicon, results in only the singular noun ‘construction’ being visible to the syntax for phrase/ sentence formation. Crucially, the internal morphological structure becomes opaque to syntactic operations, supporting the lexicalist claim about the autonomy of word formation.

This view that some nominalizations (derived ones) take place in the lexicon is known as the Weak Lexicalist Hypothesis in the literature (Spencer, 1991, p. 179). Proponents of a more

radical version, the Strong Lexicalist Hypothesis, propose that all types of word formation occur in the lexicon without any interference from syntactic structure by including inflectional word formation in the lexicon. While strengthening the lexicalist agenda, Halle (1973) proposes a morphological component for word formation in opposition to the transformationalist approach. According to him, if grammar is a formal representation of human knowledge of the language, there must be something to account for the lexical knowledge of a speaker. A pre-syntactic component decides the distinction between a word and a non-word in a language; if a word is composed in a particular manner (13a), it cannot be concatenated in other forms (13b, 13c).

- (13) a. trans-form-at-ion-al
 b. *ion-trans-al-at-form
 c. *al-form-at-ion-trans

One of the most significant diagnostics explored in this section regarding the placement of LCs (particularly nominals) in either the lexicon or syntax is the distinction between idiosyncratic and compositional properties. Lexicalist approaches suggest that elements exhibiting idiosyncratic properties should be listed in the lexicon, whereas those displaying regularity or compositionality are better accounted for through syntactic formation. Halle (ibid.) further posits that if inflectional components exhibit idiosyncrasy, this provides strong evidence to treat inflection as part of the lexicon's listed elements.

According to Halle, the pre-syntactic component (lexicon) not only takes care of derivational word formation by changing the categorical features of a word (i.e., Chomsky's nominalization of verbs) but also accounts for the idiosyncrasies of inflectional word formation. In Russian, the instrumental case markers *-om* or *-oj* attached to a noun designating a particular time of year (summer, winter, etc.) or time of day (morning, night, etc.) behave differently when attached to other nouns in the language (14).

- (14) a. *let-om* (in summer)
 b. *zim-oj* (in winter)
 c. *avgust-om* (*in August)
 d. *obed-om* (*at dinner)

The adverbial property in (14a, 14b) is absent in (14c, 14d). The lexicon takes care of this idiosyncratic inflectional information.

The lexicalist framework thus presents a comprehensive view of categorization where both derivational and inflectional word formation occur within the lexicon. The evidence from phenomena like Russian instrumental case markers (15), following Halle (1973), strengthens the lexicalist position that even seemingly regular morphological processes can exhibit unpredictable semantic and syntactic behaviors that are best captured through lexical listing rather than syntactic derivation. This treatment of both derivational and inflectional morphology within the lexicon provides a unified account of categorization while addressing the complex interplay between regular and idiosyncratic properties in word formation.

Having established the theoretical foundations of the lexicalist position, it is crucial to understand the specific lexical categories listed in the lexicon and their representation.

1.3.1.1 A Feature-Based Model

Chomsky (1970) interprets lexical categories in terms of abstract binary features. Two binary features, [N] and [V], along with their $\pm$ values, can be used to characterize the four primitive lexical categories in the lexicon: noun, verb, adjective, and preposition (Table 1.3).

Category	[N]	[V]
Noun	+	–
Verb	–	+
Adjective	+	+
Preposition	–	–

Table 1.3: Feature-based classification of lexical categories.

The description of the lexical categories in Table 1.3 forms four sets of natural classes based on a common $+$ or $-$ value of the features (Stowell, 1981, pp. 21–22). Chomsky’s feature system includes only the natural classes (Table 1.4), as all the morphological and syntactic rules can refer to natural classes.

Feature	Natural Classes
[+N]	Nouns, Adjectives
[-N]	Verbs, Prepositions
[+V]	Verbs, Adjectives
[-V]	Prepositions, Nouns

Table 1.4: Natural classes of lexical categories (Stowell, 1981; Panagiotidis, 2015).

A pair like a noun and an adjective would form a natural class, but a noun and a verb wouldn't. Syntactic generalizations like [-N] categories are case assigners, or [+N] categories have 'of' insertion in English (15), and cannot be applied to a set of unnatural classes (Noun and Verb; and Adjective and Preposition) (Stowell, *Ibid.*, p. 23).

- (15) a. The enemy's destruction of the city.
b. John's fear of heights.

To summarize, lexical categories include nouns, verbs, adjectives, and prepositions in Chomsky's feature-based category system, where P was included in a revised version to include PP in the X-bar system (Stowell, 1981). Among other feature-based analyses of lexical categories, Baker (2003) proposes a reduced set of lexical categories following Chomsky's ideas of the lexicon.

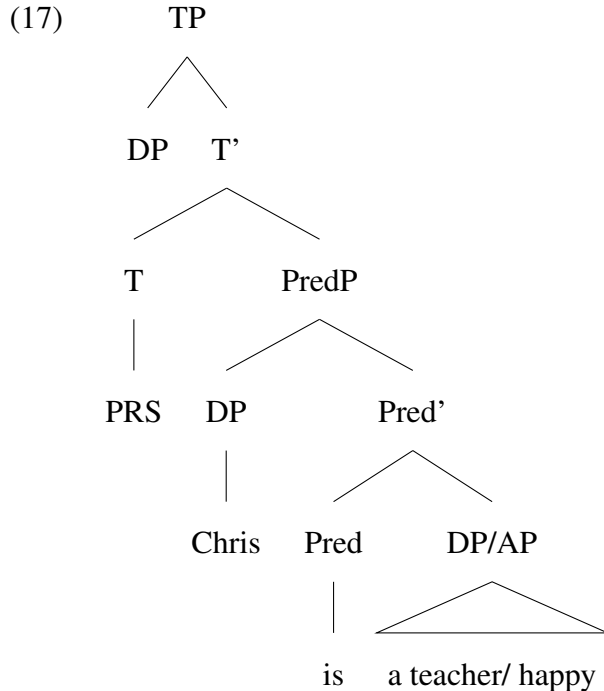
1.3.1.2 The Core Lexical Categories

While Chomsky (1970) treated nouns, verbs, adjectives, and prepositions as lexical categories based on $[\pm N]$ and $[\pm V]$ features, Baker (2003) redefines lexical categories structurally, excluding prepositions from this class. Baker considers only nouns, verbs, and adjectives as the core lexical categories. He provides the content to the features [N] and [V], which were merely taxonomic earlier (Panagiotidis, 2015).

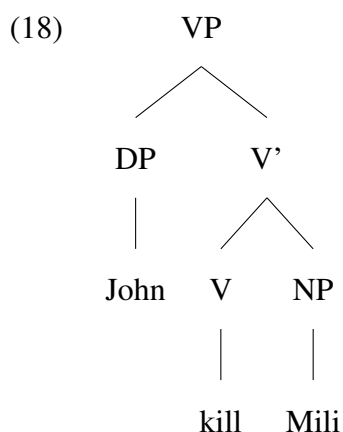
According to Baker, nouns are the only lexical category with a criterion of identity. N can be antecedents to anaphors (16a, 16b) and pronouns (16c). NPs can provide the antecedents for pronouns in discourse. Nouns behave as canonical arguments of a structure and receive theta roles (agent, patient, etc.). They are not the assigners of theta roles and require a functional head 'Pred' to assign theta roles (16).

- (16) a. [John_i] likes [_{reflexive} himself_i].

- b. [_{NP} John and Mary_j] like [_{reciprocal} each other_j].
- c. [John_i] said that [_{pronoun} he_i] would come to the party.



A verb, on the other hand, is a distinct category that has a specifier. According to Baker, the verbness of the verb is decided by its ability to have a specifier; no other category may do so (18). Verbs can assign theta role to its arguments. They are also known as the licensors of a subject.



Adjectives have neither referential indices like a noun nor specifiers like a verb. In Baker's system, adjectives are –N and –V, and he places adjectives in the “elsewhere condition” as they lack both a referential index like nouns and a specifier like verbs. They do not have a

defining syntactic property. Adjectives, like nouns, rely on a functional head to assign theta roles (11). Baker defines the abstract $[\pm N/V]$ features of Chomsky through the concepts of ‘nounness’ and ‘verbness’ that he assigns to lexical categories (LCs). Panagiotidis (2015) summarizes the semantic interpretations and the syntactic behaviours of this $[N]$ and $[V]$ in (Table 1.5).

Features	Semantic Interpretation	Syntactic Behaviour
$[N]$	Sortality	Referential index
$[V]$	Predication	Specifier

Table 1.5: Features and interpretations in Panagiotidis’ (2015) analysis.

In Baker’s system, prepositions, which belonged to lexical categories in the Chomskyan system, are part of the functional category. According to Baker, the number of prepositions in a language is quite small, forming a closed class. Even though languages like English have more than fifty prepositions in their vocabulary, this is very few compared to the uncountable number of nouns and verbs. Conversely, languages like Mohawk and Chichewa have only four and two to five prepositions, respectively. Baker argues that prepositions do not participate in derivational processes to form new words like nouns, verbs, and adjectives do.

Even though Baker started with the lexicalist binary features $[N]$ and $[V]$, he acknowledges that his idea of categories is structural (Baker, 2003, p. 267). For a verb to have verbness or $[V]$, it must have a specifier in syntax. Similarly, nouns must have referential indices in syntax. Baker considers nouns, verbs, and adjectives the only LCs, cross-linguistically by associating them with fundamental cognitive divisions in human perception. Nouns correspond to objects, verbs to actions, and adjectives to properties. He suggests that children naturally adopt these distinctions from an early stage, and all languages consistently represent them (p. 300).

In spite of the widespread acceptance of the lexicalist hypothesis in the literature, several theoretical and empirical challenges emerged that questioned the foundational assumptions of the lexicalist view. One of them was the assumption that only a particular subclass of nominals (derived nominals²) are listed in the lexicon, and the other subclass is derivationally formed in the syntax. This motivated researchers to question if both the types of nominals, or more broadly words can be formed only in syntax. Such inconsistencies in maintaining a strict sep-

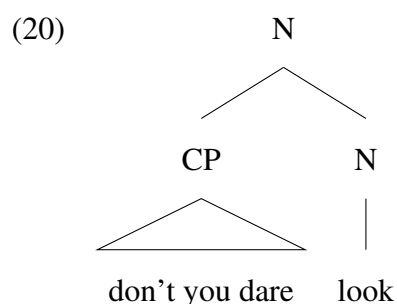
²See Section 2.4 for a detailed distinction between the two major subclasses of nominals.

aration between lexical and syntactic operations, particularly listing of words and sentence building in different workspaces, led subsequent researchers to identify further problems in the lexicalist assumption. The non-lexicalist assumptions, developed to address such issues, attempted to revisit the listedness of some words in the lexicon and formation of others in syntax, the unidirectional relation between the lexicon and the syntax, and the arbitrary distinction between word-level and phrasal idiosyncrasies, etc. In the subsequent subsection, we understand how non-lexicalist approaches propose a unified system for both word formation and sentence construction using the syntactic workspace and offers a more parsimonious understanding for categorization.

1.3.2 The Non-Lexicalist Assumptions

The recent generative literature argues against the lexicalist model, dispensing with the view that two different systems are required for word formation and sentence building. The argument is that only one of the systems (syntax here) can take care of both word and sentence formation. It is methodologically economical because one system is better than two. The lexicalist position that the output of the lexicon is always the input of syntax is disregarded. Bruening (2018) argues that the output of the phrasal system can feed into word formation through compounding, as shown in (19), represented by Bruening (ibid.) in (20).

(19) I gave her a *don't you dare* look.



Here, *don't you dare* is the first member of the compound: *don't you dare look*, implying that compound formation (word formation in general) is not a lexical process. The phrase as the first member of a compound is undoubtedly a phrase, not a lexicalized element, since it obeys syntax rules (21a). Violation of syntactic rules generates ill-formed structures (21b).

(21) Bruening (2018)

- a. She had that *I'm-so-proud-of-myself* look.
- b. She had that **myself-is-so-proud-of-me* look. (Reflexives are bad as subjects.)

Bruening claims that phrasal syntax can feed into word formation, illustrating how phrases can behave like nouns (22), when nouns are zero-derived forms of words:

- (22)
- a. *A know it all*
 - b. *A wanna be*
 - c. *A good for nothing*

The lexicalist condition that syntax cannot access the sub-word units, as words are formed in the lexicon (Lapointe, 1980, p. 8; Selkirk, 1982, p. 70; Bruening, *ibid.*, p. 13), is also discarded. ‘Ellipsis’ in coordination is taken as an example to observe that a single system can take care of both phrases and sentences. According to Bresnan and Mchombo (1995), ellipsis is a phrasal process that cannot be seen in sub-word units. Chaves (2008, 2014), however, argues differently. According to Chaves, ellipsis in coordination can target both phrases (23a) and sub-word units (23b, 23c).

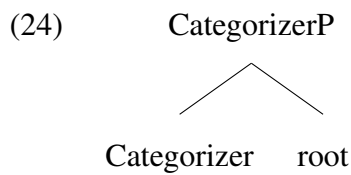
- (23)
- a. The break-in on Monday was a [~~rare breach of royal security~~] but [not unheard-of breach of royal security].
 - b. [~~Over application~~] and under application. (Chaves, 2014, p. 836)
 - c. Pre-~~revolutionary~~ and post-revolutionary France (Chaves, 2008, p. 264)

Ellipsis, in coordination, when applied to the sub-word level and phrasal level, refutes the whole idea of lexicalism, which claims that there should be no process that deals with both phrases and morphemes (Bruening, *ibid.*). Recent generative frameworks, like distributed morphology and exoskeletal frameworks, align with this non-lexicalist idea for both word formation and phrase building.

1.3.2.1 Distributed Morphology

Halle and Marantz (1993) and Marantz (1997) provide an alternative framework, called Distributed Morphology (DM) for word formation based on the non-lexicalist assumption that word formation and categorization occur in the syntax. DM, a framework within the Minimalist Program (Chomsky, 1995) argues against the idea of a separate Lexicon that acts as a

morphological component according to Chomsky (1970). According to this view, lexical items do not have any categorial feature in the lexicon (or its equivalent). They are stored as roots and receive their categorial values by merging with a categorizer, which is a functional head such as *n*, *v* or *a*. The syntactic environment (the complement position) of *n*, *v*, and *a* categorizes the acategorial roots as nouns, verbs, and adjectives, as shown in (24), demonstrated by Don and van Lier (2013, p. 59) in (25). To summarize, according to DM, there are only two types of terminal nodes, namely the roots and the categorizers.³



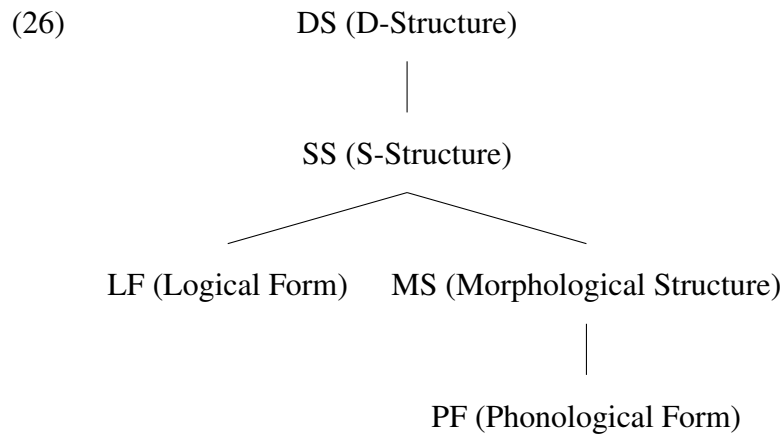
(25) Don & van Lier (ibid)



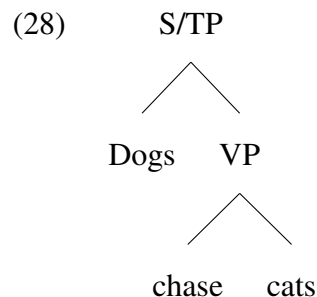
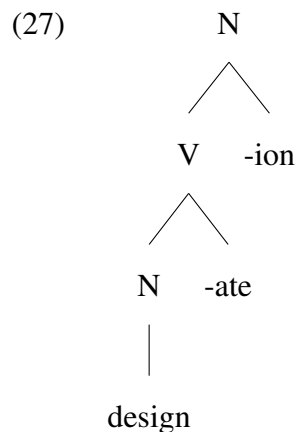
The lexicon in DM is divided into multiple modules like ‘the narrow lexicon’, ‘vocabulary’, and ‘encyclopedia’, which Marantz (1997) refers to as list 1, 2, and 3, respectively.

The narrow lexicon is a generative component that provides the acategorial roots and the grammatical features to the syntactic component for word formation. The 2nd list - vocabulary is crucial in assigning phonological forms after the syntactic structure is built. Lastly, the encyclopedia is a store to the special idiosyncratic meanings of the roots for particular syntactic contexts. It takes care of the idioms. Halle and Marantz (1993) propose the structure in (26) for word formation in DM. In (26), the morphological structure (MS) before the PF assigns the most compatible morphological form of the Vocabulary item (VI).

³Harley and Noyer (2000) refer to the roots and the categorizers as *l*-morphemes and *f*-morphemes, respectively.



The emergence and development of DM are based on both theoretical and empirical motivations. First, as Siddiqi (2010) discusses, the commonality between morphology and syntax, highly fusional languages like Latin and Greek are more morphology-dependent. On the other hand, more isolating languages like English and Mandarin depend on syntax. Grammatical features like case, tense, aspect, mood, voice, etc., are syntactic features with morphological realizations. Siddiqi also shows the parallel structure between morphology and syntax, since both words and phrases follow hierarchical structures (27-28).



A parallel between (27) and (28) also claims that the role of the word in the lexicalist tradition is now replaced by the role of the root. Roots are the most minimal units that work as inputs to the syntactic derivation. Both complex words and complex phrasal structures are the result of syntactic derivations. This commonality between the formation of words and sentences/phrases also reduces the need for a separate lexicon.

Secondly, 'Late Insertion' is a major theoretical motivation for DM. It is one of the key tenets of DM, which argues that phonological forms are inserted post-syntactically (Marantz, 1997, p.3). This opposes the lexicalist assumption in which words enter the derivation with specified phonological forms. The late insertion assumption explains the context-sensitive allomorphy. Since phonological form is assigned post-syntactically, it can refer back to syntactic and morphological structures before inserting the appropriate form. This explains the English plural allomorphy, where plural is just a feature [PL] in syntax. The feature gets its phonological form following the late insertion.

The third major motivation for DM to be considered as an architecture for word formation is 'Underspecification'. This principle of DM states that all grammatical features are not fully specified in the VI. They are minimally specified and compete for insertion. DM suggests that VIs do compete for insertion, where the most specified and compatible form of the VI wins the competition (subset principle) for the insertion at the terminal node. This explains the occurrence of a single morphological form in multiple grammatical environments (see Kramer, 2016). This also explains why irregular forms or specific rules block general rules while the insertion of the VI. Commonly known as the 'blocking effect', the preference for specific rules over general rules can be understood by the plural marking on English nouns like 'cow' and 'ox'. The irregular plural marker '-en' (specific for the noun 'ox') is chosen during Vocabulary Insertion by blocking the default plural marker '-s', which can act as a plural marker for the noun 'cow'. These motivations for DM challenge the lexicalist system that proposes a separate engine for word formation.

Marantz also argues against the Lexicon that treats word level idiosyncrasies as special and puts it in the lexicon. Words are not special in this sense because special meaning can be seen at any level. Idiomatic phrases also carry special semantics, which are also built in the syntax following the syntactic rules. Following Jackendoff (1996), Marantz argues that treating word meanings (idiosyncratic) as truly special by ignoring the special idiomatic meaning of phrases is redundant. In the lexicalist literature, idiosyncrasy has been treated as a part of the lexicon, whereas anything that is predictable or regular is encoded in the syntax. This is followed at the word level, treating idiosyncratic words as a part of the lexicon while ignoring the phrasal idioms. Marantz argues that there is not a clear distinction between the special meaning of words and the special meaning of phrases. Units of any size can have meanings that are not

composed of the meanings of the parts they are derived from. This makes Marantz claim that the only elements that must have special meanings are roots. Both words and phrases are built using the similar combinatorial rules in syntax.

Empirically also, if we obtain the assumption that regularity or predictability should be encoded in syntax, we should follow a non-lexicalist approach like DM to get rid of the listedness seen in (5, 6). Also, in a language where cross-categorical conversion is extremely regular, DM can work as one of the best frameworks for studying category formation.

1.3.2.2 Exoskeleton

A more radical assumption of the non-lexicalist view, known as the exoskeletal approach (Borer, 2003, 2005) proposes that roots not only do not have categorial features in the lexicon, but they also do not require categorizer heads like *n* or *v* in the syntax. Roots are categorized in the functional environment during the derivation by functional heads like D and T. The roots, also known as the encyclopedic items ⁴ (EI), are free from grammatical information such as category features, argument information, and word formation. Along with these EIs, the grammar also includes a functional lexicon embedded with grammatical formatives like $[\pm pl]$, $[\pm pst]$, etc., and category-labeled morphemes like *-ize*, *-al*, *-ful*, etc. (in the case of English). The grammatical formatives merge with the roots (EIs) to categorize the roots in a functional structure.

More specifically, a root becomes a VP when it is merged with a T (29a) and an NP when it is merged with a D (29b). The heads of the VP and the NP are verbs and nouns, respectively.

(29) Borer (2003, 2005)



We also need a phonological representation after the derivation. A structure like $V + \langle pst \rangle$ is

⁴Borer's (2003) Encyclopedia is the repository of the acategorial roots, which is equivalent to Chomsky's lexicon. Roots, the components of the encyclopedia, are called EIs.

phonologically represented in (30a, 30b), where the roots in the functional structure are *sink* and *dog*, respectively.

- (30) a. $[T[_v\text{sink}]] - \langle \text{pst} \rangle \rightarrow (\text{sank})$
 b. $[T[_v\text{dog}]] - \langle \text{pst} \rangle \rightarrow (\text{dogged})$

In addition to categorization, argument structure is also managed by the merger of the functional heads. Considering the rationale of this exoskeletal approach, the noun-verb distinction is taken to be purely configurational (Panagiotidis, 2015).

Now that there are two approaches to word formation, i.e., in the lexicon or in the syntax, we see that the syntactic word-formation models like DM can resolve the concerns that the lexicalist model had. Now, the question is whether the distinction among categories is universally present. The next section deals with the fluidity of categories in languages in order to address the arguments for and against the existence of distinct categories.

The non-lexicalist approaches discussed in this section, particularly DM and the Exoskeletal model, provide a theoretical framework that can account for languages with fluid categorization systems. Both models propose that roots are the most minimal units in a derivation and are fundamentally acategorical, acquiring their categorial status through syntactic operations rather than lexical specification. This theoretical position is particularly valuable when examining languages that demonstrate extensive categorial fluidity, where the same root can freely function across traditional lexical categories.

To summarize, we saw that both DM and the exoskeletal approach provide crucial theoretical frameworks to analyze categorial fluidity. In this dissertation, I use DM to study the formation of categories in Santali. Even though the core assumption of both the frameworks deals with syntactic formation of categories from acategorical roots, I show a number of motivations to adopt the DM approach in the current study. First, DM's categorizer heads (*n*, *v*, *a*) give us a more explicit mechanism to trace the derivational history of categorial stems. This is crucial because as we will see in Chapters 2, 3, and 4, Santali categories are not only formed from roots, they are also formed from already categorized stems through recategorization. Santali shows highly regular patterns in formation of nouns, verbs, and adjectives. The DM architecture captures the systematic regularity. Secondly, we will also see that the idiosyncratic or compositional interpretation of Santali categories are sensitive to locality constraints

and phasehood. DM takes the first categorial head as the first phase (Marantz, 2000) or closed interpretation domain (Arad, 2003) in the derivation, which predicts the idiosyncratic meaning or irregular morphology. The merge of the second categorizer, then yields compositional or transparent meanings. Such a phase-based analysis is not a design feature of the exoskeletal approach. The exoskeletal account primarily takes care of argument and event structures using the functional structure, which also deals with the formation of categories directly using the functional structures. In this dissertation, adopting the DM approach, I assume that categories are formed by categorizing heads within the functional structure, and not by the functional structure alone.

While an underspecificational approach like DM explains categorization as a syntactic phenomenon, universally, its full potential can be visualized when we consider languages with a fluid categorial status. Although much of syntactic theory (using a lexicalist position) has been built on languages with relatively rigid categorial distinctions, there is growing evidence that many languages have fluid categorial systems (Rijkhoff and van Lier, 2013). This brings us to a crucial empirical question: How widespread is categorial fluidity across languages, and what implications does this have for our theoretical models? As we will see in the following section, Munda languages like Santali present empirical evidence that strongly supports these non-lexicalist assumptions, showing that any root can take on different categorial functions through syntactic operations. The categories, formed syntactically, are regular, and carry compositionality.

1.4 A Fluid Categorization System

The distinction among lexical categories, more specifically, between N and V, is considered to be the most robust categorial distinction. If any language has a distinction among its lexical categories at all, it is between N and V (Sapir, 1944; Whaley, 1997; Evans, 2000; Croft, 2003). The universality of the distinction between N and V is attested in both typological and generative literature. Also, Baker (2003) considers A to be one of the universally present lexical categories.

A conflicting claim that has long been reported in the literature states that some languages do not have dedicated categories for basic communicative functions like reference, predication,

and modification. These languages, instead, have fluid word classes, the members of which can carry out more than one of these communicative functions (Rijkhoff and van Lier, 2013). Table (1.6) presents a list of languages with fluid categorial distribution. These are a few of the languages that are reported as the ones without a clear distinction among the LCs.

Language Family	Languages	Reference
Malayo-Polynesian	Tongan	Churchward (1953), Broschart (1997)
	Samoan	Churchward (1951)
	Tagalog	Foley (1998)
	Riau Indonesian	Gil (2013)
	Sri Lanka Malay	Nordhoff (2012)
Wakashan family	Nootka	Hockett (1958), Mithun (1999)
Turkic languages		
Australian languages		Rijkhoff and van Lier (2013)
Salishan languages		
Austroasiatic languages	Santali	MacPhail (1953), Rau (2013)
	Kharia	Peterson (2003, 2011)

Table 1.6: Languages with a categorial fluidity

According to Gil (2005), on the other hand, not having a distinction among categories does not mean lacking a distinction among functions like reference, predication, and modification. Rather, when we assume that there is no N-V distinction, it means that there is no distinction in the mental lexicon, which aligns with the non-lexicalist assumption discussed above. The next step is to understand this lack of categorial distinction through empirical data and see how much fluidity among LCs these flexible languages have.

The categorial fluidity exhibited by these languages (Table 1.6) provides compelling empirical support for non-lexicalist approaches to word formation. In these languages, roots can freely participate in both nominal and verbal environments, aligning with the core claims of underspecificational approaches, like DM that assumes that roots enter syntax without inherent category features. Among these languages, this study focuses on the Munda family, particularly Santali, to examine in detail how this categorial fluidity manifests and what it reveals about the nature of lexical categories cross-linguistically.

The next section will uncover this by providing empirical support from Santali, one of the North Munda languages, which is reported as a language having a flexible word class.

1.4.1 Santali

The Munda languages, which are the central focus of the current study through the empirical lens of Santali, belong to the Austroasiatic (AA) family and are primarily spoken in central and eastern India, including Odisha, Jharkhand, Chhattisgarh, Madhya Pradesh, Uttar Pradesh, West Bengal, and Maharashtra (Anderson, 2008, p. 1), as well as in eastern Nepal and western Bangladesh (Peterson, 2008).

This study focuses specifically on Santali, one of the most widely spoken languages from the North Munda sub-group and the third most spoken AA language overall. Santali is predominantly spoken by the Santal community across the same central and eastern Indian states mentioned above, with additional speaker populations in eastern Nepal and western Bangladesh (Anderson, 2008; Peterson, 2008). The empirical support for this study comes from the variety of Santali spoken in Odisha and the border regions between Odisha and Jharkhand (1.1). Due to its geographical distribution, Santali exists in close contact with the dominant Indo-Aryan (IA) languages of each region, resulting in a substantial population of bilingual speakers. This sustained contact has led to significant linguistic borrowing, with Santali borrowing various lexical items and grammatical features, including classifiers and word order patterns, from IA languages. Like other Munda languages, Santali exhibits agglutinating morphology.



Figure 1.1: Santali spoken regions in India

The highlighted regions in Figure (1.1) include districts like Saraikela, Purbi Singhbhum, Paschimi Singhbhum, Gumla, and Simdega in the state of Jharkhand and Mayurbhunj, Sundargarh, and Jharsuguda in Odisha where Santali is predominantly spoken.

1.4.2 Fluid Categorization in Munda Languages

Munda languages are well known in the typological literature for their weak distinction between noun and verb. The broader classification substituting the rudimentary noun-verb classification is the contrast between an argument and a predicate. Any morpheme can take on a case marker, number marker, and sometimes a definite marker and behave like an argument, or take TAM, phi, and voice markers and behave as a predicate. Along with words denoting actions, events, or conditions (prototypical verb-like words in other languages), almost every possible word in Kharia, another Munda language spoken in the same region, can behave like a predicate with verbal markers if a proper context is provided (31).

(31) Kharia (Malhotra, 1982)

- | | |
|--|---|
| a. <i>lebu Del-ki</i>
man come-PST.MID
'The man came.' | b. <i>bhagwan lebu-ki</i>
God man-PST.MID
'God mann-ed.' (became a man) |
|--|---|

The tense and voice marker *-ki*, which is attached to an action word come in (31a), is also attached to a referent word man in (31b). Therefore, *lebu* 'man', which behaves as a noun in (31a), functions like a verb in (31b). Similar examples in Santali (32) contribute to the non-lexicalist assumption that every morpheme is acategorical in the lexicon.

(32) Santali ⁵

- | | |
|--|--|
| a. <i>raajaa sen-en-a-e</i>
King go-PST.MID-FIN-3SG
'The king went.' | b. <i>uni raajaa-en-a-e</i>
He king-PST.MID-FIN-3SG
'He kinged.' (He became a king.) |
|--|--|

The middle voice, past tense marker *-en*, encoded with the finite marker, and the phi feature of the subject is marked on both an action word (32a) and a referent (32b), showing a similar pattern to Kharia. Apart from the verbalization of non-action words, voice plays a major role in carrying the transitivity of the clause. The active voice behaves as a transitivizer with a causative function (33a), and the middle voice as an intransitivizer with inchoative functions (33b).

- (33) a. *uni in-ke raajaa-ked-e-a-e*
He I-OBJ king-PST.ACT-OBJ-FIN-3SG.SUBJ
'He kinged me.' (He made me a king.)
- b. *in raajaa-en-a-i*
I king-PST.MID-FIN-1SG
'I kinged.' (I became a king.)

From the pilot data collected from the field, Santali is seen to have an extreme case of categorial fluidity in the case of verbalization; not only referent words like nouns (33), but also any word

⁵Uncited examples are from my own fieldwork data

can inflect for a verbal marker like tense and voice and get verbalized in syntax (34). Adjectives like *maaraang* ‘big’ (34a), demonstratives like *noa* ‘this’ (34b), kinship DPs like *ini bohya* ‘my sister’ (34c), animal names like *seta* ‘dog’ (34d), and even the most restricted kind of noun, i.e., proper names like *binit* ‘Vineet’ (34e) also get verbalized by taking the tense and voice clitics.

(34) Santali

- a. *daare maaraang-en-a*
Tree big-PST.MID-FIN
‘The tree bigged.’ (The tree became big)
- b. *hana noa-en-a*
That this-PST.MID-FIN
‘that this-ed.’ (That became this)
- c. *uni ini bohya-en-a-e*
She my sister-PST.MID-FIN-3SG
‘She my sister-ed.’⁶ (She became my sister)
- d. *uni seta-en-a-e*
He dog-PST.MID-FIN-3SG
‘He dogg-ed.’ (He became a dog)
- e. *uni binit-en-a-e*
He Vineet-PST.MID-FIN-3SG
‘He Vineet-ed.’ (He became Vineet)

The categorial fluidity in Santali highlights how elements from diverse lexical classes can undergo categorial change. This opens up further discussion on the universality and limitations of category distinctions across languages. In the subsequent section, I show diagnostics proposed by Evans and Osada (2005) to establish that Santali belongs to the typical group of languages that are known as flexible languages (Rijkhoff and van Lier, 2013), showing a fluid categorial system.

1.4.3 Diagnostics for Categorial Fluidity (Evans and Osada, 2005)

From the above data (32 - 34), it appears on the surface that Santali displays a high degree of flexibility in verbalization. Santali also fulfills almost all the requirements set by Evans and

⁶Context: My father married someone who had a daughter.

Osada (2005) to be considered to have categorial fluidity.

These diagnostics are particularly relevant for evaluating non-lexicalist approaches to word formation. If languages like Santali truly demonstrate systematic categorial fluidity, they would provide strong empirical support for frameworks like DM, which propose that roots enter syntax without inherent categorial features.

The first condition by Evans and Osada is **Bidirectionality**, where a language with a fluid categorization should have morphemes that can be used both as arguments and predicates. In Santali, a single morpheme can be used as both a predicate (V) and an argument (N). It is an impossible task to find complete symmetry in terms of the frequency of occurrence and is not necessarily required, according to Van Lier and Rijkhoff (2013). They suggest that despite whatever categorization system a language follows, it is commonly found that in all languages, certain functions are more actively connected to a word than some other functions. According to Croft (2001), English adjectives do occur as attributes and predicates, but not with an identical distribution. This does not mean that English lacks adjectives as a word class. Also, determining whether every morpheme in Santali can have nominal and verbal counterparts is beyond the scope of the current thesis.

This bidirectional flexibility in Santali aligns precisely with DM's assumption that roots are acategorial until they enter the syntactic derivation. The fact that the same root can take either nominal or verbal morphology based on its syntactic environment supports the Late Insertion principle of DM, where morphological forms are determined post-syntactically. However, roots never remain categorial post-syntactically following the categorization assumption (Marantz, 1997).

The second condition is **Semantic Compositionality**, meaning the semantic function of a morpheme in any two syntactic positions (argument and predicate) to be compositionally derivable from the meaning of the morpheme. The Santali example in (32) shows that *raajaa* (king_v) in (32b) has the same semantics as *raajaa* (king_n) in (32a).

This diagnostic test provides crucial support for the claim that categorization is purely determined by functional structure. When *raajaa* 'king' maintains its core semantics whether functioning as a noun or verb, it demonstrates that meaning is not tied to lexical category but rather interpreted compositionally within its functional environment - exactly as predicted by non-lexicalist approaches.

The third condition is **Exhaustiveness**, which states that all words in a language should have lexical overlap for the language to be considered a fluid language. Since it's not feasible to check all the lexemes of a language, Evans and Osada suggest testing some nominal subclasses like proper names, kinship terms, and animal and plant names that are restricted nominals and do not usually go through verbalization. Santali fulfills exhaustiveness, as proper names, kinship terms, animal and plant names, etc., can have a predicative use in the language (34).

The fact that even traditionally 'rigid' nominal classes in Santali (proper names, kinship terms, etc.) can function predicatively challenges lexicalist assumptions about inherent categories. Instead, it supports the non-lexicalist view that categorical restrictions in languages like English arise from syntactic constraints rather than lexical specification. This makes Santali's categorical system pertinent for testing the predictions of underspecificational frameworks like DM.

Even if categorial alternation (using the same root as a noun or a verb) is not an uncommon phenomenon in major Indo-European (IE) languages like English, there are some constraints that do not allow every root to be used flexibly. Hale and Keyser (1993) show that objects can be verbalized in restricted cases. According to Hale and Keyser (*ibid*), verbalization in (35) produces impossible constructions in English, unlike the noun incorporation structures in (36).

(35) Impossible constructions in English (Hale & Keyser, 1993, p. 61):

- a. *She churched her money.
(She gave a church her money.)
- b. *He bushed a trim.
(He gave a bush a trim.)
- c. *They housed a coat of paint.
(They gave a house a coat of paint.)

(36) Possible constructions in English:

- a. She shelved her books.
(She put her books on a shelf.)
- b. He saddled the horse.
(He provided the horse with a saddle.)

English denominal verbals (in terms of fluid verbalization) display idiosyncratic properties. In intransitive structures, they are restricted to certain unergative verbs (37a), not to the verbalization of other nouns (37b).

- (37) a. He danced/sneezed/laughed.
 b. *He manned/John-ed/stoned.

Hindi, a primary language spoken in the Santali-speaking region, does not have any categorial fluidity at all (Panagiotidis, 2015), whereas Odia, the other dominant language spoken in that region, has much lesser categorial fluidity, which is different from Santali.

1.4.4 Why Santali Categorization?

Studying Munda languages like Santali as the focus of my thesis has two primary reasons. The first reason is that these languages are crucial because of the fluidity of their lexical categorization (Peterson, 2011; Rau, 2013). Taking typological literature like Peterson (ibid), and Rau (ibid), who opine that there is no clear distinction between nouns and verbs in Munda languages, as a departure point will make this study contribute to the universality of word categories. Universally, languages are known to be carrying LCs. Whereas while discussing the lack of categorial distinction in Kharia, another Munda language spoken in the Santali-speaking region, Peterson (2011) states that languages need categories only when they have a necessity for certain categories. A categorial distinction does not need to be present obligatorily in all languages. Kharia, therefore, does not possess lexical categories like nouns, verbs, and adjectives. Rau (2013) has a similar line of thought for Santali.

On the other hand, supporting the universality of the presence of LCs, in the current proposal, I assume that LCs are present across languages with cross-linguistic variation stemming from whether the process takes place in the lexicon or in syntax. My assumption that concepts in languages like Santali are not left uncategorized is based on empirical diagnostics: if two roots are the only contentive morphemes in a sentence (38), they cannot take the same grammatical marker (nominal marker like case and verbal marker like tense/voice) in a sentence.

- (38) a. **hoR* *sen*
 man.NOM go.NOM

- b. **gidRa hoR*
boy.NOM man.NOM
- c. **hoR-en(-a-e) sen-en(-a-e)*
man-PST.MID go-PST.MID
- d. **gidRa-en(-a-e) hoR-en(-a-e)*
boy-PST.MID man-PST.MID

Since the grammatical marker *-en* carries tense and voice information, which are prototypically attached to verbs, and the null nominative marker carries the case information that is attached to nouns, it is possible for us to make a syntactic distinction between N and V. In (38), when two roots in a sentence take either case marker only or tense/voice marker only, ungrammatical structures are produced.

This constraint on grammatical marking provides evidence for how categorization operates in the syntax. While roots may be acategorical as proposed by DM, the syntactic system still enforces categorial distinctions through functional structure - roots cannot simultaneously occupy multiple categorial positions by becoming only noun or only verb in the same structure. This pattern aligns with both DM's view of post-syntactic morphological realization and the Exoskeletal approach's emphasis on functional structure determining category.

The view in the typological literature that Munda languages lack LCs is unwarranted. Therefore, even seemingly acategorical languages have categorial distinctions if we consider the possibility of categorization taking place syntactically.

Secondly, from the pilot data, it is seen that there is a compositional semantics of another category (N/A) in verbalization, which makes Santali different from languages like English, in which category-changing operations carry idiosyncrasy. An interesting contribution to the categorization literature would be to understand the structural head responsible for this compositionality. Also, another reason to study Santali is to explore how categorization like nominalization, verbalization, and adjectivization happens in the syntax and what the repercussions are on the grammar of the language.

1.5 Research Questions

This dissertation addresses three major questions. First, from the empirical perspective, it investigates how Santali categories, such as, nouns, verbs, and adjectives, are formed in Syntax. Second, it examines how a syntactic account like DM can account for Santali categorization. Third, and more broadly, what insights Santali contributes to the theoretical framework of DM itself.

This dissertation does not only contribute cross-linguistic evidence for syntactic categorization from an understudied language, like Santali. It expands the theoretical scope of DM by examining category formation configurationally and understanding what kind of complements categorizers can select in the derivation. These general issues are operationalized in the following specific research questions which are addressed in three dedicated chapters.

1. How are Santali nouns formed in the syntactic workspace? What are the structural positions for nominalization in Santali? How can morphologically zero nominals (ZNs), analyzed as root-derived in the literature, exhibit complex internal structure showing stem-level derivation in Santali?
2. How are verbs syntactically formed in Santali? How does verbalization, claimed to be an idiosyncratic operation cross-linguistically, demonstrate systematic syntactic compositionality yielding productive change-of-state semantics in Santali?
3. What is the internal structure of adjectives in Santali? Within stem-derived adjectives, how does the structural asymmetry between denominal and deverbal forms relate to categorizer height in Santali's derivation of adjectives?

1.6 Questionnaire and Methodology

The data for this research was collected through linguistic fieldwork conducted in Santali-speaking regions of Odisha and Jharkhand (Figure 1.2). The data for this study was collected from a total of 12 Santali speakers representing diverse backgrounds. The participant sample consisted of two distinct groups: field participants and remote participants. The preliminary data was collected from 8 Santali speakers (3 men and 5 women) residing in the

Santali-speaking regions of Odisha and Jharkhand. All field participants were bilingual, with secondary proficiency in either Odia or Hindi. The participants ranged in age from 25 to 60 years, representing various educational backgrounds and occupations within the community. To supplement the field data and provide additional perspectives, 4 additional participants were recruited online. These participants were graduate students at the University of Delhi whose mother tongue is Santali. These participants represented a younger, more formally educated demographic (age range 22-28) and possessed multilingual competence in Santali, Odia, Hindi, and English. The data collection sessions were conducted in accordance with ethical guidelines by the Indian Institute of Technology Delhi for linguistic fieldwork.

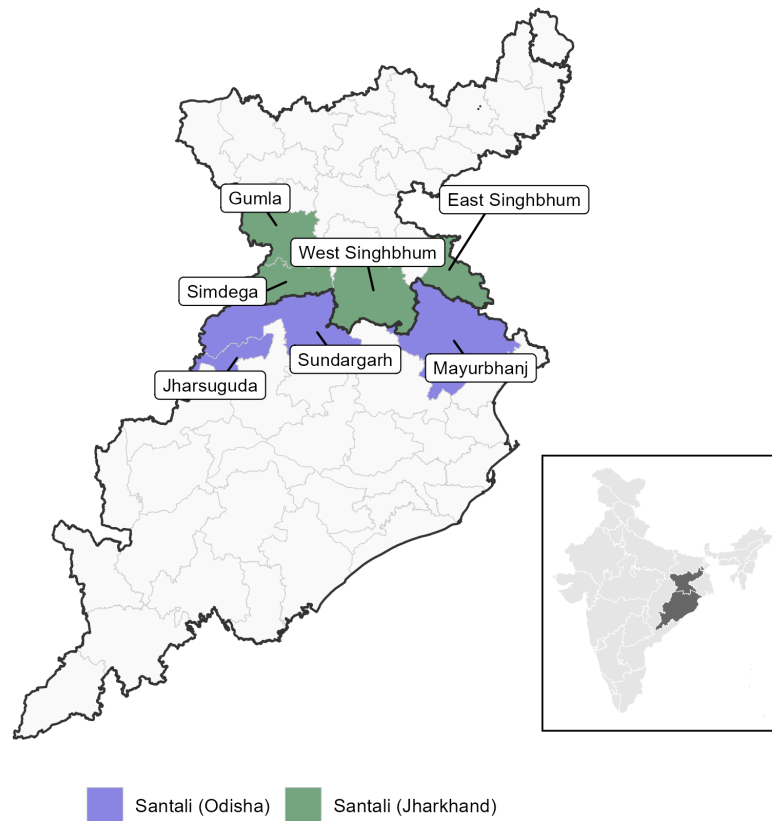


Figure 1.2: Fieldwork Region in Jharkhand and Odisha

The data were collected in three structured phases: (1) a Translation Phase, (2) an Acceptability Judgment Phase, and (3) a Categorical Fluidity Phase. Each phase is described below.

Phase 1: Translation In the first phase, the participants were presented with sentences in Hindi and Odia, since the Santali speakers involved in this study are bilinguals, speaking ei-

ther Odia in Odisha or Hindi in Jharkhand (Ghosh, 2008). The sentences covered prototypical nouns, verbs, and adjectives across various sentence types, like intransitive, transitive, and ditransitive, and across different tenses, aspects, moods, persons, and numbers. The participants were asked to provide Santali translations of these sentences. Since Santali does not have grammatical gender (Ghosh, 2008), the questionnaire was designed to avoid gender-sensitive constructions. Using Bodding (1929) as a reference for verification, this phase established the baseline paradigm of Santali grammatical categories.

The translations were then used to construct a grammatical template of Santali sentences across tense and argument structure types. A representative example of the kind of data elicited in this phase is given in the first line of (39) in Odia. The sentence was presented to the participants in Odia, and the translation was obtained in Santali (40).

(39) Odia

- a. *jon ga-l-aa*
John go-PST-3SG
'John went/left'
- b. *jon amba khaa-il-aa*
John mango eat-PST-3SG
'John ate a mango'

(40) Santali

- a. *jon sen-en-a-e*
John go-PST.MID-FIN-3SG
'John went/left'
- b. *jon uul jom-ke-a-e*
John mango eat-PST.ACT-FIN-3SG
'John ate a mango'

Phase 2: Acceptability Judgment In the second phase, the participants were presented with morphologically complex and syntactically novel structures, including structures that placed nouns in verbal paradigms and vice versa, to test their intuitions about the acceptability of categorial alternations. Both a sentence rating forced-choice task and a yes/no task were employed. In the sentence rating task, two sentences were presented simultaneously, one using a canonical verb and one using a noun in the same verbal inflectional frame. The participants

were asked to choose the sentence they found more acceptable, or to indicate that both were equally acceptable (41).

- (41) a. *uni sen-en-a-e* ☐
 he go-PST.MID-FIN-3SG
 ‘He went/left.’
- b. *uni setaa-en-a-e* ☐
 he dog-PST.MID-FIN-3SG
 ‘He dogged.’ (magical context)
- Lit: He became a dog

The yes/no task, on the other hand, presented individual phrases and sentences to the informants, who were asked to mark them as grammatical or ungrammatical (42) - (43). There were also some filler sentences used in the questionnaire to prevent participants from giving strategic or repetitive responses.

- (42) a. *uni-a sen baa-ij kusi adaa*
 he-GEN go NEG-1SG good feel
 ‘I did not like his leaving’
- b. *jon-rena eneh nyel-ke-a-ij*
 John-GEN play see-PST.ACT-FIN-1SG
 ‘I saw John’s playing.’
- (43) a. *gidRa hoR-en-a-e*
 kid man-PST.MID-FIN-3SG
 ‘The kid man-ed.’ (The kid became a man.)
- b. *bhit hariaal-en-a*
 wall green-PST.MID-FIN
 ‘The wall green-ed.’ (The wall became green.)
- c. *uni raska-en-a-e*
 he happy-PST.MID-FIN-3SG
 ‘He happied.’ (He became happy.)

The two tasks, while similar in design, differed in their elicitation goals: the sentence rating task revealed the relative acceptability of categorial alternations in a comparative context, whereas the yes/no task tested the absolute grammaticality of a given structure in isolation. In this way, the tasks were complementary, targeting different dimensions of the informants’ linguistic judgment.

Phase 3: Categorical Fluidity In the third phase, the acceptability judgment tasks were specifically targeted at testing the limits of categorical fluidity in Santali. The questionnaire contained sentences designed to place restricted nominal classes, including kinship terms, animal names, and proper names, into verbal inflectional paradigms to test the most extreme cases of verbalization. Sentences used in this phase were presented to the informants in Hindi and Odia. The example of the kind of input sentences are shown in Hindi in (44).

(44) Hindi

- a. *woh meri bahen ban-i*
(s)he I.GEN sister become-3SG.F
'She became my sister.'
- b. *woh raam ban-aa*
he Raam become-3SG.M
'He became Raam.'

The sentences to elicit verbalization of restricted elements like kinship terms and proper names (44) were collected in Santali (45), where the become interpretation was a result of the verbalization process (see Chapter 3 for the analysis).

(45) Santali

- a. *uni ini bohya-en-a-e*
(s)he I.GEN sister-PST.MID-FIN-3SG
'She my sister-ed.' (She became my sister.)
- b. *uni raam-en-a-e*
he Raam-PST.MID-FIN-3SG
'He Raam-ed.' (He became Raam.)

This demonstrates that in Santali, there is no morphosyntactic barrier between a canonical verb like *sen* 'go' and a proper noun like 'Raam' when the latter appears in a verbal inflectional paradigm or the former in a nominal one. The verbal markers, like past tense, middle voice, finiteness, and phi-agreement, attach freely to both, yielding grammatical and interpretable sentences. This observation is central to the arguments in all three content chapters that categorization in Santali is syntactically conditioned rather than being lexically listed elements.

1.7 Roadmap to the Thesis

Following the discussion of theoretical frameworks on categorization in Chapter 1, this dissertation examines specific categories such as nouns, verbs, and adjectives in Chapters 2, 3, and 4, respectively. Chapter 2 focuses on nominalization. After reviewing the relevant literature on nominalization, I analyze how nouns are formed in Santali within a non-lexicalist framework. The chapter discusses various types of nominals, including prototypical nominals, affixal nominals, and zero nominals, as well as their syntactic formation. Notably, zero nominals in Santali exhibit intriguing patterns in terms of high nominalization point that challenge the canonical root-derived assumption of zero nominal formation.

The third chapter focuses on verbalization in Santali. In this chapter, I first develop diagnostics to demonstrate that the fluidity observed in categorization through verbalized structures indeed represents the verbalization of other categories, such as noun and adjective. These diagnostics establish that the verbal markers are not merely surface-level affixes attached to nouns or adjectives, nor do they imply an underlying null copula. Following this, I explore the phenomenon of extreme verbalization in Santali, analyzing its structural features and addressing the semantic consequences of such verbalization in the language.

Subsequently, I move to a discussion of adjectives in Chapter 4, where I examine this third core lexical category in greater detail. In this chapter I show that adjectives, one of the very less-studied categories in the literature, are also formed derivationally in Syntax. I begin the chapter with the categorial status of adjectives and how it is puzzling across languages. Further I move to prove using diagnostics that adjectives constitute a composite category in Santali. Like both nouns and verbs, adjectives in Santali are also formed from both acategorial roots and already categorized stems. Adjectives are formed configurationally at multiple structural heights, showing asymmetry between nominal inputs and verbal inputs.

Chapter 2

Structure and Derivation of Nouns

2.1 Introduction

Nominalization, which involves the derivation of nominals from roots, verbal bases, or adjectival bases, has remained an ongoing debate for linguistic theory despite extensive research over the past few decades. Seminal studies by Lees (1960) and Chomsky (1970) established crucial preliminaries in understanding nominalizations, focusing on their overlapping categorial nature that exhibits characteristics of both nominal and verbal structures.

In Chapter 1, the pilot data reveal that categories in Santali exhibit significant fluidity. This fluidity of categorization can be extended to LCs, like the formation of nominals, verbals, and adjectivals. Focusing on nominalization, this chapter aims to examine whether nominalization in Santali demonstrates categorial fluidity. Since it has already been discussed in typological literature (Neukom, 2001; Ghosh, 2008) that there is no restriction for a word to behave like a noun, intuitively, it seems that a strong lexicalist position to analyze the formation of nominals in Santali will not work. Peterson (2003, 2008, 2010, 2011), states that in some Munda languages, any lexical element can behave like both nouns and verbs. He shows it in (1) by taking an example from Kharia, another Austroasiatic (AA) language spoken in the same geographical regions where Santali is spoken. Peterson's explanation is that there is no clear distinction between a noun and a verb in languages like Kharia. The difference comes from the positions a word occupies in a sentence, like that of an argument or a predicate.

(1) Kharia

- a. *lebu Del-ki*
man come-PST.MID
'The man came.'
- b. *bhagwan lebu-ki*
God man-PST.MID
'God mann-ed.' (God became a man)

In this chapter, I argue that categories in AA languages like Santali are formed derivationally in the syntactic workspace, the process is more complex than Peterson's idea of the difference between nouns and verbs depending on their position in a sentence. Not all nominals in Santali exhibit idiosyncratic meanings, making it unlikely that they are all formed directly from an acategorical root. Instead, nominals in Santali involve larger categorial structures during the process of nominalization. Santali exhibits three types of nominals: (i) prototypical nominals, (ii) affixal nominals, and (iii) zero nominals. I propose that zero nominals differ in their behavior and formation compared to prototypical and affixal nominals.

The broad research question I address in this chapter is, how are nominals formed in Santali? Since there is a lot of cross-categorial overlap among the categories in Santali, do all the nominals in Santali have other categorial properties, like verbal or adjectival properties?

Following a non-lexicalist account, I show in this Chapter that Prototypical nouns and affixal nominals are formed directly from the roots by merging with a nominalizer. On the other hand, zero nominals in Santali display verbal properties like gerunds. They are, therefore, formed with a verbal structure to accommodate Argument Structures, contradicting the 'root-derived' structure proposed for zero nominals in the literature (Borer, 2013).

In this chapter, I begin by defining nouns and nominalization. Following a survey of various types of nominals across the world's languages, I focus on Chomsky's (1970) seminal work, examining the derived-gerundive distinction in nominalization. I then explore the further decomposition of derived nominals before turning to a detailed analysis of nominal types in Santali. Specifically, I investigate the distribution and formation of prototypical nominals and affixal nominals in Santali, highlighting the innovative formation of zero nominals in the language. Using diagnostic tests, I demonstrate that zero nominals in Santali are stem (verb)-derived nominals rather than root-derived ones, retaining both the syntactic and semantic prop-

erties of the input verbs after nominalization.

2.2 What is a Noun?

2.2.1 Underlying Structure of a Noun

Traditionally, nouns are defined as entities that refer to a person, a place, or a thing. This semantic approach of defining nouns has been widely adopted in classical grammar. In traditional terms, nouns are defined based on their meaning, not their structural properties. The definition remains inadequate when we include nouns like ‘moment’, ‘theory’, ‘mind’, ‘calculation’, ‘happiness’, etc. These abstract nouns demonstrate that the traditional definition is too narrow and fails to capture the full range of elements that function as nouns cross-linguistically. The traditional notion also recognized morphological properties of nouns, like their ability to form plurals. However, these observations have been made from a purely descriptive point of view, without making any theoretical predictions.

Modern typological literature defines nouns in distributional terms. Nouns are defined depending on the position in which they occur in a sentence in relation to the other categories in the sentence. Nouns can appear with determiners, adjectives, possessors, etc.

In the generative tradition, according to Baker (2003), nouns are not primitive in the way generative theories assumed earlier (Chomsky, 1970). Baker also doesn’t argue for a purely syntactic underspecification approach to understand nouns to be formed as a product of syntax. His theory creates a balance between the lexicalist and the non-lexicalist approaches of defining nouns. As seen in chapter 1, instead of being formed by using primitive binary features, such as [+N-V], nouns are formed by the structural criterion of having [N]. This [N] for Baker is the noun’s ability to have a ‘referential index’ and to participate in various binding relations (Baker, 2003, p. 21; Baker, 2011, p. 1167). This ‘referential index’ is a syntactic tag that is connected to the ‘criterion of identity’, Baker’s semantic criterion of defining nouns. The criterion of identity shows a noun’s ability to define something that counts as the same entity or individual across different contexts. This sameness, according to Baker, is a precondition for a noun to hold a referential index in its syntactic version (pp. 95-96). The referential index allows the noun to serve as an argument in a sentence, work as a referent for pronouns,

etc. Baker points out that the criterion of identity becomes important when we consider quantificational expressions (2). Quantifiers like ‘every’, ‘some’, ‘each’ and ‘no’ are restricted to nouns only, or according to Baker, to the elements with criterion of identity. (Baker, 2003, p. 97) shows that the nouns in (2) do not refer to anything, but they still have the ‘criterion of identity’.

(2) Nouns as restrictors for quantifiers (Baker, 2003, p. 97)

- a. No [_{NP}letter(s)] arrived today.
- b. No [_{NP} wine] is served during Lent.
- c. *No [_{AP} rude] is tolerated here.
- d. *No [_{VP} pay(ing) parking fees] is pleasant.

Baker strengthens his defining criteria of nouns by proposing the Noun Licensing Condition, which states that a referential index must be co-indexed with an element it c-commands, such as a thematic role, a pronoun, or a trace. Baker further explains the referential index by stating that any element cannot simultaneously have a referential index and predication properties like licensing a specifier.

There are certain cross-linguistic generalizations for nouns. They can appear with articles to have a definite vs. indefinite distinction. They occupy the complement position of D in a DP and head their own phrasal projection: NP.

They display cardinality by appearing with numerals or plural markers. They behave as antecedents for pronouns and reflexives. Nouns constitute canonical argument phrases since they behave as the subject and object of a clause and are also theta role receivers.

Following Baker (2003), Prasada (2008), and Acquaviva (2009) define nouns from the sortality point of view. Nouns express ‘sorts’ or kind-level expressions to define entities in a speaker’s understanding of the world. Sortality defines nominals exclusively since it is associated with classification, number, quantification, individuation, etc. Prasada (2008) expands Baker’s understanding of sortality, which focuses on the criterion of identity only. Sortality, according to Prasada, incorporates application, identity, and individuation. Unlike verbs, nominals are not encoded with a temporal dimension. Nouns, according to Acquaviva, are time-stable. Verbs, on the contrary, change in time.

I align with Baker in the current work by opposing the feature-based definitions of nouns and arguing that nouns are defined based on structural criteria. However, there are some fundamental differences between my approach and Baker's way of defining nouns. I differ from Baker's approach in various ways. First, I show that nouns are always formed from the nominalization process. This nominalization process forms not only prototypical nouns, but also nominals with various degrees of verbal properties. Following the DM approach, and aligning with Ross (1973), I argue that there are different degrees of nominal properties various types of nominals carry, and all these nouns are formed in syntax by attaching the nominalizer at different heights in the derivation. Secondly, Baker doesn't decompose the nouns further. For Baker, nouns are the categories with the privative [N] feature, which are defined in the structure by the referential index. He doesn't discuss how a noun gets an idiosyncratic or compositional interpretation depending on the way it is formed. Unlike Baker, I do not take nouns or verbs to be the atomic units in a language. However, I understand acategorial roots, not a category like a noun, to be the atomic units that work as building blocks of syntactic operations. Nouns are formed from these acategorial roots in the derivation. The categorizers assign an idiosyncratic or compositional nominal value to the roots depending on the nominalization heights in the structure.

By analyzing nominals in Santali, a language known in the literature for its categorial fluidity, I show that Santali nouns are formed by merging a nominalizer with either a root or a stem (verb or verbal extended structure). Since morphological criteria are not an ideal way of seeing the nominal property in Santali nominals, as without any overt nominalizer, Santali can have nominals that structurally differ, all nominals in Santali do not carry the same level of nominal property.

However, before going in detail to the empirical domain, I present various types of nominals seen cross-linguistically following Aikhenvald (2011) mainly because of one reason. The structural properties seen by the modern typological literature and Baker in the current section are not restricted to prototypical nouns only. There are nouns cross-linguistically that show properties of other categories, like verbs and adjectives. However, what Baker's system fails to address is that even if these nominals have a criterion of identity, how much property of another category, like verbal or adjectival property, they have. Also, how they are different in their structure and formation from prototypical nominals. The next section discusses the nouns

and the properties they exhibit based on a cross-linguistic survey done by Aikhenvald (2011). Then I move to the nominalization process.

2.2.2 Nouns Cross-linguistically: Challenges in Baker's System

According to Aikhenvald (2011), nouns cross-linguistically can have verbal and adjectival properties. Aikhenvald discusses five types of nouns, among which three types have verbal properties, and the fourth and fifth types deal with adjectival properties in nouns.

The first type of nouns carries the activity, state, or property of a verb. Activity and process nouns like (3), collective or mutual action nouns, like (4, 5), state nouns, like (6), property nouns, like (7), are the sub-parts of the first type of verb-derived nouns, according to Aikhenvald.

- (3) a. English (Dixon, 2005)

Your **shouting** woke me up.

- b. Urarina (Aikhenvald, 2011)

ate-su-na
fish-kill-NMLZ

'Fishing'

- c. Apalai (Aikhenvald, 2011)

j-oe-py-ry
1sg-come-NMLZ

'My coming'

- (4) Tagalog (Rubino, 1998)

- a. *hábul* 'chase'

- b. *habul-án*
chase-NMLZ

'rumble (many people chasing one another)'

- (5) Motuna (Onishi, 1994)

- a. *taapu* 'help'

- b. *taa-taapu*
NMLZ-help
'helping (each other)'

(6) English (Dixon, 2005)

'hate' → 'hatred'

(7) Turkana (Dimmendaal, 2001)

a. *ro* 'be bad'

b. *a-ro-n-ì-si* 'badness'

The second type of nouns possesses an identical argument structure to the verb. Aikhenvald presents agentive nouns, like (8, 9), and object nouns, like (10), as the second type of verb-derived nouns.

(8) English (Aikhenvald, 2011)

'Employ-er', 'kill-er', etc.

(9) Quechua (Weber, 1989)

a. *pishta(ku)* 'slaughter'

b. *pishtaku-w* 'slaughterer'

(10) Apalaí (Koehn and Koehn, 1986)

y-ny-mero-ry
1-NMLZ-write-NMLZ

'the thing I am writing'

The third type of nouns represents semantic properties like instrument, location, destination, manner, reason, or time. Instrument nouns, like (11); location nouns, like (12, 13); manner nouns, like (14); reason nouns, like (15); and time nouns, like (16), belong to this type.

(11) Movima (Haude, 2006)

iwani-wamba:-ni
speak-instr:bound.root.round-process

'telephone'

(12) Yukaghir (Maslova, 2003)

modi-be
reside-LOC.NMLZ

‘Place of residence’

(13) Urarina (Olawsky, 2006)

eno-ala-naha
cook-purp-LOC

‘A place for cooking food’

(14) Supyire (Carlson, 1994)

a. *jiile* ‘cross (a river)’

b. *jiile-ŋka*
cross-NMLZ

‘manner of crossing’

(15) Sundanese (Robins, 1959)

a. *dataŋ* ‘arrive’

b. *paŋ-dataŋ*
NMLZ-arrive

‘reason for arriving’

(16) Supyire (Carlson, 1994)

a. *kaan* ‘give’

b. *tèè-kaan*
NMLZ-give

‘time to give/pay’

In the examples from different languages in (3 - 16), we see that all the nouns have internal characteristics of a verb, which are turned into nouns by using a nominalizer. While Baker’s framework accounts for the criterion of identity of nouns through their referential index, the cross-linguistic data presented by Aikhenvald demands focus beyond these semantic and syntactic definitions. Many of the noun types in (3 - 16) exemplify nominal forms that may have referential index in their external structure, but Baker’s system doesn’t address the internal structures of these nominals.

Also, like verbal properties, since we can find adjectival properties in nouns cross-linguistically as well, Aikhenvald shows instances of adjective-derived nouns. The fourth type of nouns in Aikhenvald’s survey involves properties or states associated with properties (17, 18).

(17) Maale (Amha, 2001)

a. *dalgi* ‘wide’

b. *dalg-um-ó* ‘width’

(18) Somali (Tosco, 1999)

a. *cád* ‘white’

b. *caddáan* ‘whiteness’

Aikhenvald includes a person characterized by a property in the fifth type of nouns. These types of nouns have [+Human] properties, which also have adjectival characteristics (19, 20).

(19) Indonesian (Sneddon, 1996)

a. *besar* ‘big’

b. *pembesar* ‘important person’

(20) Lango (Noonan, 1992)

a. *mwòl* ‘soft/ humble’

b. *à-mwóló* ‘humble person’

We have seen from Aikhenvald’s cross-linguistic survey that nouns show properties of other grammatical categories, particularly verbs and adjectives. Like verbs, the adjectival characteristics of the nouns show that the class of nouns is much more structurally heterogeneous than Baker’s system discusses.

Based on this, I infer that there is more structure to nouns than what is seen on the surface. In this chapter, I show that this structure includes a nominalizer and that all nouns can be viewed as products of nominalization processes. As mentioned in Chapter 1, I adopt the DM-based account for this analysis. The verbal and adjectival properties of nouns in Aikhenvald’s survey point toward the assumption that nouns do not form a rigid, homogeneous category, rather, “nouniness” appears to be a fluid concept that exists on a continuum. We’ll explore the nominalization process and how different nouns can be derived through this process in the next section. We’ll also see how different nominals in a scalar approach of word classes are formed at different nominalization heights.

2.3 Nominalization

Studies on nominalization have received much attention in the generative literature since (Chomsky, 1970). Before going into detail about Chomsky’s distinction, I will explain what nominalization is in the generative literature.

2.3.1 Defining Nominalization

According to Aikhenvald (2011), nominalization is a type of derivation that results in an end product that carries the characteristics of a noun. The process includes an input, which is a verb or any non-nominal category, which produces an output defined as a ‘noun’ (Crystal, 1980, p. 328; Payne, 1997, p. 223, a.o.). This definition lacks the correspondence between the input and the output. Shibatani (2019) adds to this definition by positing nominalization as a metonymic process. According to Shibatani, any random verb or non-nominal category cannot work as an input to produce a noun. For example, the noun ‘singer’ cannot be derived from the verb ‘walk’, and ‘driver’ cannot be derived from ‘kill’. Hence, the definition that a noun is derived from a verb is incomplete. Therefore, X_V can function as X_N in nominalization (with or without suffixation), where X is the identical verbal or nominal form. In summary, in the process of nominalization, when a V turns into an N, there has to be a common semantic base. The take that nominalization is the process of forming any kind of nouns, indirectly defies Chomsky’s (1965) way of listing nouns as a rigid category listed in the lexicon. While nouns may seem fundamentally different from verbs, natural language allows for overlaps between these categories through nominalization processes, creating a spectrum of noun-like elements that retain varying degrees of their verbal origins. According to Alexiadou (2020), the definition of nouns is not absolute. The distinction between N and V is gradual in nature. Nouns include nominalized elements like derived nominals, mixed nominals, gerundive nominals like *poss-ing*, *acc-ing* (Abney, 1987), etc., along with the prototypical nouns.¹

In the next subsection, we see how, depending on the nouniness they carry, these nominals fall on the Noun-Verb continuum proposed by Ross (1973).²

2.3.2 The Scalar Properties of Nominals

Ross (1973) proposes ‘nouniness’ as a scalar property of linguistic expressions. It implies that some linguistic expressions are more nouny than others. This view rejects the idea that LCs like nouns and verbs are discrete units. Following Ross, Abney (1987) proposes a spectrum of

¹I use the term prototypical or lexical noun to refer to a ‘simple noun’ (Abney, 1987) or ‘concrete noun’ (Berger, 2017). Iordachioaia (2020) calls it a ‘lexical noun’.

²Explanation on the structures and the formation of all these constructions cross-linguistically is beyond the scope of this chapter. See Ross (1973) and Berger (2017) for examples and explanation on the noun-verb continuum.

nouniness (Figure 2.1) because some nominals carry less nominal properties (nouniness) than others. Languages can vary based on the types of nominals they can have from the continuum.

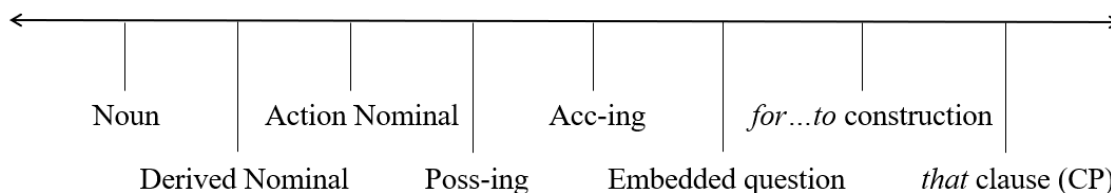


Figure 2.1: Ross's (1973) Noun-Verb Continuum

In (21), Ross shows that the nouns or expressions with nominal properties have the most nouniness in (21a), and the least nouniness in (21h).

(21) Ross (1973)

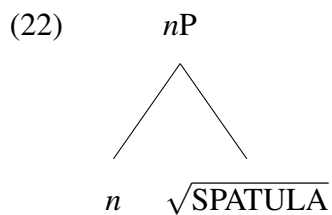
- a. Spatula (Noun)
- b. The gift of the letters to Frieda (Derived nominal)
- c. The giving of the letters to Frieda (Action nominal)
- d. Max's giving the letters to Frieda (Poss-ing)
- e. Max giving the letters to Frieda (Acc-ing)
- f. How willingly Max gave the letters to Frieda (embedded question)
- g. For Max to have given the letters to Frieda (for...to construction)
- h. That Max gave the letters to Frieda (that clause/CP)

This continuum approach to categorizing nominal structures provides a more nuanced framework than traditional binary classifications. As we saw in this subsection, rather than viewing LCs as discrete, the continuum acknowledges the gradient nature of nominalization processes across languages. After having this gradient understanding, when we define 'nouns', we do not only define 'Noun' in (Figure 2.1). Rather, we need to identify what fundamental conceptual properties distinguish different points of nouns in the continuum and what differences they have in a derivational structure. I propose that these points on the scale (Figure 2.1) can be understood derivationally, as nominalizer height. By nominalizer height, I refer to the level at which the nominalizer attaches in the derivation. Further, I also propose that the nominalizer height is subject to cross-linguistic variation. Both of these points are elaborated in the next section.

2.3.3 Nominalizer Heights and Structural Complexity

In this subsection, I propose that the different positions on Ross’s continuum correspond to different heights of nominalization and degrees of syntactic complexity. When we analyze nouns as a product of the nominalization process, following a DM framework, we can see that different points in Figure 2.1 have different positions in the derivation where the nominalizer (*n*) merges. Lower nominalization corresponds to lesser verbal properties, and higher nominalization retains more verbal properties, like the argument licensing properties of the verbal counterpart.

‘Nouns’ or prototypical nominals, listed as the first element in Figure 2.1, along with all other types, are the outputs of the nominalization process. In the purview of DM, following Iordachioaia (2020), Alexiadou et al. (2007), Marantz (1997), a.o., I posit that prototypical nominals are categorization of roots. A prototypical nominal like ‘spatula’ can be formed by merging a zero nominalizer (*n*) with the root $\sqrt{\text{SPATULA}}$ (22).

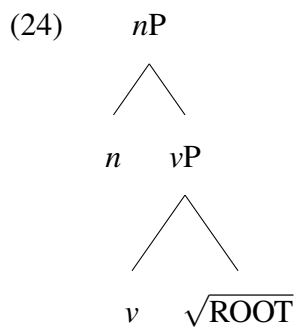


Prototypical nouns like ‘spatula’ lack verbal functional layers (*vP*, *VoiceP*, etc.) and derive their interpretation solely from encyclopedic knowledge associated with the root. Comparing the categorization of such a prototypical nominal ‘cat’ and a deverbal nominal ‘examination’, Iordachioaia (2020) calls the root-based categorization like (22) ‘inner’ derivation, as it only involves one layer of word formation. Derivation like (22) underscores one of the important principles of DM, i.e., non-productivity/ idiosyncrasy. Like other root-derived forms, prototypical nouns exhibit idiosyncratic meanings and resist productive argument structure inheritance. Thus, prototypical nouns occupy the baseline of nominal categorization (Figure 2.1), distinct from deverbal nominalizations³ that require syntactic event structure. ‘Derived nominals’, which lack the verbal properties, are also formed in a similar manner like (22). The ‘-ing’ nominals, which are divided as ‘action nominals’, ‘Poss-ing’, and ‘Acc-ing’ by Ross, can have

³see sections 2.4.1 and 2.4.2 for a detailed discussion on the deverbal nominals in Ross’s Noun-verb continuum.

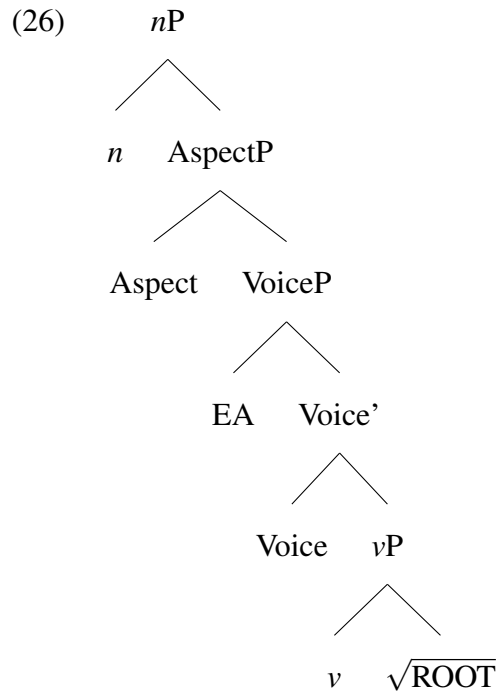
different structures depending on their verbal properties. According to Alexiadou (2005), nominals like (21c), which she refers to as nominal gerunds, do not carry accusative objects, and can have determiners, making themselves nounier than other ‘-ing’ nominals. Although they lack verbal extended structures, they have more verbal properties than the derived nominals (22). To form the action nominal (or what Chomsky calls a mixed nominal) in (23), repeated from (21c), the nominalizer, therefore, attaches to a verbalized root (24).

(23) The giving of the letters to Frieda (Action nominal)



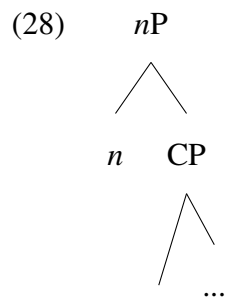
Verbal gerunds, such as Poss-ing (21d) and Acc-ing (21e) constructions, involve even more complex structures where nominalization occurs at different heights of the verbal extended structure, preserving different properties of the verbal base. These are what Chomsky (1970) calls gerundive nominals that differ from mixed nominals (action nominals, according to Ross) like (21c) due to the lack of an article and the ‘of’-phrase. The structure in (26) can explain the gerunds like the poss-ing (25a repeated from 21d), and the acc-ing (25b repeated from 21e), where the VoiceP introduces a specifier position for the EA. The AspectP layer can be added to the structure depending on the verbal properties of the nominal, such as the presence of aspectual modification.

- (25) a. Max’s giving the letters to Frieda (Poss-ing)
 b. Max giving the letters to Frieda (Acc-ing)

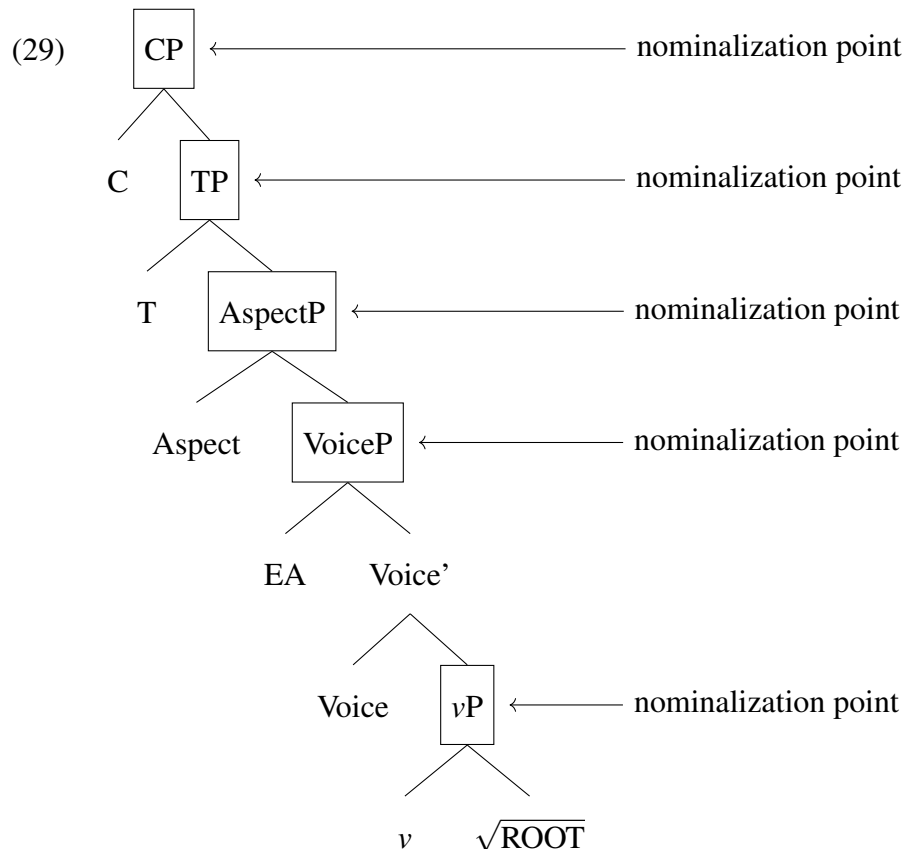


There are also languages in the world that can have a nominalizer attached to the structure at a point as high as the CP level (28). Languages like Japanese (Alexiadou, 2020) and Tamil belong to this category of languages. A ‘that clause’ in Ross’s continuum in (27), repeated from (21h)

(27) That Max gave the letters to Frieda (that clause/CP)



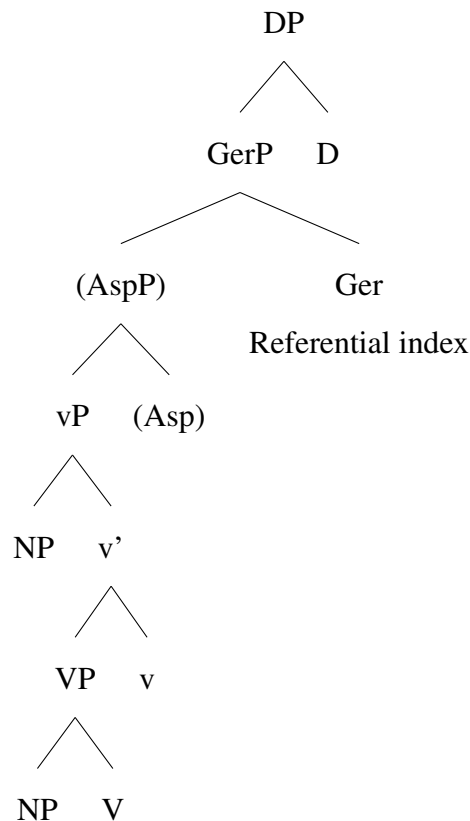
After establishing that N is not a rigid category and languages can have nominals corresponding to different points in the continuum proposed by Ross, we need to understand what these structural counterparts of the different nominals actually mean for the current work in Santali. We can summarize Ross’s division of nominals in the continuum and their formation in the syntax by merging an *n* in different levels in (29).



Now, since we understand that nouns cannot be defined using a rigid categorial status as opposed to verbs, and we can analyze nouns based on their degree of nouniness, it is important to ask what this means to Baker's way of defining nouns. A pivotal question here is whether Baker's criterion of identity and referential index change depending on the nouniness of a noun in Ross's continuum. In other words, if Baker means that only nouns (prototypical) and DNs have the criterion of identity, and nominals with lesser nouniness lack or have less criterion of identity. This question is directly addressed by Baker (2011). Baker's position is that there is no fundamental difference in the criterion of identity between the types of nouns based on their nouniness. This means that all constructions that are nominal show an entity with a referential index, and the criterion of identity that determines 'the sameness' of nouns across different referential contexts. Baker states that even if there is no difference in terms of the criterion of identity and referential index among the nominals we saw in Ross's continuum (figure 2.1), the difference among the nominals lies in the functional structure that introduces the referential index. To simplify, Baker suggests that both prototypical nouns and gerunds have referential index, where the prototypical nominals are fully nominal, and have referential index as they can function as an argument, like subject, object, etc. in a structure. Gerunds

also have referential index because they can occupy the same argument positions, like subject and object, take case, etc. However, according to Baker, the nominality of the gerunds comes from a functional head ‘Ger’ being nominal in (30). Even if it is less nominal than prototypical nominals, it bears the referential index and criterion of identity.

(30) Referential index on Ger (Baker, 2011, p. 1166)



The structural difference I show in my analysis, however, is that instead of a GerP for gerundive nominals, following DM, I show that nominals are formed by merging a nominalizer (*n*), which remains uniformly unchanged for any kind of nominalization in Santali. In my analysis, the referential index can be introduced to assign nominality to the structure at the *n*-merge position.

At this point, after seeing the potential positions in the structure for nominalization and checking what it means for Baker’s referential index, we move to briefly see the distribution of Santali nouns. By understanding the types of nouns that can be seen in the language, we can have a broader goal to check how we can place Santali nouns in a structure like (29).

2.3.4 Distribution of Santali Nouns

As we have seen in Chapter 1, Santali is a language well known for its weak distinction of categories. As a result, nouns in Santali are not restricted to the prototypical nominals we see in other languages. Santali also has affixal nominals and zero nominals. To see if all nominals, like affixal nominals and zero nominals, have the same external distributional properties as that of prototypical nominals, we have to see what positions they occupy in the matrix clause. In (29), the prototypical nominals or proper nouns in Santali can occupy the position of a subject of a matrix clause (31a), an object of a verb (31b), and an object of a pre-/post position (31c).

- (31) a. *hoR/jon uul jom-ke-a-e*
man/John mango eat-PST.ACT-FIN-3SG
'The man/John ate mango.'
- b. *uni hoR/jon-ke Daal-ked-e-a-e*
he man/John-OBJ hit-PST.ACT-OBJ-FIN-3SG.SUBJ
'He hit the man/John.'
- c. *uni aRaa-re japid-ke-a-e*
he house-in sleep-PST.ACT-FIN-3SG
'He slept at home.'

Affixal nominals, being very rarely used in the present-day Santali, can also occupy every position that a prototypical nominal can occupy. In (32), the affixal nominal appears as the subject of a matrix clause (32a), as the object of a verb (32b), and as the object of a post-position (32c).

- (32) a. *al-ic uul jom-ke-a-e*
write-NMLZ mango eat-PST.ACT-FIN-3SG
'The writer ate mango.'
- b. *uni al-ic-ke Daal-ked-e-a-e*
he write-NMLZ-OBJ hit-PST.ACT-OBJ-FIN-3SG.SUBJ
'He hit the writer.'
- c. *uni al-ic-salaa sen-en-a-e*
he write-NMLZ-with leave-PST.MID-FIN-3SG
'He left with the writer.'

Similarly, zero nominals (ZN) also occupy the same positions in Santali. In (33), they take the position of a subject of a matrix clause (33a), object of a verb (33b), and object of a post-position (33c).

- (33) a. *uni-a sen baa-ij kusi adaa*
 He-GEN leave not-1 SG happy feel
 ‘His leaving didn’t make me happy.’
- b. *ij uni-a sen nyel-ke-a-ij*
 I he-GEN leave see-PST.ACT-FIN-3SG
 ‘I saw his leave(ing).’
- c. *uni japid re kusi bujho-ke-a-e*
 He sleep in happy feel-PST.ACT-FIN-3SG
 ‘He felt happy in sleeping.’

Even though the Santali nominals in (31-33) show the same external distribution, I claim in this chapter that they differ from one another in their structure and formation in the syntax. The difference lies in the level at which the nominalizer is attached in the derivation. I claim that the prototypical and the affixal nominals in Santali are formed at the lowest nominalization point (see 29). Therefore, they have only nominal properties by showing an idiosyncratic nominal interpretation. On the other hand, ZNs in Santali are formed at a higher point, i.e., by merging a nominalizer with the VoiceP and AspectP. Therefore, the ZNs in Santali are stem-derived entities encoded with verbal properties. My claim means that the prototypical and affixal nominal in Santali are formed and share properties with Derived Nominals (DN) or Result Nominals (RN) in syntax, whereas ZNs in Santali are formed and share properties with Gerundive Nominals (GN) or Argument Supporting Nominals (ASN). Before going to the analysis and showing how the three types of nominals are formed in the syntax in Santali, we need to go back to the most influential works on nominalization associated with Chomsky (1970), and followed by Grimshaw (1990) to understand the formation and function of DN and GN, and ASN and RN in the next section.

2.4 The Origin of the Nominalization Debate

In the generative tradition, before Chomsky (1970), nominalization, more generally categorization, was considered to be a syntactic phenomenon (see Lees, 1960). Words were considered to be formed using transformational rules in the syntactic workspace. By the introduction of the lexicon (Chomsky, 1965), word formation, more specifically, categorization, becomes a part of the lexicon where it is treated equally with the listed properties, such as subcategorization,

selectional restrictions, and phonological representation of the morphemes, etc. (Alexiadou, 2020). This section examines the seminal literature on nominalization in order to distinguish between the various subtypes of nominals and to comprehend their nouniness within the noun-verb continuum.

2.4.1 The Derived and Gerundive Nominals (Chomsky, 1970)

The most notable source for the discussion on Nominalization can be traced back to Chomsky (1970). Chomsky posits two different engines for the formation of two different kinds of nominals, namely, derived (DN) and gerundive nominals (GN). DNs like *destruction*, which are formed with non-*ing* affixes, are no longer under the jurisdiction of syntax and, therefore, moved to the lexicon, the non-derivational engine for word formation (see Alexiadou, 2020). On the other hand, GNs are syntactically formed and comprise specifically the ‘Poss-ing’ (34a) and ‘Acc-ing’ (34b) constructions. Chomsky also identifies a third intermediate type, ‘-ing-of’ nominals (34c), which he terms mixed nominalization, noting that they share properties of both DNs and GNs.

- (34) a. John’s destroying the city
b. John destroying the city
c. The destroying of the city

The primary motivation for Chomsky to differentiate between DNs and GNs (Poss-ing and Acc-ing) is that the DNs are more nominal than the GNs, the type of nominals that have embedded verbal properties. Since both the nominals do not behave identically, their formation also differs from one another. According to Alexiadou (2020), DNs, unlike GNs, do not have verbal structures like gerunds and, therefore, are posited by Chomsky to be listed in the lexicon.

The contrast between DN and GN is dependent on structural criteria. Alexiadou et al. (2007) highlights the following diagnostics used by Chomsky to differentiate between the two types of nominals. DNs (35a) have restricted productivity, whereas GNs (35b) don’t.

- (35) a. militate → *militation
b. militate → militating

The DNs and the associated propositions have varied semantics and carry some idiosyncrasies. According to Chomsky, GNs transparently carry the properties of the underlying verbal element. On the other hand, deverbal nominals like ‘laughter’, ‘marriage’, ‘construction’, ‘action’, and so on have their individual ranges of meanings and varied semantic relations to the underlying form.

DNs carry the internal structure of a noun phrase, while GNs do not. Even if GNs have external structures of nominals, they carry less nouniness than DNs, following the Noun-Verb continuum by Ross (1973) (Figure 2.1). *John’s* in (36a) cannot be substituted by any other determiner (37a), whereas *John’s* in (36b) can be replaced by other determiners like ‘the’ (37b).

- (36) a. John’s destroying the house annoyed us. (GN)
- b. John’s destruction of the house annoyed us. (DN)
- (37) a. *The destroying the house annoyed us. (GN)
- b. The destruction of the house annoyed us. (DN)

Auxiliaries cannot be present with the DNs (38a), while the presence of auxiliaries is possible with GNs (38b).

- (38) a. *John’s having criticism of the play annoyed us.
- b. John’s having criticized the play annoyed us.

Another diagnostic test used by Chomsky to distinguish between the two subtypes of nominals is the modification test. DNs can be modified by adjectives and GNs by adverbs, not vice versa.

- (39) a. John’s careful restoration of the book surprised us. (DN)
- b. *John’s careful restoring the book surprised us. (GN)
- c. *John’s carefully restoration of the book surprised us. (DN)
- d. John’s carefully restoring the book surprised us. (GN)

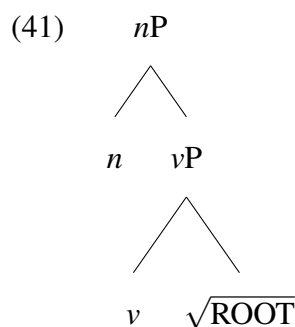
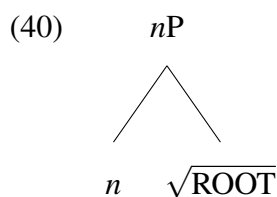
For the current work, along with the diagnostics by Alexiadou et al. (2007), I consider one of the pertinent non-lexicalist differences between the DN and the GN depending on how they are formed in the syntax. Non-lexicalist literature (see Halle and Marantz, 1993; Marantz, 1997;

Arad, 2003; Embick, 2007; Don and van Lier, 2013; Bruening, 2018, a.o.) takes both types of nominals to be formed in the syntax from an acategorical root. However, depending on how much nouniness a noun carries, the formation of both DN and GN differs from one another.

I posit structures like (40) and (41), following Marantz (1997), and Arad (2003) for the formation of DN and GN, respectively. In (40), the acategorical root directly merges with the nominalizer. This is how a DN like *destruction* is formed from a categorially and semantically underspecified root $\sqrt{\text{DESTROY}}$ and the nominalizer with the overt suffix *-tion*. Prototypical nouns or lexical nouns are also formed in a similar manner (Iordachioaia, 2020). To illustrate this process more concretely, when we have a DN like *destruction*, the derivation proceeds as follows: We begin with the acategorical root $\sqrt{\text{DESTROY}}$, which lacks syntactic category features. This root merges directly with the nominalizing head *n* (realized as *-tion*). This direct merge results in the deverbal nominal ‘destruction’. When we form a prototypical noun like *cat* using a structure like (40), the nominalizer lacks *-tion* type overt morpheme on the *n*.

In (41), however, a verbal gerund (see Abney, 1987) like *destroying* is formed through a different process. The acategorical root $\sqrt{\text{DESTROY}}$ first merges with a verbalizing head *v*. This creates a verbal projection (*vP*). Only then does the nominalizer *n* merge with this already verbalized structure, making GNs stem-derived.

This structural difference accounts for the more verb-like properties observed in GNs compared to DNs. For instance, while *destruction* behaves predominantly like a noun, *destroying* retains verbal characteristics.



The distinction between DN and GN has been pivotal in driving further research on nominalization. However, one of the major issues in the division was treating DN as a homogeneous subcategory. In the next subsection, I show further development in the nominalization research

by discussing further decomposition of the DNs.

2.4.2 Decomposition of the Derived Nominals

Following Chomsky (1970) and Abney (1987), Grimshaw (1990) in her seminal work, decomposes the DNs to differentiate between two types of DNs, i.e., Complex Event nominals (CEN) and Result nominals (RN). This binary classification challenges the previously held assumption that DNs constitute a unified nominal subclass. The distinction depends on whether the morphosyntax and interpretation of the DNs are compositionally related to the verbal counterpart's event structure or whether they behave like prototypical nouns. Borer (2013) refers to the CEN and RN as Argument Supporting nominals (ASN) and Referential nominals (RN), respectively. I will refer to both CEN (Grimshaw, *ibid.*) and ASN (Borer, *ibid.*) as ArgStr-N⁴ since they carry the argument structure of the base verb (42a, I also refer to both Result nominals and Referential nominals as Ref-N (42b), following Iordachioaia (2020).

(42) Iordachioaia (2020)

- a. The instructor's (intentional) examination of the papers took a long time. (ArgStr-N)
- b. The instructor's examination/exam (*of the papers) is on the table. (Ref-N)

In (42), '*examination*' behaves and is formed differently depending on its ability to take arguments. Since '*examination*' takes a PP argument in (42a), showing a complex event, it behaves more like a GN or the verbal counterpart of the nominal, i.e., '*examine*' in (43).

(43) The instructor examined the papers for a long time.

However, '*examination*' in (42b) cannot display argument structure; therefore, it behaves like a prototypical noun. I present the major differences between the ArgStr-N and Ref-N in (Table 2.1), following Grimshaw (1990), Alexiadou et al. (2007), Borer (2013), among others.

- (44) a. The 'examination' by the teacher in three hours (ArgStr-N)
- b. *The 'examination' in three hours (Ref-N)

⁴I use terms like ArgStr-N and Ref-N to avoid confusion and maintain terminological clarity in this work following Iordachioaia (2020)).

ArgStr-N	Ref-N
• Eventive reading • Obligatory argument structure • Built on verbal bases • Compatible with aspectual modifiers like ‘in three hours’ (44a) • <i>by</i>-phrase is an argument (45a) • Can take adjectives like ‘frequent’ and ‘constant’ with singular number feature (46a) • Cannot be selected by predicates that require an entity noun (47a)	• Result reading, no eventive reading • Argument structures are not obligatory • Not built on verbal bases; composed of roots directly • Not compatible with aspectual modifiers like ‘in three hours’ (44b) • <i>by</i>-phrase is not an argument (45b) • Can take adjectives like ‘frequent’ and ‘constant’ only with plural number feature (46b) • Can be selected by predicates that require an entity noun, not an eventive noun (47b)

Table 2.1: Differences between ArgStr-N and Ref-N

- (45) a. The ‘examination’ of the students by the teachers (ArgStr-N)
b. The ‘examination’/ ‘exam’ (Ref-N)
- (46) a. The frequent ‘examination’ of the students by the teacher (ArgStr-N)
b. The frequent ‘exams’/ *‘exam’ (Ref-N)
- (47) a. *The ‘examination’ of the patients was on the table (ArgStr-N)
b. The ‘examination’/ ‘exam’ was on the table. (Ref-N)

According to Grimshaw’s proposal, the distinction between DN and GN is eliminated, since DN doesn’t form a uniform subclass. A finer way of classifying nominals is to understand if the nominals have been derived from the roots directly, or if they have been verbs in their derivational history. GNs and ArgStr-Ns have structural and semantic features of a verb, whereas Ref-N, lexical N/ prototypical N, and Zero nominals (ZN) do not carry verbal properties⁵ and are directly derived from roots. Following Grimshaw (1990) and Alexiadou and Borer (2020), (48) shows the typology of nominal suffixes.

⁵In contrast, in section 2.6.3, I argue that ZNs have verbal properties and pattern with verb-derived nominals.

- (48) a. *-ing* nominalizers: Event argument/ Argument structure only
 b. $\emptyset$ (Zero) nominalizers: Reference/ Result only
 c. *-ation, -ance, -ment, etc.*: Both Event argument and Reference/Result

This decomposition of the DN and the distinction between ArgStr-N and Ref-N can be better explained using a non-lexicalist framework, like DM, because we can form all types of nominals in the syntax. The distinction between nominals without verbal properties (Ref-N and lexical N) and nominals with verbal properties (ArgStr-N, encompassing both Grimshaw's CENs and Chomsky's GNs) is dependent on structures like (40) and (41), which refer to root-derived and verb-derived nominals, respectively.

In this section, we have examined the principal distinction between derived nominals (DN) and gerundive nominals (GN) proposed by Chomsky (1970), which differentiates nominals based on their verbal properties. Following DM, the non-lexicalist approach adopted for the current dissertation, the structural differences proposed for the composition of DN and GN are root-derived and stem-derived, respectively. Grimshaw's (1990) further decomposition of DNs into ArgStr-N and Ref-N refined this framework by demonstrating that DNs themselves are not a homogeneous class and ArgStr-N, a part of DN in the Chomskyan distinction, also carries extended verbal structures. These distinctions between root-derived and verb-derived nominals provide essential theoretical tools for analyzing the Santali nominalization system in the subsequent section, particularly its Zero nominals, which appear to challenge conventional categorization.

2.5 Interim Summary

So far, before diving deep into direct empirical evidence from Santali, we have explored the key distinctions between nouns (as results) and nominalization (as the process of forming nouns). Following a survey of various definitions, we concluded that nouns are sortal entities, defined as elements incorporating application, identity, and individuation (Baker, 2003; Prasada, 2008; Acquaviva, 2009). Nouns, as a result of the nominalization process, have a referential index and a criterion of identity according to Baker (2003, 2011). However, we could also see through the scalar properties of nominals that Baker's system doesn't address

the internal structure and the formation strategies of the nominals in the noun-verb continuum. While discussing nominalization in line with Aikhenvald (2011) and Shibatani (2019), we have also reviewed various types of deverbal and deadjectival nominals as discussed by Aikhenvald (2011).

Additionally, we examined the derived nominal (DN) and gerundive nominal (GN) distinction, following Chomsky (1970), which has been fundamental to the lexicon-syntax debate in word formation. Chomsky's (1970) conclusion that a syntactocentric analysis applies to GN formation paved the way for non-lexicalist approaches to investigate whether all categories can be derived syntactically. We then decomposed the DNs in line with Grimshaw (1990) and Borer (2013), to understand that we needed a finer distinction like ArgStr-N and Ref-N to differentiate between nominals based on their verbal properties. This will be crucial for the current work because, in Santali, I claim that Zero nominals behave like both GN and ArgStr-N, which will be discussed in detail in (section-2.6.3).

With this theoretical groundwork on nouns and nominalization established, we now turn to Santali. Renowned for its fluidity of categorization, Santali offers an intriguing context to examine these phenomena. Section 2.6 will explore the extent of this fluidity, particularly in the formation of nominals. While GN formations in many languages exhibit a degree of fluidity due to their verbal characteristics, we will assess whether Santali accommodates similar categorial values in its nominals. In Santali, however, to analyze the fluidity of categorization, we'll have to first see if Santali has nominals like those of other languages.

2.6 Nominalization in Santali

Opposing the fundamental traditional definition, entities that denote person, place, thing, etc., are not the only ones to be considered as nouns in Santali. Nominal markers like case markers, definitive markers, and number markers can be attached to turn any lexeme into a noun. The morpheme *jom*, which behaves as a verb in (49a), also behaves like a noun (action nominal) in (49b) by carrying a null nominative case. It can also have a referential nominal correspondence in (49c) by carrying a nominalizer suffix *-a*.

- (49) a. *uni jom-ke-a-e*
He eat-PST-FIN-3SG

‘He ate.’

- b. *tehen-a jom bes geyaa*
Today-GEN eating good COP
‘Today’s eating is good.’

- c. *tehen-a jom-a bes geyaa*
Today-GEN eat-NMLZ good COP
‘Today’s food is good.’

Before delving into the specifics of zero (derived) nominals, it is essential to first examine how nominals are formed in Santali. In Santali, nominal formation can produce prototypical nouns, affixal nominals, and zero nominals. This section focuses on these three types of nominals, categorized based on their internal structures.

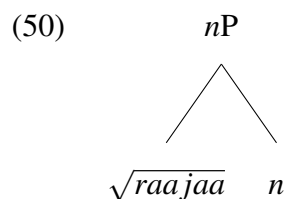
Nominalization in Santali can apply at various levels of a root or stem, leading to the formation of distinct nominal types. Following Alexiadou and Borer (2020), nominal markers may attach to acategorial roots, verbal stems, verbal extended projections with internal arguments, verbal projections with external arguments (VoiceP level), higher levels in the verbal projection, such as the AspectP level, etc. The affixation height of the nominalizer is pivotal in determining the verbal nature of the resulting nominal (Alexiadou, 2001, 2009; Alexiadou et al., 2010; Sichel, 2010; Alexiadou et al., 2011). A higher attachment point often correlates with stronger verbal properties, while lower attachment, such as root-level nominalization, tends to exhibit more nominal properties, aligning with Ross’s continuum (Figure 2.1), summarized in (29).

In the following sections, I will examine each type of Santali nominal in detail, starting with prototypical nouns in section 2.6.1, followed by affixal nominals in section 2.6.2, and finally focusing on zero nominals in section 2.6.3. The analysis of zero nominals will be particularly significant as they demonstrate properties of both GN and ArgStr-N, challenging traditional categorization.

2.6.1 Prototypical Nouns/ Lexical Nominals

This type of noun belongs to the first element ‘Noun’ in Ross’s continuum, which carries extreme nominality (see Figure 2.1). As already seen in (22) in DM (Halle and Marantz, 1993; Marantz, 1997), this class of nominals can be formed by merging a nominalizer *n* with the

acategorial root (50).



This, however, doesn't stop the root $\sqrt{raajaa}$ from becoming a verb (51) in the derivation (see Chapter 3).

- (51) *uni raajaa-en-a-e*
 He king-PST.MID-FIN-3SG
 'He king-ed.' (He became a king)

This type of nominal includes traditionally defined common nouns such as *hoR*, 'man' *puthi* 'book', proper nouns like *binit*, mass nouns like *hassa* 'sand', etc.

2.6.2 Affixal Nominals

These are the nominals reported by Neukom (2001), Ghosh (2008), formed by affixation. Suffixation, infixation, and prefixation are the derivational strategies used to form nominals from roots, verbal or adjectival inputs. However, this type of nominal is not seen in the present-day Santali. Even if some affixal nominals are acceptable to the Santali speakers, affixation to form nominals is not a natural process in Santali.

There are two suffixes *-ic* and *-ak* that result in animate and inanimate nominalization. Following Neukom (2001), Ghosh (2008) lists the verbs that can be nominalized through suffixation (see Table 2.2).

Roots/ Verbs/ Adjectives	Animate nouns	Inanimate nouns
<i>dal</i> 'beat'	<i>dal-ic</i> 'one who beats'	<i>dal-ak</i> 'the beaten thing'
<i>jam</i> 'eat'		<i>jam-ak</i> 'food'
<i>gac</i> 'die'	<i>gac-ic</i> 'dead one'	
<i>al</i> 'write'	<i>al-ic</i> 'writer'	<i>al-ak</i> 'written thing'

Table 2.2: Deverbal Nominalization by suffixation

In Santali, the nominalizing suffixes like *-ic* and *-ak* are used to form deadjectival nominals, as well (Table 2.3).

Adjectives	Animate nouns	Inanimate nouns
<i>ponD</i> ‘white’	<i>ponD-ic</i> ‘white one (animate)’	<i>ponD-ak</i> ‘white thing (inanimate)’
<i>boge</i> ‘good’	<i>boge-ic</i> ‘good one’	<i>boge-ak</i> ‘good thing’
<i>arak</i> ‘red’	<i>arak-ic</i> ‘red one’	<i>arak-ak</i> ‘red thing’
<i>jaha</i> ‘any/some’	<i>jaha-ic</i> ‘anyone/ someone’	<i>jaha-ak</i> ‘any thing/ something’

Table 2.3: Deadjectival Nominalization by suffixation

The most productive affixation process to derive nominals morphologically is infixation (Ghosh, 2008). There are four infixes *t*, *n*, *m*, and *R*, which are inserted along with a vowel⁶ into the verb to form deverbal nominals (Table 2.4).

Infix	Verbs	Nouns
<i>-t-</i>	<i>bar</i> ‘to fear’	<i>ba-t-ar</i> ‘fear (N)’
	<i>rak</i> ‘sew’	<i>ra-t-ak</i> ‘seam’
<i>-n-</i>	<i>jak</i> ‘sweep’	<i>ja-n-ak</i> ‘broom’
	<i>bet</i> ‘cover with palm’	<i>be-n-et</i> ‘stopper/lid’
	<i>tal</i> ‘bind’	<i>ta-n-al</i> ‘knot’
<i>-m-</i>	<i>lak</i> ‘peel’	<i>la-m-ak</i> ‘scraper’
	<i>han</i> ‘child/ to child’	<i>ha-m-an</i> ‘nephew’
<i>-R-</i>	<i>gam</i> ‘to name’	<i>ga-R-am</i> ‘grand old/ namesake’
	<i>ce</i> ‘squeak’	<i>ce-R-e</i> ‘bird’

Table 2.4: Deverbal nominalization by infixation (Ghosh, 2008)

The last affixal form of nominal derivation, according to Ghosh, is prefixation. The prefix *ma-* is also used to nominalize non-nominal elements in Santali (Table 2.5), although it is not as productive as the suffixation and the infixation.

Verbs	Nouns
<i>arsal</i> ‘to light’	<i>marsal</i> ‘light (N)’
<i>rak</i> ‘to cry’	<i>marak</i> ‘peacock’
<i>cet</i> ‘to teach’	<i>macet</i> ‘teacher’

Table 2.5: Nominalization by prefixation

After discussing the affixal nominals as noted in the literature by Neukom and Ghosh, we can proceed to examine zero nominals, the result of the most productive process of nominalization in Santali.

⁶The first vowel of the input gets inserted in the output along with the infix due to phonological reasons

2.6.3 Zero Nominals

The third type of nominal seen in Santali is zero nominals (ZNs). Santali nominalization becomes a topic of extensive empirical investigation due to the zero nominal morphology that makes the nominal behave externally like a DN and internally like a GN (52). In (52), *eneh*, which behaves as a verb in (52a), also behaves as a noun in (52b). It doesn't take any morphological nominalizer and takes an N position, structurally, i.e., the possessum position in a DP.

(52) Santali

- a. *uni eneh-ke-a-e*
He play-PST.ACT-FIN-3SG
'He played.'
- b. *uni-a eneh bes geyaa*
He-GEN play good COP
'His playing is good.'

Santali possesses ZNs like (52), and I show that these ZNs have an elaborate verbal structure embedded inside them. Structurally, they behave like GNs and ArgStr-Ns (see Section- 2.4.2) contrary to canonical ZNs in languages (see Borer, 2013). The following subsection deals with ZNs according to the existing accounts.

2.6.3.1 Zero-Nominals in the Literature

In the generative literature, ZNs are considered to be a part of DNs. DNs are faithful to roots and display idiosyncratic properties, unlike GNs (Borer, 2013; Levin and Rappaport Hovav, 2013; Iordăchioaia, 2020; a.o). Grimshaw (1990, p.67) considers ZNs to be Ref-Ns, a part of the DNs that is more nominal than other types of deverbal nominals. A similar conclusion is assumed by Borer (2013), where using the Exoskeletal (XSM) model, she shows the derivation of 'walk' (53).

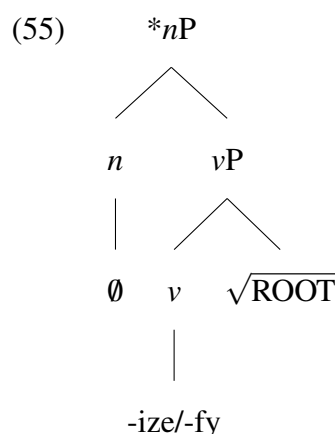
(53) $[D_{[C=N]} \sqrt{WALK}]$

In (53), Borer's model lacks a categorizer head since the ZN 'walk' is indirectly categorized by extended functional heads like D. ZNs contain neither argument structure (Borer, 1999, 2013) nor complex events (Grimshaw, 1990). This can be evidenced in (54a). Since the nominal in

(54a) contains a complex event or argument structure, the ZN ‘salute’ is not allowed. ZNs also cannot be modified by aspectual adverbials like ‘for three hours’, according to Borer (54b).

- (54) a. The salutation/ saluting/ *salute of the officers by the subordinates
b. The walking/ *walk of the dog for three hours

This entails that ZNs do not carry verbal properties and are never derived with an overt or a covert verbal suffix from verbs and, therefore, cannot display a structure like (55).



Like DNs, ZNs have idiosyncratic properties and lexical gaps (Cetnarowska, 1993, p. 19), and they do not show regularity or compositionality in nominalization. According to Cetnarowska (1993), the idiosyncratic properties and lexical gaps involve the restriction of certain ZNs in a language. In other words, the presence of ZNs like ‘permit’ and ‘flow’ is possible in a language like English, whereas ‘submit’ and ‘grow’ are not allowed. ZNs also have idiomatic meanings like in (56).

- (56) Cetnarowska (1993, p. 19)
- a. **spin** ‘a short journey in a car or other vehicle for pleasure’
b. **split** ‘a dessert made from a banana or other fruit cut into two pieces with ice cream on top’

Therefore, ZNs or DNs are derived by merging a root with an overt or a null nominalizer, projecting a structure like prototypical nouns (50). According to Borer (2013), ZNs are always formed directly from the roots and behave like lexical nouns (prototypical nominals). Borer (2013, p. 331) lists some exceptions, where some ZNs in English like, ‘(ex)change’, ‘release’,

‘abuse’, ‘discharge’, ‘murder’, ‘(mis)use’, ‘endeavor’, ‘consent’, ‘ascent’, ‘descent’, ‘resolve’, etc. carry argument structures (see also Harley, 2009; Lieber, 2016; Iordachioaia, 2020).

In the next section, using empirical evidence from Santali, I claim that arguments carrying ZNs are not really exceptions. Languages with a fluid categorial system, like Santali, possess ZNs that always display the argument structures of their verbal bases. ZNs in Santali can be intransitive, transitive, and ditransitive, depending on the valency of their verbal counterparts. By going a step further, I claim that Santali doesn’t only have ArgStr ZNs, but it also carries Gerundive ZNs by showing accusative-marked internal arguments. Santali doesn’t have morphological overt infinitival markers that are usually used as gerundive markers in GNs in most languages in the world. Santali also doesn’t have grammatical gender like Hindi, a dominant language spoken in the Santali-speaking regions. Therefore, the well-known DN-GN distinction is not so apparent. In the next section, the central claim I make is that ZNs are derived at a higher stage of nominalization, containing verbal properties in the nominalization height.

2.6.3.2 Zero-Nominals in Santali

As seen in (52), and repeated below as (57), ZNs like *eneh* in (57b) are seen in Santali. The presence of ZN in Santali is a productive process (58-60), which supports the categorial fluidity in the language.

(57) Santali

- a. *uni eneh-ke-a-e*
He play-PST.ACT-FIN-3SG
‘He played.’
- b. *uni-a eneh bes geyaa*
He-GEN play good COP
‘His playing (game) is good.’

(58) Santali

- a. *uni sen-en-a-e*
He leave-PST.MID-FIN-3SG
‘He left.’
- b. *uni-a sen baa-ij kusi adaa*
He-GEN leave not-1SG good feel
‘I didn’t like his leave(ing).’

(59) Santali

- a. *uni puthi paDhaao-kan-a-e*
He book read-COP.PST-FIN-3SG
'He was reading a book.'
- b. *uni-a puthi paDhaao baa-ij kusi adaa*
He-GEN read book not-1SG good feel
'I didn't like his read(ing) a book.'

(60) Santali

- a. *uni taaDaam-ke-a-e*
He walk-PST.ACT-FIN-3SG
'He walked.'
- b. *uni-a taaDaam bes bujho-ij*
He-GEN walk good feel-1SG
'I like his walk(ing)/ the way he walks.'

The ZNs in (57b, 58b, 59b, 60b) give the identical representation as the verbs (57a, 58a, 59a, 60a), respectively, on the PF level. They behave like English ZNs, such as 'salute (N)' and 'exchange (N)', etc. We should note that there is a difference between 'salute (N)' and 'exchange (N)', where the former is a Ref-N, and the latter is an ArgStr-N. On the surface, the Santali ZNs look like prototypical nominals (see 61a for comparison).

- (61) a. *uni-a chehraa bes bujho-ij*
He-GEN face good feel-1SG
'I like his face.'
- b. *uni-a eneh bes bujho-ij*
He-GEN play good feel-1SG
'I like his play(ing).'

The morphologically null nominalizer in (57b, 58b, 59b, 60b) and (61b) creates the illusion of the nominals being formed directly from the root.

In this section, I present diagnostics like Adverbial modification (Chomsky, 1970), presence of argument structure (Grimshaw, 1990), aspectual modification (Grimshaw, 1991; Borer, 2013; Alexiadou et al., 2007; Iordachioaia, 2020), and productivity of ZN to claim that ZNs like (57b, 58b, 59b, 60b) are verb-derived, and do not behave like canonical DNs. They have a verbal internal structure, and therefore, they behave both as ArgStr-N and GN in Santali. I

also claim that Santali doesn't have a clear DN-GN distinction. Therefore, an efficient way to divide the nominals is by the way they are formed. I posit two types of nominals, the first one is Root-derived nominals, which include prototypical nominals with a null nominalizer and affixal nominals with a nominalizer affix. The other one is stem-derived nominals (or equivalently word-derived nominals) to cover both Grimshaw's CENs and Chomsky's GNs — all nominals whose derivational input is a verbal or otherwise categorized stem rather than an acategorical root.

I list the diagnostics to show that Santali ZNs have embedded verbal properties.

Adverbial Modification: One of the important tests Chomsky (1970) uses to differentiate between DN and GN is the modification test. Since DNs behave like prototypical nouns, they can have adjectival modifiers (39a repeated as 62a), and GNs are modified by adverbs, carrying verbal properties (39d repeated as 62b). To state this in a more non-lexicalist way, root-derived nominals tend to be modified by adjectives, while stem-derived nominals show a greater compatibility with adverbial modifiers. However, this should not be taken as a categorical distinction. Stem-derived nominals can also take adjectival modification, as Grimshaw's (1990) ArgStr-Ns illustrate. The tendency towards adverbial, rather than adjectival modification is a direct effect of the nominalization height in the verbal functional structure. When nominalization happens at a higher level in the verbal functional structure (GN), the nominals reject adjectival modifiers and accept adverbial modifiers, retaining verbal properties.

If ZNs are a subpart of the DNs in Santali, they should be able to be modified by adjectives like (62c).

- (62) a. John's careful **restoration** of the book surprised us. (DN)
 b. John's carefully **restoring** the book surprised us. (GN)
 c. John's quick **run** to the market.

However, Santali displays contrary evidence. ZNs in Santali are modified by adverbs, which are also comparable with the verbal counterparts of the ZNs (63). The manner and time adverbs *dhirte* and *jotodin*, respectively, modify the ZN *paDhaao* in (63a) and *eneh* in (63b). These time and manner adverbs are verbal modifiers that also modify the verbal counterparts of the ZNs in (64a) and (64b).

(63) Santali

- a. *uni-a dhirte puthi paDhaao baa-iŋ kusi adaa*
He-GEN silently book read not-1SG good feel
'I didn't like his read(ing) a book silently.'
- b. *uni-a jotodin chendu eneh bes geyaa*
He-GEN everyday football play good COP
'His playing football every day is good.'

(64) Santali

- a. *uni dhirte puthi paDhaao-kan-a-e*
He silently book read-COP.PST-FIN-3SG
'He was reading a book silently.'
- b. *uni jotodin chendu eneh-ke-a-e*
He everyday football play-PST-FIN-3SG
'He played football every day.'

The verbal characteristics of the ZNs in (63) are further supported by their inability to accept adjectival modifiers. In (65b), the use of an adjectival modifier like *sean* for the ZN results in an ungrammatical structure. However, the ZN allows adverbial modification in (65a).

(65) Santali

- a. *uni-a seante puthi paDhaao baa-iŋ kusi adaa*
He-GEN cleverly book read not-1SG good feel
'I didn't like his read(ing) a book cleverly.'
- b. **uni-a sean puthi paDhaao baa-iŋ kusi adaa*
He-GEN clever book read not-1SG good feel
'I didn't like his clever read(ing) (of) a book.'

The adverbial modification tests in (63, 65a) show that ZNs have verbal properties in Santali and are formed at a higher level in the nominalization height. The claim becomes more evident when we apply a similar test to Hindi, a dominant language with a clear GN-DN divide, spoken in the Santali-speaking regions (66). The adverbials 'everyday' and 'silently', which modify the verb in (66a), also modify the GN in (66b), but not the DN in (66c).

(66) Hindi

- a. *vah roz/ dhire dhire kitaab paDh-taa hai*
He everyday/ silently silently book read-3SG.M COP.3SG

‘He reads a book every day/ silently.’

- b. *us-kaa roz/ dhire dhire kitaab paDh-na mujhe achhaa nahi laga*
He-GEN everyday/ silently silently book read-INF I.DAT good not feel
‘I didn’t like his reading the book every day/silently.’

- c. *us-ki *roz/*dhire dhire kitaab-ki paDhaai mujhe achhi nahi lagi*
He-GEN everyday/silently silently book-GEN read I.DAT good not feel
‘I didn’t like his reading (of) the book every day/silently.’

Based on the ability to be modified by adverbs, (63a, 63b) and (66b) behave similarly. The only major difference between the two is that the nominal in (66b) has an infinitival marker as a gerundive nominalizer, whereas the nominals in (63a) and (63b) are ZN. Therefore, based on the adverbial modification test, ZNs in Santali are stem-derived and are formed at a higher n-attachment point.

Presence of Argument Structure: The presence of nominal or verbal property on a nominal can be checked by the nominal’s ability to have its argument structure. The decomposition of the DN by Grimshaw (1990) as ArgStr-N and Ref-N divides the DNs depending on their ability to carry argument structures. ArgStr-Ns, like GNs, behave identically to their verbal bases by carrying argument structure, and the Ref-N behaves like the prototypical nominals. To make the distinction clear, let’s refer to the nominals with argument structures (ArgStr-N and GN) as partial nominals and the nominals without argument structures as complete nominals⁷. If the hypothesis that Santali ZNs behave like ArgStr-Ns and GNs (partial nominals) is correct, then we should see argument structures with ZNs in Santali productively. According to Alexiadou (2010), only nouns that carry a verbal structure in their derivational history have argument-bearing properties. The presence of arguments in Santali ZNs would evidence that these ZNs have verbal properties. The verb *eneh* (67a) and *nyel* (67b) have two arguments each. The ZNs *eneh* in (68a) and *nyel* in (68b) also carry the identical argument structure as their verbal bases in (67). The external arguments (EA) remain EAs by taking a genitive form, and the internal arguments (IA) retain their accusative forms in (68), making the construction in (68) behave like GNs in other languages (see English from Iordachioaia (2020) in (69)).

⁷See Iordachioaia (2020) for a similar distinction between defective and full nominals. Iordachioaia’s distinction mainly depends on the nominal’s ability to take the types of determiners.

(67) Santali

- a. *uni chendu eneh-ke-a-e*
He football play-PST-FIN-3SG
'He played football.'
- b. *jon uni-ke nyel-ked-e-a-e*
John he-OBJ see-PST.ACT-OBJ-FIN-3SG.SUBJ
'John saw him.'

(68) Santali

- a. *uni-a chendu eneh bes geyaa*
He-GEN football play good COP
'His playing football is good.'
- b. *jon-rena uni-ke nyel*
John-GEN he-OBJ see
John's seeing him

(69) English (Iordachioaia, 2020)

- a. John's refusing the offer
- b. John's performing the song

The equivalence of the Santali ZN *eneh* in (68a) to the English GNs 'refusing' and 'performing', and their argument-bearing ability, shows that Santali ZNs are similar to GNs. The only difference here is that Santali ZNs lack an overt morphological nominalizer like *-ing* in English, or *-na* in Hindi. Therefore, according to this test, Santali ZNs are partial nominals, which are formed at a high n-attachment point carrying verbal characteristics.

Aspectual Modification: The third diagnostic test examines whether Santali ZNs can take aspectual modifiers like 'in three hours'. According to Grimshaw (1990), Alexiadou et al. (2007), Borer (2013), a.o., nominals with verbal properties are compatible with aspectual modifiers. The presence of Santali ZN with aspectual modifiers would further attest that they have verbal properties and act like partial nominals.

To establish a baseline, (70) first shows that verbs in Santali accept 'in-adverbials'. In (70a), the verb *paDhaao* 'read' combines with the temporal modifier 'in three days', and in (70b), the verb *jom* 'eat' combines with 'in three hours', demonstrating that both predicates have telic interpretations available.

- (70) a. *uni peyaa din-re puthi paDhaao-ke-a-e*
 He three day-in book read-PST-FIN-3SG
 ‘He read the book in three days.’
 b. *uni peyaa gharonj-re uul jom-ke-a-e*
 He three hour-in mango eat-PST-FIN-3SG
 ‘He ate a mango in three hours.’

Crucially, the ZNs *paDhaao* ‘read’ in (71a) and *jom* ‘eat’ in (71b) are also compatible with these same aspectual modifiers, mirroring the behavior of their base verbs.

- (71) a. *uni-a peyaa din-re puthi paDhaao ba-iṅ kusi adaa*
 He-GEN three day-in book read not-1SG good feel
 ‘I didn’t like his reading a book in three days.’
 b. *uni-a peyaa gharonj-re uul jom ba-iṅ kusi adaa*
 He-GEN three hour-in mango eat not-1SG good feel
 ‘I didn’t like his eating mango in three hours.’

The parallel compatibility of both the verbs and their corresponding ZNs with ‘in-adverbials’ indicates that the ZNs inherit the telic aspect of their base verbs. The presence of the aspectual modifiers with Santali ZNs makes them equivalent to the English ArgStr-Ns in (44).

Productivity of ZN: Productivity is considered a language’s ability to create infinite grammatical and meaningful expressions by using grammatical rules. In classical generative tradition, it is encoded in syntax, whereas idiosyncratic rules are listed in the lexicon. Extending this idea further, non-lexicalists posit that idiosyncrasy is locally encoded and productivity has an intervening categorizer in the derivation (Arad, 2003; Marantz, 2013).

The GN formation from verbal bases is a productive process across languages, as GNs are not root-derived. The productive GN formation using the *-ing* nominalizer in English (72), and the *-naa* nominalizer in Hindi (73) displays the structural and semantic properties of a V in the derivation of the Ns.

- (72) English
 a. $[_n\text{-ing}[_v[\sqrt{\text{GROW}}]]]$
 b. $[_n\text{-ing}[_v[\sqrt{\text{WALK}}]]]$
 c. $[_n\text{-ing}[_v[\sqrt{\text{INVITE}}]]]$

(73) Hindi

- a. $[_n\text{-naa}[_v[\sqrt{\text{PADH}}]]]$ ‘studying’
- b. $[_n\text{-naa}[_v[\sqrt{\text{CHAL}}]]]$ ‘walking’
- c. $[_n\text{-naa}[_v[\sqrt{\text{BULAA}}]]]$ ‘calling’

The nominals in (72, 73) are stem-derived nominals because of the verbal properties they carry. They are productive, compositional from their verbal counterparts, and regular in nature. The root-derived nominals differ from their stem-derived nominal counterparts by lacking productivity and the presence of idiosyncrasy in nominalization (74, 75).

(74) English

- a. $[_n\text{-th}[\sqrt{\text{GROW}}]]$
- b. $[_n\emptyset[\sqrt{\text{WALK}}]]$
- c. $[_n\text{-ation}[\sqrt{\text{INVITE}}]]$

(75) Hindi

- a. $[_n\text{-aai}[\sqrt{\text{PADH}}]]$ ‘study’
- b. $[_n\text{-waa}[\sqrt{\text{BULAA}}]]$ ‘call’

If our prediction that Santali ZNs are partial nominals is true, we should expect productivity of ZN formation from the verbal bases in Santali. As already shown in (57b, 58b, 59b, 60b) and in (76-83) below, we can see that the ZNs lack idiosyncrasy, carry compositional semantics to their verbal counterparts, and can be productively formed from their verbal counterparts.

The formation of unergative ZNs (76) from unergative verbs (77), unaccusative ZNs (78) from unaccusative verbs (79), transitive ZNs (80) from transitive verbs (81), and ditransitive ZNs (82) from ditransitive verbs (83) evidence that Santali ZNs are not limited idiosyncratic elements. They carry the compositional meaning of their verbal equivalents.

(76) Unergative ZNs

- a. *gaaDi-a* *rokaao*
car/train-GEN stop
‘The car’s/ train’s stop’
- b. *binit-rena* *eneh*
Vineet-GEN dance
‘Vineet’s dance’

(77) Unergative verbal structures

- a. *gaaDi rokaao-ke-a-e*
car/train stop-PST-FIN-3SG
'The car/ train stopped.'
- b. *binit eneh-ke-a-e*
Vineet dance-PST-FIN-3SG
'Vineet danced.'

(78) Unaccusative ZNs

- a. *jaahaaj-rena boDuj*
ship-GEN sink
'The ship's sink(ing)'
- b. *kaanch-rena rapud*
glass-GEN break
'The glass's break(ing)'

(79) Unaccusative verbal structures

- a. *jaahaaj boDuj-en-a-e*
ship sink-PST.MID-FIN-3SG
'The ship sank.'
- b. *kaanch rapud-en-a-e*
Glass break-PST.MID-FIN-3SG
'The glass broke.'

(80) Transitive ZNs

- a. *jon-rena uni-ke Daal*
John-GEN he-OBJ hit
'John's hitt(ing) him'
- b. *jon-rena in-ke nyel*
John-GEN I-OBJ see
'John's see(ing) me'

(81) Transitive verbal structures

- a. *jon uni-ke Daal-ked-e-a-e*
John he-OBJ hit-PST.ACT-OBJ-FIN-3SG.SUBJ
'John hit him.'

- b. *jon in-ke nyel-ked-e-a-e*
 John I-OBJ see-PST.ACT-OBJ-FIN-3SG.SUBJ
 ‘John saw me.’

(82) Ditransitive ZNs

- jon-rena mili-ke puthi em*
 John-GEN Mili-DAT book give
 ‘John’s give(ing) the book to Mili’

(83) Ditransitive verbal structure

- jon mili-ke puthi em-ked-e-a-e*
 John Mili-DAT book give-PST.ACT-OBJ-FIN-3SG.SUBJ
 ‘John gave the book to Mili.’

Having established that Santali ZNs are stem-derived nominals that carry verbal semantic and structural properties, we now turn to the formation of the Santali ZNs in the derivational workspace.

2.6.3.3 Formation of Santali ZNs

After establishing that Santali ZNs are partial nominals, which behave like GNs or ArgStrNs, we can move to the formation of ZNs in the derivation. Based on how they are formed, nominals in Santali can be divided into root-derived and word-derived nominals. Among the 3 types of nominals seen in Santali, both prototypical nominals and affixal nominals are root-derived. They carry an idiosyncratic meaning directly derived from the root and do not contain a verbal structure like the Santali ZNs. Both the prototypical nominals and affixal nominals are complete nominals in Santali and are formed like (84) by merging a nominalizer (*n*) with an acategorial root.

- (84)
- $$\begin{array}{c}
 nP \\
 \swarrow \quad \searrow \\
 \sqrt{\text{ROOT}} \quad n \\
 \quad \quad \emptyset / \text{-affix}
 \end{array}$$

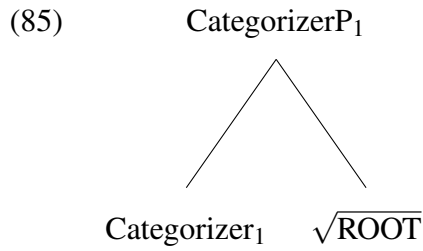
On the other hand, ZNs, carrying verbal structure, are not formed directly by merging an *n* with a root. Since they are formed by the nominalization of an already categorized root, they are word-derived, unlike the structure in (84), which is root-derived. ZNs have an extended structure by being word-derived, and they carry a compositional semantics to the base they are derived from. To explain this phenomenon, I use the Locality Constraint on the Interpretation of Roots (LCIR), following Arad (2003).

Arad, building on Marantz (1997), posits that the categorizer head accesses the semantics of the most local head during the category formation. In other words, according to the LCIR (Arad, 2003), the interpretation of the root is constrained by its interaction with the first categorizer. This “first categorizer position” corresponds to what Marantz (2007) and Panagiotidis (2015) describe as the first phase position in the derivational structure. To understand why the interpretation of the root is limited only to the first categorizer position, we will have to go back to cyclicity in syntax (Chomsky, 2000, 2001).

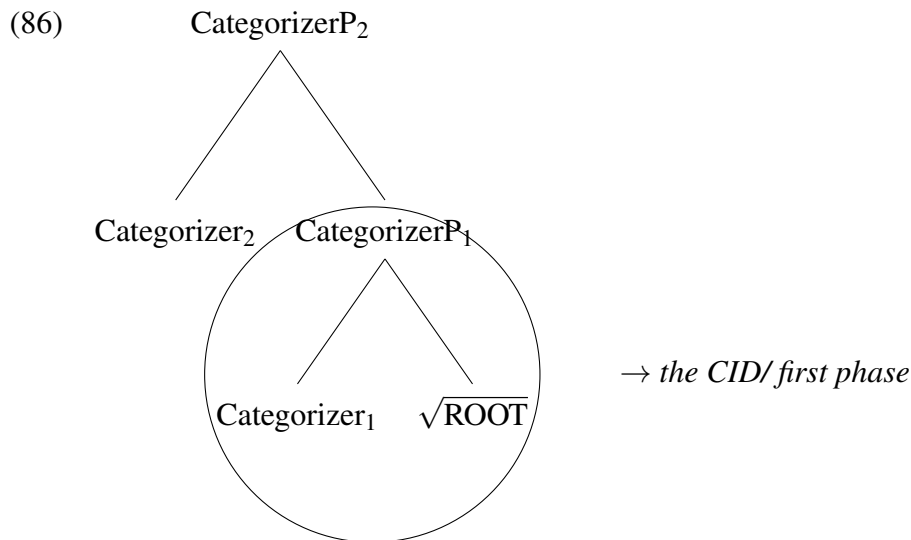
Since syntactic operations are cyclical and local, according to Chomsky, we will apply the phase theory to the derivation of ZNs in Santali, considering the primary assumption that words are syntactically formed in Santali. Marantz (2007) considers categorizer phrases (*nP*, *vP*, etc.) as phases, the other phases being the VoiceP⁸ and the Complementizer Phrase (CP). Coming back to the cyclicity in syntactic operations, when a phase head merges in the derivational space, it activates the semantic realization and phonological spell-out of a chunk. (Chomsky, 2000; Marantz, 2007).

In (85), *categorizer*₁ is the first categorizer, which is equivalent to the first phase, and the semantically underspecified root gets the interpretation when it merges with *categorizer*₁. Since, by definition, the root is impoverished, following the DM account, we cannot expect the first categorizer to have compositional meaning. After the first cycle, the root’s semantic interpretation and phonetic form are fixed. The domain of the first categorizerP forms a closed interpretation domain (CID), which is similar to the phase boundary, according to Chomsky (ibid.). Anything outside the CID does not have access to any atomic unit inside the domain, such as the ROOT. The CID is the idiosyncratic/ non-productive domain.

⁸I use *voice* to refer to the *v* in Chomsky (2000), which is same as Kratzer’s (1996) Voice, a head that deals with causativity and transitivity and the licenser of the external argument. The *v* used in this work, however, is the categorizer head.



Once the first cycle is complete, the domain is sent to the Morphology/ PF and LF. When the second categorizer (second phase) merges in the derivation (86), it can access only the phase edge, which is the *Categorizer₁* head (and its specifier). Everything below the *Categorizer₁* is not visible. Therefore, any categorization that happens outside the CID doesn't categorize the ROOT but re-categorizes the category on the *Categorizer₁*. In (86), the *Categorizer₂*, which merges with the *CategorizerP₁*, can only access the fixed interpretation of the root on the *Categorizer₁*. It creates the compositional/ regular domain since it carries the semantics of the first categorizer head.



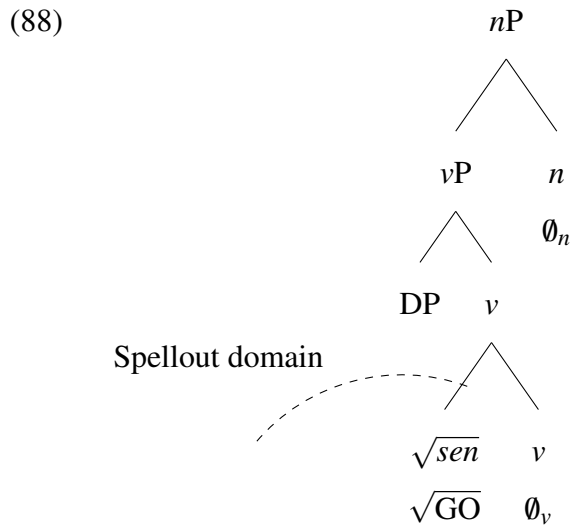
This evidence shows that categorial alteration of a word can carry multiple phases/ CID in the process of derivation of a word (Marantz, 2007, p. 1).

Taking this forward to Santali, I posit that Santali ZN formation follows this cyclicity in the derivation of categories, and the interpretation of ROOTs is restricted to the first categorizer position. Therefore, in ZNs like *sen* 'leave' in (58b, repeated here as 87a), which carries the verbal property of the verb *sen* 'leave' in (58a, repeated here as 87b), can be formed by the merge of the second phase/ second categorizer (*n*).

(87) Santali

- a. *uni-a sen baa-iŋ kusi adaa*
 He-GEN leave not-1SG good feel
 ‘I didn’t like his leave(ing).’
- b. *uni sen-en-a-e*
 He leave-PST.MID-FIN-3SG
 ‘He left.’

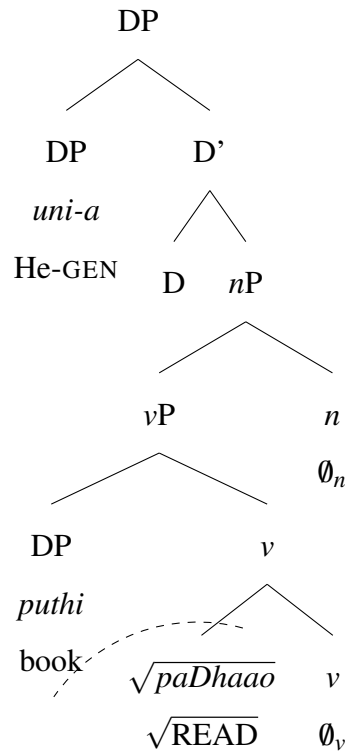
In (88), when the ROOT $\sqrt{sen}$ merges with the first categorizer v , it creates a CID or a phase boundary. The categorial status and the semantic interpretation are now fixed as a verb for further derivations. When the second categorizer (n) merges in the structure, it doesn’t have access to any atomic unit (ROOT) inside the CID and can only access the verbal interpretation from the v . The n then nominalizes the verb sen_v . Since the n is not local to $\sqrt{sen}$, it doesn’t access the idiosyncratic meaning of the ROOT.



The whole nP (88) is a ZN in Santali. Since the nP (N) carries the entire verbal structure of sen_v ‘leave_v’, it can contain the internal argument (IA) of the base verb when the base verb is a transitive one. In (89), structurally represented as (90), the ZN $paDhaao_n$ ‘read_n’, which is formed by nominalizing $paDhaao_v$, carries the argument structure of the verb, as well.

- (89) *uni-a puthi paDhaao*
 He-GEN read book
 ‘his read(ing) book’

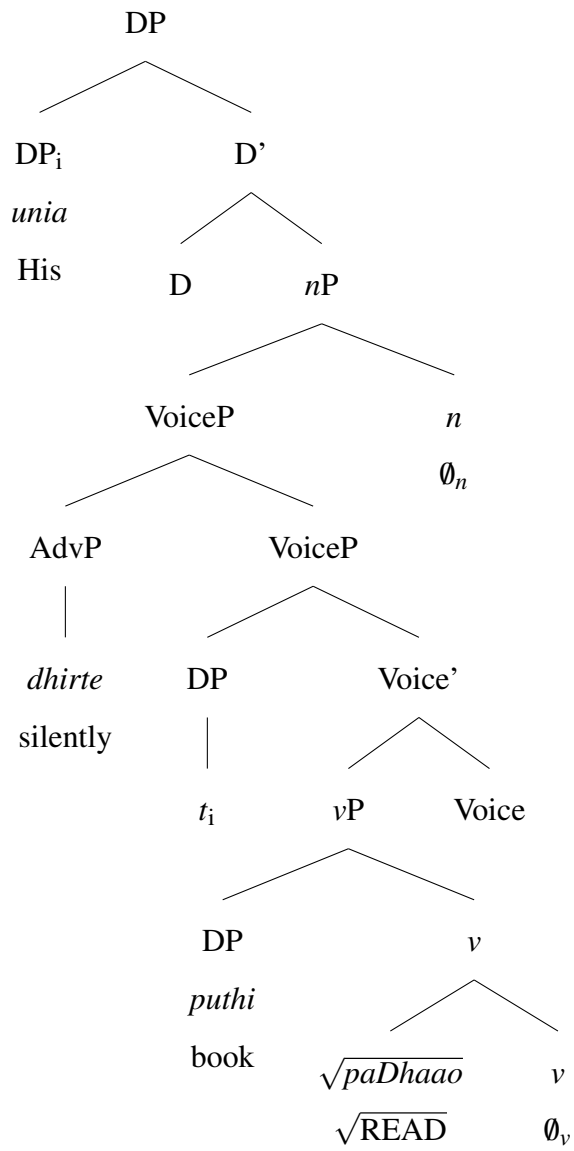
(90)



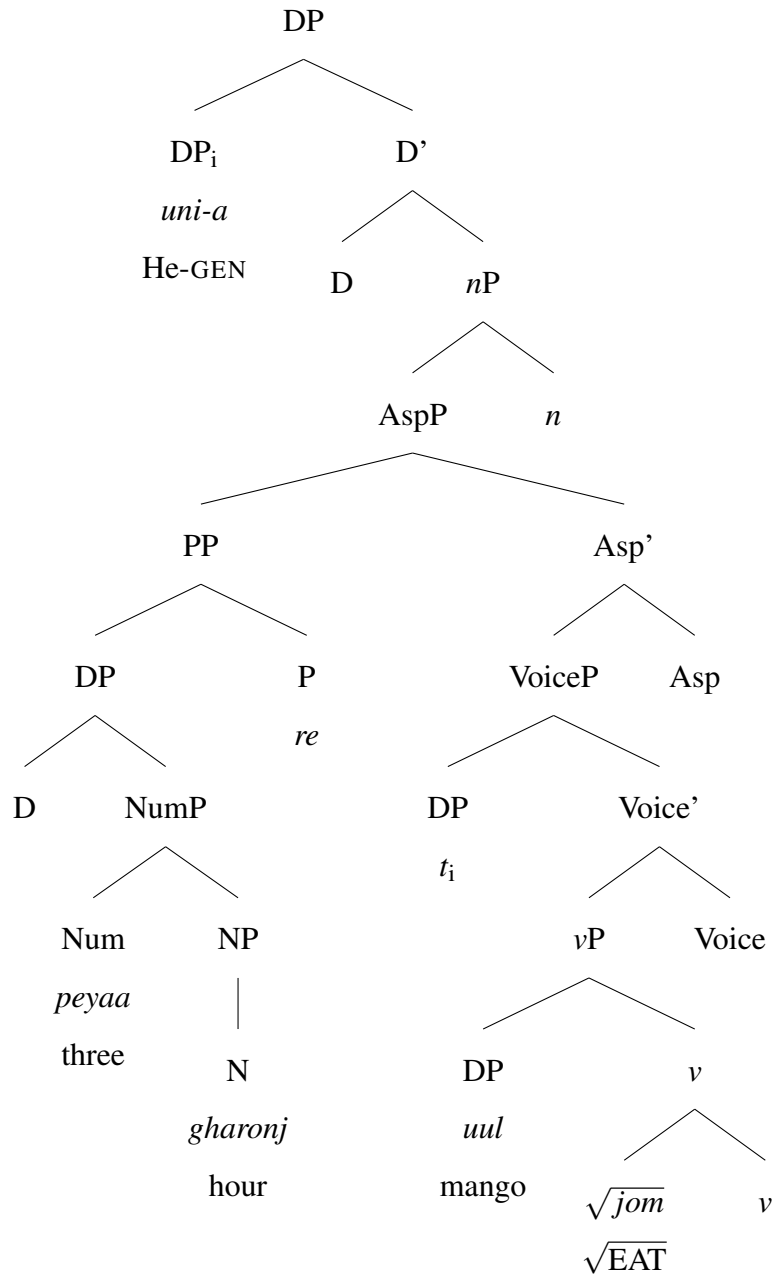
The ZNs in structures like (88, 90) qualify all the diagnostic tests used to identify a verbal structure. As already seen in (63a) and repeated here as (91a), the ZNs can be modified by adverbials like *dhirte* ‘silently’ (92). They can also have an aspectual modifier, like ‘in three hours’ (91b), structurally represented as (93).

- (91) a. *uni-a dhirte puthi paDhaao*
He-GEN silently book read
‘his reading a book silently’
- b. *uni-a peyaa gharonj-re uul jom*
He-GEN three hour-in mango eat
‘his eating a mango in three hours.’

(92)



(93)



The adverbial modification (92) and the aspectual modifier (93) further support that Santali ZNs are not complete nominals and carry an elaborate verbal structure.

In this subsection, we saw that even if Santali has three types of nominals, such as prototypical nominals, affixial nominals, and zero nominals, they can form two classes depending on how they are formed in the derivation. The first two can be formed by nominalizing the root, and the last one can be formed by nominalizing a verbalized root, not a root directly.

2.6.4 A note on Nominalizer Height

In this chapter, we saw that nominalization in Santali happens at various points in the derivation. The level a nominalizer can attach to, can be too low in the derivation, like the first merge position of the root. It can also be as high as the AspectP (AspP) level in Santali to project deverbal ZNs. When a nominalizer attaches to the AspP level, the nominal structure accommodates the aspectual modifiers and argument structures by including the VoiceP level below the AspP in the derivation. However, even if Santali has AspP as the upper limit of nominalization, the level of the merge of a nominalizer can go up to TP and CP levels cross-linguistically. According to Alexiadou (2020), and Kornfilt and Whitman (2012), languages like Japanese (94) and Turkish (95) display nominalization of TPs.

(94) Japanese (Alexiadou, 2020)

haruo ga [[zyotyuu-no soozisi-ta] heya] o mi-ta
Haruo NOM maid-GEN clean-PST room ACC see-PST

‘Haruo saw the room that the maid cleaned.’

(95) Turkish (Alexiadou, 2020)

hasan [uſağ-in oda-yı temizle-diğs-in]-i söyle-di
Hasan servant-GEN room-ACC clean-FN-3.SG-ACC say-PST

‘Hasan said that the servant cleaned the room.’

Similarly, following Zu (2009), Borsley and Kornfilt (2000), Alexiadou also shows that Greek contains nominalization of the CP level (96).

(96) Greek (Alexiadou, 2020)

to oti o Nikos efige me stenahorise
the that the Nikos.NOM left me.ACC upsetted

‘That he left upset me.’

However, Alexiadou claims that the higher-level nominalizations like CP, the nominalization happens by the merge of D with the CP. In (96), the CP is nominalized by the D because D takes the CP as its complement.

2.7 Summary and Conclusion

In this chapter, I have presented three types of nominals in Santali: prototypical nominals, affixal nominals, and zero nominals. The first two types are formed by merging a null or affixal nominalizer with an acategorial root. Zero nominals, however, exhibit intriguing patterns that deviate from what has been reported in the literature. Unlike traditional accounts, which suggest that zero nominals are derived from roots, my findings indicate that they originate from already categorized elements. Santali zero nominals (ZNs) exhibit a clear verbal extended functional structure. Through multiple diagnostics, such as adverbial modification, argument structure preservation, aspectual modification, and high productivity across verb classes, I have shown that Santali ZNs align more closely with argument-supporting nominals (ASNs) and gerundive nominals (GNs) than with result/referential nominals. This challenges the cross-linguistic generalization that zero-derived nominals are necessarily root-derived and idiosyncratic. This evidence suggests that zero nominals are generated within the syntactic workspace, characterized by high semantic compositionality and productivity.

The findings of this chapter have crucial theoretical implications for studies on categorization and word formation. They suggest that zero morphology in Santali can encode complex functional structures typically associated with overt morphemes in other languages.

Chapter 3

Structure and Derivation of Verbs

3.1 Introduction

A central debate in the non-lexicalist theories of the generative tradition concerns whether denominal verbalization exhibits systematic compositionality or is restricted to idiosyncratic root-derived meanings, with significant implications for our understanding of word formation and semantic interpretation.

Within the architecture of word formation, verbalization (derivation of verbs) has occupied a core theoretical and empirical significance, paralleling the well-studied process of nominalization. Verbalization¹, also known as ‘denominal verb formation’ or ‘conflation’ (Hale and Keyser, 1993), ‘noun to verb conversion’ (Iordăchioaia, 2023), can refer to the formation of verbs from other lexical categories like nouns (N) through conversion (Hale and Keyser, 1993), adjectives (A), (Embick, 2004; Aikhenvald, 2011), and roots (Marantz, 1997; Halle and Marantz, 1993; Harley and Noyer, 1999; Embick, 2007; a.o.).

In this chapter, using empirical evidence from Santali, I discuss the structure and formation of both root-derived and stem-derived verbals, demonstrating that Santali exhibits both types of verbalization processes systematically. As we have observed in Chapter 2, verbs in Santali can be turned into other categories, specifically zero nominals (ZNs) that retain extended structural

¹I use the term ‘verbalization’ for consistency in this dissertation. In the literature, there are different terms used for which I call verbalization. However, since I will be showing evidence for productive compositional semantics of nominals in verbalized elements, the term ‘Verbalization’ cannot mean creating new verbs without any related meaning to their nominal bases. Also, the use of the term ‘verbalization’ to refer to formation of verbs out of roots or other categories is well accepted in the generative literature (Marantz, 1997; Arad, 2003; Marantz, 2013; Alexiadou, 2020).

and semantic properties of a verb. In this chapter, by showing that verbs are also formed from other categories, I propose that categories in Santali are not only formed from acategorial roots, they are also formed from already categorized element, indicating a bidirectional nature of category formation.

I argue against the long-standing claim in the generative literature that verbalization is always idiosyncratic. By providing evidence from Santali, I claim that a verbalized structure can accommodate more categorial structures in the complement of the verbalizer (*v*), even to form verbs out of nominals by retaining the nominal semantics in the verbalized structure. Crucially, when verbs are derived from other categories, they consistently exhibit a regular change of state (become) semantics, providing strong evidence for systematic compositionality in Santali verbalization processes. Before discussing the process of verbalization and how the structure accommodates systematic semantic compositionality, I present some diagnostic tests to prove that the phenomenon of nominals inflecting for verbal markers in the structure is the process of verbalization, not a structure with the presence of a null semi-copula (Pustet, 2003; Hengeveld, 1992, p. 36) between the nominal base and the inflectional markers. To provide further evidence in favor of the verbalization analysis, I compare Santali structures with Hindi/Urdu, a dominant Indo-Aryan language spoken in the Santali-speaking regions.

The chapter is structured as follows. I start by providing an introduction to the chapter in Section 3.1. Section 3.2 begins by understanding the definition of verbs from the traditional approach to the generative account, followed by an overview of Aikhenvald's cross-linguistic survey of verbal patterns. This section further introduces the verbal structures in Santali. Section 3.3 constitutes the theoretical core of the chapter, examining the process of verbalization. Within this section, I critically evaluate two major theoretical frameworks on verbalization before turning to an analysis of verbalization in Santali in Section 3.4, with particular attention to both root-derived and stem-derived verbal structures. I employ three diagnostic tests to demonstrate that Santali exhibits stem-derived verbalization, a typologically marked phenomenon that distinguishes it from the majority of the world's languages. Subsequently, I propose a feature-based model to account for the productive nature of stem-derived verbalization in Santali. The chapter concludes with an examination of argument-alternating operations, using empirical evidence from Santali data in Sections 3.5 and 3.6.

3.2 What is a Verb

3.2.1 Understanding the Structure of a Verb

Traditionally, verbs were defined semantically as actions, states, or processes that are predicative in nature. From a morphological viewpoint, they were considered to be the categories that are inflected for tense, aspect, and mood. Verbs are also inflected for phi features, like person, number, and gender. In the Hindi/Urdu example in (1) the verb *aa* is inflected for tense, aspect, and phi features.

(1) Hindi/Urdu

jon aa-yaa th-aa
John come-PST.3SG.M be.PST.PERF-3SG.M
'John had come.'

Verbs can also be inflected for honorificity in some languages (2).

(2) Hindi/Urdu

pandit-ji aa-ye
Priest-HON come-PST.3SG.HON
'The priest came.'

Verbs are defined as semantic role assigners. In a structure like (3), the verb 'kill' assigns agent and patient semantic roles to 'John' and 'the snake', respectively.

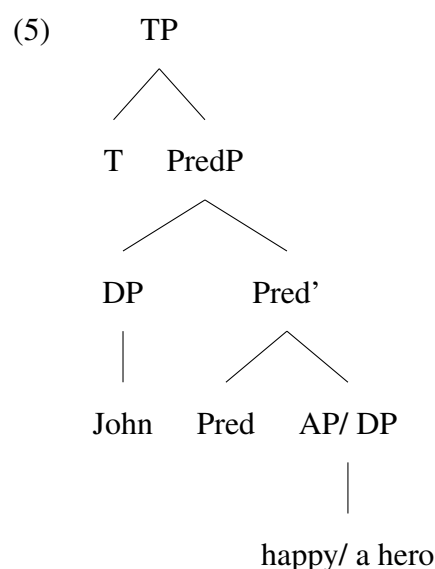
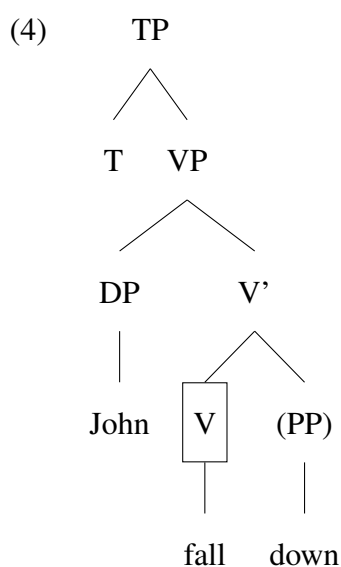
(3) [[_{Agent}John] [_{VP}killed [_{Patient}the snake]]].

Verbs have valency, as they can be intransitive, transitive, and ditransitive depending on the number of arguments they take.

Even in the generative tradition, Chomsky (1970) provides a feature-based formalism where he defines a verb, using binary features [+V, −N] (see Section-1.3.1.1), that lack substantive content according to Baker (2003). However, these kinds of traditional definitions are flawed for many reasons. One of them is the lack of mapping of the empirical evidence to the definition. According to Panagiotidis (2015), there are verbs like 'exist', 'emanate', and 'consist (of)' that challenge the action/state-kind of characterization. Although verb's ability

to be inflected for tense is a good morphological definition, non-finite verbs and imperatives are tenseless, contrasting with the morphological definition.

According to Baker (2003), Baker and Gondo (2020) a verb is the only lexical category (LC) that projects a specifier, along with a complement. This makes verb distinct from other LCs like nouns and adjectives. Baker defines verbs as intrinsic predicates that can license subjects. Baker and Gondo (2020) show that in (4) the subject ‘John’ is generated at the specifier position inside the VP, whereas the subject ‘John’ is generated outside the NP/AP, in the specifier of Pred in (5).



In this way, Baker distinguishes verbs from other LCs, showing that LCs like a noun or an adjective cannot have a specifier. They need a functional head (Pred) to have the subject in the specifier position (5).

My analysis of verbs in this chapter differs from Baker’s understanding of verbs in several ways. Baker (2003) focuses on defining the identity of a verb as a part of core LCs, which includes nouns and adjectives along with verbs. He defines a verb as an LC separate from LCs like nouns and adjectives by showing the structural property of having a specifier. Baker’s privative feature [+V] corresponds to this structural property of having a specifier or licensing a subject. That means if X is an LC and it can project a specifier, it must be considered a verb in a language. Baker does not discuss the internal structure of a verb, focusing on how verbs are formed. In this chapter, on the other hand, I talk about how verbs are formed, assuming

that verbs are the results of a verbalization process. They can be formed both from a root and an already categorized element in the verbalization process. The feature [+V] and the ability to project a specifier are assigned to verbs, the result of the verbalization process. Even if verbs constitute a distinct LC, according to Baker, I show that there are bigger structures to the derivation of verbs. In the next subsection, I present cross-linguistic data to show that there are verbs cross-linguistically that show properties of other categories like nouns and adjectives.

3.2.2 Verbs cross-linguistically

According to Aikhenvald (2011), verbs cross-linguistically have properties of other LCs like nouns and adjectives. There are different semantic classes listed by Aikhenvald that show the derivation of verbs from nouns or adjectives.

The first semantic type in Aikhenvald's list is inchoative verbalization. In this type, the result verb has an interpretation of 'become N' or 'acquire the property of N' (6 - 9).

- (6) Hausa (Newman, 2000)
 - a. *Mùsùlmī* 'Muslim'
 - b. *Mùsùluntà* 'become a Muslim'
- (7) Movima (Haude, 2006, p. 493)
 - a. *jo'me:* 'bird'
 - b. *jo'me:-ni* 'turn into a bird'
- (8) Dyirbal (Dixon, 1972, p. 86)
 - a. *barɣan* 'youth'
 - b. *barɣan-bi* 'become a youth'
- (9) Nyangumarta (Sharp, 2002, p. 19)
 - a. *karukaru* 'nausea'
 - b. *karukaru-jarri* 'become nauseated'

The second semantic type is propriative. In this type, the verbalized element has a 'have N' or 'have a property of N' meaning (10 - 14).

- (10) Urarina (Olawsky, 2006)

- a. *lana* ‘husband’
 - b. *lana-oka* ‘she has a husband’
- (11) Indonesian (Sneddon, 1996)
 - a. *anak* ‘child’
 - b. *ber-anak* ‘have children’
- (12) Ainu (Tamura, 2000, p. 218)
 - a. *tum* ‘strength’
 - b. *tum-asnu* ‘be strong’
- (13) Cupeño (Hill, 2005, p. 279)
 - a. *-ash* ‘pet’
 - b. *ash-lyu* ‘have a dog’
- (14) Tamambo (Jauncey, 1997)
 - a. *bwero* ‘ear’
 - b. *bwero-bwero* ‘be deaf’

Aikhenvald also shows examples that show the causative interpretation in a verbalized element showing the meaning of ‘produce or manufacture N’ (15 - 16).

- (15) Indonesian (Sneddon, 1996)
 - a. *telur* ‘egg’
 - b. *ber-telur* ‘lay an egg’
- (16) Movima (Haude, 2006, p. 486)
 - a. *juve* ‘dugout canoe’
 - b. *juve-ni-ti* ‘construct a dugout’

There are also semantic subclasses of verbalization which Aikhenvald calls manipulative verbs. These verbs have an ‘apply/use N’ interpretation (17).

- (17) Indonesian (Sneddon, 1996)
 - a. *topi* ‘hat’
 - b. *ber-topi* ‘wear a hat’

There are cross-linguistic examples of denominal verbs with an interpretation of ‘deprive of N’ (18 - 19).

(18) English (Aikhenvald, 2011)

- a. *caffeinate*
- b. *de-caffeinate* ‘remove caffeine from’
- c. *arm*
- d. *dis-arm* ‘deprive of arms’

(19) German (Aikhenvald, 2011)

- a. *Waffe* ‘arms’
- b. *ent-waffnen* ‘disarm’

The cross-linguistic data presented by Aikhenvald above demonstrate that verbs may have complex structures in the way they are formed, and verbalization processes involve either covert or overt verbalizers. Also, languages can vary depending on the types of verbalization they show. As we saw in Aikhenvald’s survey, the verbalizers in certain languages provide a change of state meaning, whereas they give owning properties or the semantics of produce/manufacture in other languages.

3.2.3 Verb Stems in Santali

Having an agglutinative morphology, Santali finite verb stems carry various grammatical markers along with the verb. As seen in Chapter 1, verbs in Santali are not only action words, but any morphological element can appear as a verb and take the verbal inflectional markers. A Santali verb stem consists of the verb, tense and aspect markers, voice marker corresponding to transitivity, subject and object clitics, and the phi inflectional markers. The voice marker can be middle or active, where the middle voice shows intransitivity and the active voice marker shows transitivity (Ghosh, 2008; Neukom, 2001; Anderson and Jora, 2023). In (20a), the middle voice and past tense marker *-en* is attached to the verb giving an intransitive structure, where as in (20b), the active voice and past tense marker *-ked* is attached to a transitive verb. There is always an obligatory finite marker in the verb stem (20) when the verb is finite (Ghosh, 2008). Apart from this, both the object and the subject markers are seen in the verb stem in

Santali. The object marker follows the tense and the active voice marker and precedes the finite marker in a transitive sentence. Covertly nominative marked subjects agree with verbs in person and number. There is no grammatical gender, gender marking on pronominals, or gender inflection on the verb in Santali.

- (20) a. *uni sen-en-a-e*
 He go-PST.MID-FIN-3SG.SUBJ
 'He left.'
- b. *jon uni-ke nyel-ked-e-a-e*
 John he-OBJ see-PST.ACT-OBJ-FIN-3SG.SUBJ
 'John saw him.'

The animacy feature of the subject and the object DPs interacts with the phi agreement. The verb doesn't inflect for the person or number feature of the subject or the object when the subject or the object is inanimate (21). There is no agreement marking on the verb with the subject in (21a), because the subject *arsi* 'mirror' is inanimate. Similarly, in (21b), there is no object marking on the verb since the object *puthi* 'book' is inanimate.

- (21) a. *arsi rapud-en-a-∅*
 mirror break-PST.MID-FIN-∅
 'The mirror broke.'
- b. *Arel puthi nyel-ked-∅-a-e*
 Arel book see-PST.ACT-∅-FIN-3SG.SUBJ
 'Arel saw the book.'

The order of the verb and other grammatical morphemes in a verb stem in Santali can be summarized in (22).

- (22) verb-tense.voice/transitivity-(OBJ)-FIN.PERSON.NUMBER.SUBJ

As I have already mentioned, Santali verbs are not limited to action words or prototypical verbs only, and the verbness can be defined on a configurational basis in this language. The kind of constructions where any LC can be verbalized in Santali, showing the categorial feature of the base LC on the verb, which I call stem-derived verbs, are the primary focus of this study. The constructions in (23) that the Santali verbs show properties of other LCs, like a noun, after verbalization.

- (23) a. *uni **dhiri**-en-a-e*
 He stone-PST.MID-FIN-3SG.SUBJ

‘He stoned.’ (He became a stone.)

b. *jon uni-ke dhiri-ked-e-a-e*

John he-OBJ stone-PST.ACT-OBJ-FIN-3SG.SUBJ

‘John stoned him.’ (John made him a stone/ become a stone.)

Two observations emerge from this section. First, the cross-linguistic evidence from Aikhenvald, combined with Santali’s fluid verb formation patterns, confirms that verbs are derived through verbalization rather than being primitive grammatical elements. Secondly, verbs in (23) with zero verbalizers suggest that even if Santali verbs are morphologically simple, they are structurally complex. It is because not only are the verbs in (23) different from prototypical verbs (20) in Santali, they also carry the semantics of the base noun.

It raises questions about how verbalization should be analyzed theoretically. Below, I examine different approaches to verbalization, beginning with the incorporation account before turning to underspecification models, like distributed morphology (DM).

3.3 Verbalization

In this section, I discuss the nature of verbalization, reviewing two major accounts in the generative literature that deal with the verbalization process. I also discuss the difference between the verbalization in English and Santali. I discuss how whether English shows only idiosyncratic or idiosyncratic and compositional verbalization patterns, has remained an unsolved issue in the literature. Santali demonstrates both idiosyncratic and compositional verbalization processes, differing from English in having a productive change of state semantics.

Verbalization, according to Aikhenvald (2011), refers to the kind of derivation with an end result showing the properties of a verb. As already discussed in the introduction, a typical underspecification account like DM posits that verbalization, or any kind of categorization in general is a syntactic process. In the case of the derivation of a verb, a verbalizer (v) merges with an acategorical ROOT to result in a verb.

The verbalization process has been a topic of discussion in the generative literature, focusing on whether the verbalizer takes an acategorical root or an already categorized noun/adjective as a complement. Using this, I address the question of verbalization height. In underspecification approaches, the categorizer height corresponds to the idiosyncratic or the compositional

interpretation of the acategorial root in category formation. In this chapter, focusing on the process of verbalization, I claim that verbalization is not always idiosyncratic as assumed in literature (Rimell, 2012; Alexiadou, 2020; a.o). Languages like Santali have verbalization that retains the interpretation of their base categories in the derivation. Therefore, I claim that Santali has both idiosyncratic and compositional verbs as products of verbalization, where the idiosyncrasy and the compositionality are subject to the merge position of the verbalizer (*v*) in the derivation.

Determining whether the verbalizer merges with an acategorial ROOT or a nominal base is tricky in languages like English. However, it is quite easy to make a distinction between root and noun in languages like Hebrew, as roots in Hebrew are usually a combination of consonants without a phonetic form (Rimell, 2012). In the case of Santali, it is difficult to say if a morphological element is a root or a nominal base just by looking at the element independently. As already seen in Chapter 1, Santali tends to show a zero/null categorizer, which makes it difficult for us to understand the categorial status of any morphological element. In (24), the morpheme *sen*, which is a root in (24a), acts as a verb in (24b), and a stem-derived denominal ZN in (24c). The verb in (24b) is derived by merging a verbalizer with a root, whereas the ZN in (24c), merges with a verbalized root to form the nominal, since the ZN carries verbal structural and syntactic properties of *sen* ‘go’.

- | | | |
|------|---|-----------|
| (24) | a. $\sqrt{sen}$ | ROOT |
| | b. <i>uni sen-en-a-e</i>
He leave-PST.MID-FIN-3SG
‘He left.’ | Verb |
| | c. <i>uni-a sen baa-ij kusi adaa</i>
He-GEN leave not-1SG good feel
‘I didn’t like his leaving.’ | Noun (ZN) |

We can understand the categorial status of a morpheme from its distributional patterns in a sentence. In English-type languages, we can observe a wide range of verbalized structures, where the verbal forms look closely related to their nominal counterparts (25).

- (25) English (Clark and Clark, 1979)
- a. Julia **summered** in Paris. (to be in Paris during summer)
 - b. Jane **blanketed** the bed. (to cover the bed)

- c. Mary **dusted** the shelf. (to remove dust)
- d. John **butchered** the cow. (to kill or cut up the cow for meat)
- e. Edward **powdered** the aspirin. (to crush the aspirin into powder)

There are certain properties that the verbalized structures in English carry. According to Rimell (2012), when nominals surface as verbs in English, they do not carry the nominal semantics after verbalization, behaving like root-derived verbals. There is also a ‘transitive bias’, where the verbalized elements in English do not usually show intransitive structures (26).

(26) English (Rimell, 2012)

- a. *Mary was **applling** at lunch today.
intended: ‘eating apples’
- b. *Beth was **hatting** when I saw her.
intended: ‘wearing a hat’

In the non-lexicalist literature, there are different assumptions for the way the verbalized structures like (25) are derived. As Clark and Clark (1979) point out, there are idiosyncratic meanings involved in the verbalization in (25). The interpretation of such verbals is dependent on particular occasions of their use and the mutual knowledge of the speaker and the hearer about a certain event, state, or process. The incorporation account by Hale and Keyser (1993) proposes that such verbs are formed by incorporating a nominal from an argument position to a verbal position, including the incorporation of a preposition on the way. A different set of proposals made by the non-lexicalists claims that verbs are derived by merging a verbalizer (*v*) with a root. The idiosyncratic or the compositional interpretation of the verb is dependent on how locally the root merges with the verbalizer head (Marantz, 1997; Arad, 2003; Marantz, 2013).

The kind of verbalization I study in this dissertation, focusing on Santali, differs from the English-type verbalization in many aspects. As already seen in (32) in Chapter 1, any morpheme can behave like a verb when it attaches to the verbal markers like tense, aspect, mood (TAM), and person, number, gender (phi) in Santali (27). They are regular, carry compositional semantics of the category they derive from, and they are not subject to the ‘transitive bias’, as we can see both intransitive (27b) and transitive (28) verbalization in Santali.

(27) Santali

- a. *raajaa sen-en-a-e*
King go-PST.MID-FIN-3SG
'The king went.'
- b. *uni raajaa-en-a-e*
He king-PST.MID-FIN-3SG
'He kinged.' (He became a king.)

(28) Santali

ij uni-ke raajaa-ked-e-a-ij
I He-OBJ king-PST.ACT-OBJ-FIN-3SG.SUBJ
'I kinged him.' (I made him a king.)

I argue in subsequent sections that these constructions (27-28) are highly productive in Santali and are formed by recategorization of nominals. However, before going into the details of the formation of structures like (27-28) and asking if the structures in the study are instances of verbalization at all, let's discuss some previous accounts on the derivation of verbs in the literature.

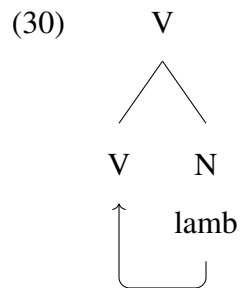
3.3.1 The incorporation account

The incorporation account, also known as 'conflation', developed by Hale and Keyser (1993), is one of the influential works on denominal verbs. According to the incorporation theory (Hale and Keyser, 1993), verbalized structures are formed by moving a nominal from an argument position to a null verbal position. The interpretation of the denominal verb is dependent on the base-generated position and the movement of the nominal (Rimell, 2012). Hale and Keyser argue that the verbal derivation happens in L-syntax, a lexical subcomponent that follows the rules of syntax.

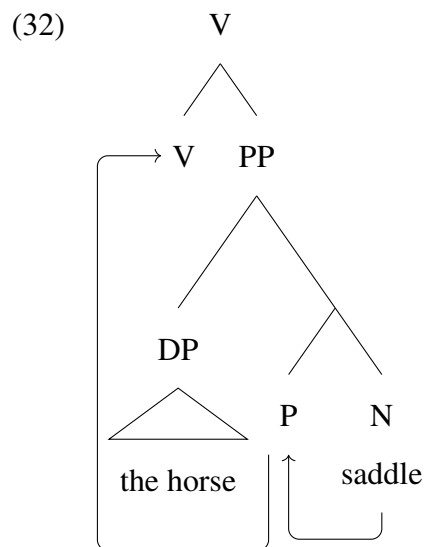
Hale and Keyser's analysis resulted in two primary structural configurations. A null verbal head takes either a nominal or a prepositional structure to generate a denominal verbalized structure. Unergative denominal verbs are formed by incorporating a nominal from a complement position to the null verbal head (29), schematically represented as (30). Location/locatum verbs are formed by the incorporation of a PP complement to the verbal head by incorporating

a nominal into a preposition and then into the verbal head (31), schematically represented as (32).

- (29) a. The ewes lambed.
b. The cow calved.



- (31) a. saddle the horse
b. shelve the books



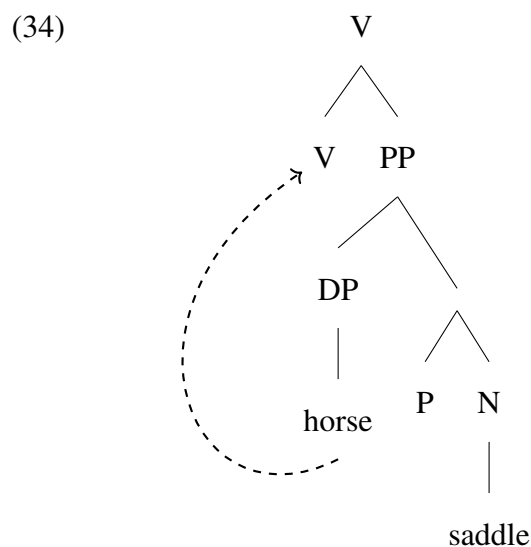
Structures in (30) and (32) are the only kind of denominal verbs English can have, according to Hale and Keyser, as verbalized structures are limited to unergative and location/locatum verbs.

Unergative examples are divided into two categories in the incorporation account. The first type includes verbs of birthing (foal, calve), and the second type includes ‘activity unergatives’ (see Rimell, 2012) that historically derive from verbal roots (laugh, run, etc.). Since the location/locatum structure generates verbs with underlying prepositional phrases, they form location verbs, like ‘corral’, ‘cage’, etc., with prepositions like ‘into’, ‘onto’, which Hale and Keyser call prepositions of “terminal coincidence”. There are also locatum verbs, like ‘sad-

dle' and 'salt,' formed through incorporation with prepositions of "central coincidence", using 'with'.

Even if the incorporation account considers the L-syntax (a lexical subcomponent that follows syntactic rules) to be a part of the lexicon, Hale and Keyser propose that categorization is fundamentally a syntactic process. They posit that verb formation through incorporation must obey syntactic rules like head-to-head movement. The unacceptability of structures like (33), schematically represented as (34) posits a constraint on incorporation from a specifier position to a null head position.

(33) *horsed the saddle

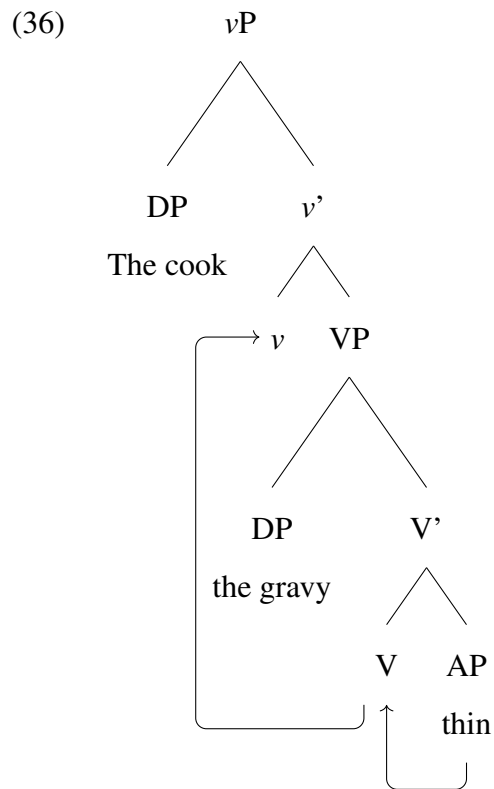


(Rimell, 2012, p. 24)

The third type of incorporation that Hale and Keyser show is for the formation of deadjectival verbs in English. Verbs in constructions like (35) are formed by incorporating an adjective into the verb position (36). These constructions are associated with a change of state where the adjective in (36) is the state, and the theme or the IA is the 'affected patient'.

(35) English deadjectival verbs

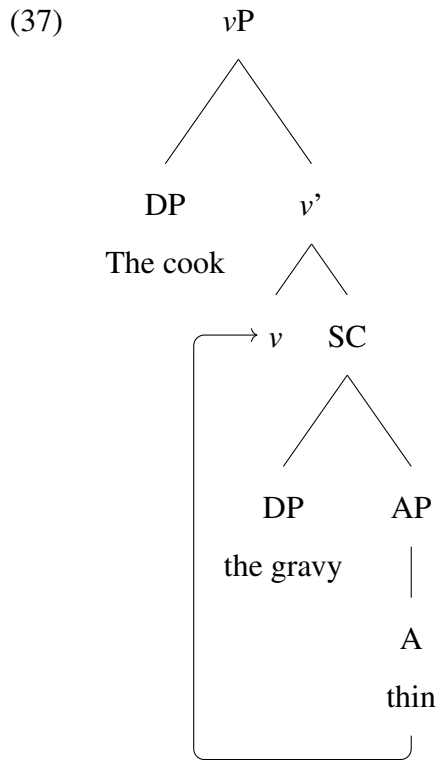
- a. The cook thinned the gravy.
- b. The oven browned the toast.
- c. The storm cleared the air.
- d. This narrows your options.



In (36),² the theme, which is the specifier of the lexical V, goes through a change of state to become the adjective, which gets incorporated into the verb. In such constructions, the state is always an AP in English, never an NP.

This analysis of deadjectival verbs has been further developed in subsequent literature involving movement to V from a small clause (Harley, 2011b; Pedersen, 2014), a.o. Both the theme DP and the state AP are taken as parts of a small clause where the theme of the verb works as the subject of the small clause that goes through change to become the state AP (37).

²The structure is modified changing the VP shell kind of structure followed by Hale and Keyser to a functional vP/VoiceP and the lexical VP for consistency.



The structure in (37), which shows incorporation from a small clause to a verb position, can have its intransitive counterparts (38), without the spec vP position (Kiparsky, 1997; Harley, 2011b; Pedersen, 2014).

- (38)
- a. The gravy thinned.
 - b. The toast browned.
 - c. The air cleared.
 - d. The sky cleared.

To summarize, the incorporation account for verbalization explains three types of verb formation in English, where two strategies are associated with denominal and the other is associated with deadjectival verbalization. The unergative denominal verbs are formed by incorporating a nominal from the complement position of the verb to the null V position, and the location/locatum verbs are formed by incorporating nominals from a complement of P to the P head, then to the V head. According to Hale and Keyser, both unergative and location/locatum verbs have an underlying transitive structure. Deadjectival verbs, on the other hand, are formed by incorporating an adjectival state predicate to the V head from a Small Clause (SC). There is also a change of state semantics associated with the SC, which shows a transition of a DP to

an adjectival state. For Hale and Keyser, the nominal and adjectival bases are lexically listed, which turn into verbs using syntactic rules, such as head movement.

Even though the incorporation account was widely accepted in the generative literature, one of its major drawbacks was the possibility of overgeneration of denominal verbs. However, such a case of overgeneration of denominal verbs is not an issue for a language like Santali, because any noun can be turned into a verb in Santali. It is also worth noting that Santali noun-derived verbs pattern with the deadjectival verbs from an SC in Hale and Keyser's approach, not the denominal verbs.

Another criticism of the incorporation approach is that it is limited to English and English-type languages. According to Mateu Fontanals (2000), the incorporation account doesn't explain resultative constructions in Romance languages. It also doesn't predict much for verbalization in languages with a richer morphology, like Santali, since Santali categorization is not unidirectional, as it shows fluidity in both N to V and V to N conversion. Hale and Keyser's approach takes nouns and adjectives to be lexically listed, which later get syntactically incorporated to form verbs. However, since nominals are not primitive to languages (Alexiadou et al., 2007; Alexiadou, 2010, 2020; Iordachioaia, 2020; Iordăchioaia, 2023), specifically to a language like Santali, as we have already seen in Chapter 2, the incorporation account fails to capture the bidirectionality of categorization. This lexical listedness of the base categories has been opposed in underspecificational accounts like DM.

Another criticism against the incorporation account is that it doesn't include denominal verbs of causative/inchoative alternation. Hale and Keyser show that the change-of-state kind of alternation is restricted to deadjectival nominals. However, Kiparsky (1997) shows that nominals, like 'carbonize', 'gasify', etc., can participate in an SC-like construction behaving like states, which then can become verbs.

The incorporation account will face several limitations when we apply it to Santali categorization. The most critical problem is related to the lexical listedness. Since this approach assumes that nouns are primitive to grammar, which are listed in the lexicon, and they get incorporated to form verbs, it won't be able to capture the Santali derivational system, where nouns themselves are derivationally formed from both roots and verbalized elements, as demonstrated in Chapter 2. In addition to this, since the incorporation approach takes verb formation to be unidirectional (from noun to verb), it will fail to account for the bidirectional fluid character-

istics of Santali categorization. In Santali, along with root-derivation, both noun-to-verb and verb-to-noun formation occur productively. Lastly, the account will fail to distinguish between verbs with idiosyncratic interpretation and those with compositional meanings.

Now, after discussing the incorporation account for verbalization, we move to DM, the other influential approach in the generative literature that analyzes verb formation using a syntactic approach different from incorporation.

3.3.2 The DM model

As already discussed in Chapters 1 and 2, categories, more specifically verbs, can be formed by merging an acategorial root with a verbalizer according to the underspecification accounts, like DM (Halle and Marantz, 1993; Arad, 2003; Don, 2005; Harley, 2009; Alexiadou and Borer, 2020). Some other subscribers of the DM account (Arad, 2003; Panagiotidis, 2015), following Kiparsky (1982, 1997), posit that verbalized structures can be derived by categorizing the roots or re-categorizing an already categorized element (N). The debate discussing whether verbalization is a root-derived process or both root-derived and stem-derived process is dependent on the point at which the idiosyncratic information of the root can be accessed.

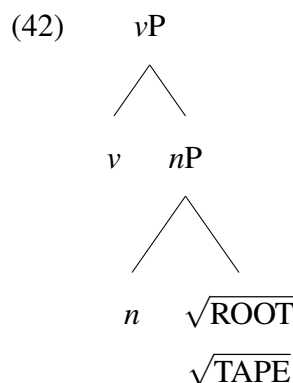
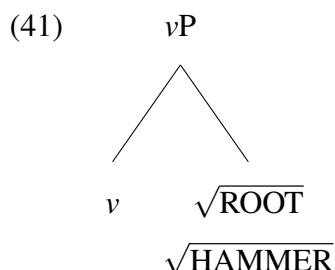
Following Kiparsky (1982), Arad (2003) points that verbs like ‘hammer’ (39) are derived directly from the acategorial root by merging a verbalizer, whereas verbs like ‘tape’ (40) are good candidates for stem-derived verbs, because they are derived by recategorizing an already categorized element. Verbs in (39) are idiosyncratic, and they do not carry the semantics of any nominal. When the roots merge with a verbalizer, the verbalizer accesses the idiosyncratic meaning from the root locally, and the semantics of the verb is decided. In case of the verbs in (40), on the other hand, ‘tape_v’, ‘chain_v’, etc., carry the semantic properties of their nominal counterparts. According to Rimell (2012), these verbs can be elaborated by hyponymous NPs.

(39) English (Arad, 2003)

- a. He **hammered** the nail with a rock. (Kiparsky, 1982)
- b. **String** him up with a rope!
- c. She **anchored** the ship with a rock.

(40) English (Arad, 2003)

- a. She **taped** the picture to the wall (with a tape/ *with pushpins).
- b. They **chained** the prisoner (with a chain/ *with a rope).
- c. Jim **buttoned** up his pants (*with a zipper).



As already discussed in (86) in section 2.6.3.3, ‘tape-type’ compositional verbals are formed by a derivation like (42), where the verbalizer merges with the Closed Interpretation Domain (CID) created by the nominalizer (*n*). The *v*, being the second categorizer, accesses the nominal categorial information from the *n*, the most local phase head, and gives a compositional nominal semantics.

The empirical motivation for the compositional verbalization seen in (42) is much older than the DM account. It can be traced back to Kiparsky’s (1982) Canonical Use Constraint (CUC), which links the verbalized element and the base nominal based on their semantic similarity (43).

- (43) **Kiparsky’s Canonical Use Constraint (CUC).** If an action is named after a thing, it involves a canonical use of the thing.

‘Tape-type’ verbs are noun-derived because, after verbalization, they can refer to the canonical use of their nominal counterpart. This explains the unacceptability of the following structures (44).

- (44)
- a. *Lola **taped** the poster to the wall with pushpins.
 - b. *They **chained** the prisoner with a rope.
 - c. *Jim **buttoned** up his pants with a zipper. (repeated from 40)
 - d. *Mary **bicycled** to work on her tricycle.

e. *He **nailed** the proclamation to the door with rivets. (McCawley, 1971)

f. ***Screw** the fixture on the wall with nails. (Kiparsky, 1982)

Arad (2003), supporting Kiparsky's CUC, proposes that languages like English have both root-derived and noun-derived verbalization. Dowd (2010), in a similar line of thought, proposes that the 'hammer-type' and the 'tape-type' distinction is dependent on the classification of nouns based on their forms and functions. 'Hammer-type' verbs are defined by their functions and therefore take hyponymous objects, which are identical in functions, not forms. On the other hand, 'tape-type' verbs are defined by their forms, which make them take instruments of identical forms as adjuncts. In the subsequent section, we'll see that Santali verbs are similar to English in this aspect. They also have both root-derived and stem-derived verbalization. However, Santali goes a step further in showing extreme productivity because verbs can be derived from any Santali stem with a productive change-of-state interpretation.

Coming back to the debate on verb formation, recent literature (Harley and Haugen, 2007; Rimell, 2012; Alexiadou, 2020; a.o.) oppose the idiosyncratic and compositional distinction by positing that verbalized structures (including the ones that look like denominal verbs on the surface), are root-derived and always idiosyncratic. They are never compositional and carry the idiosyncrasies of the corresponding roots. Structures like (45) contradict Kiparsky's CUC-based assumption which does not allow structures like (44), incorrectly predicting that these verbs take instrumental adjuncts identical to the nominal counterparts of the verbs. In (45), we can see that when the instrumental adjuncts of the verbalized structures are not too different from the manner of the nominal base of the verbalized element, the structures are perfectly acceptable in English.

(45) a. Lola taped the poster to the wall with band-aids / mailing-labels.

b. Screw the fixture on the wall with corkscrews.

Rimell also shows that many CUC verbs, which on the surface seem like they correspond to canonical event type (46). In (46), 'butchering' or 'umpiring' mean doing something a butcher or an umpire does. However, the subjects of these constructions do not need to be the exact base noun, professionally.

(46) a. **butcher** the cow

- b. **umpire** the match
- c. **jockey** the horse
- d. **referee** the game
- e. **nurse** the patient

Therefore, Rimell posits that the verbs in (46) are derived directly from a root. The DM model presents two competing observations for verbalization. One is the traditional Kiparsky-Arad-kind of approach that distinguishes between root-derived verbs ('hammer') and compositional stem-derived verbs ('tape'). I have shown that the distinction in the literature is based on semantic transparency and the CUC. The other observation is the root-derivation approach adopted by Harley and Haugen, Rimell, and Alexiadou, among others. We saw that the CUC-based distinction initially appears compelling in explaining instrument restrictions. However, the counter-evidence with directly unrelated adjuncts and semantic idiosyncrasies of the verbalized element suggests that even seemingly compositional verbalizations may be fundamentally root-derived.

Assuming stem-derived verbals are theoretically possible in languages, even if it is debatable in English, would not mean stem-derived verbals will be identical to (42). A crucial point to note here is that we would expect a structure like (42) when a language produces verbalized structure that is indeed compositional from any other categorial base, like a nominal. The system should be productive, showing regular patterns in verbalization without showing idiosyncrasies of the roots. In subsequent sections (3.4.2 and 3.4.3), we'll see that Santali has such a categorial system in terms of compositionality. Santali compositional verbs also go one step further by showing a change of state/ transition from the theme to the base category of the verb.

Now, in the subsequent section, we turn to examine whether languages like Santali exhibit truly productive or regular verbalization patterns that would support compositional derivation, or whether all verbalization ultimately reduces to root-level idiosyncrasy, or it shows both.

3.4 Verbalization in Santali

The categorial status of a word in isolation is difficult to determine in Santali because of the categorial fluidity in the language that allows any acategorial root to behave as a noun or a verb in syntax (Bodding, 1929; Neukom, 2001; Ghosh, 2008, p. 53). Morphologically, a verb in Santali is inflected for tense, aspect, mood (TAM), person, number, gender (phi), voice, and finiteness. As already discussed in chapter 2.1, action words or events are not the only category of words that behave as verbs in Santali. Rather, any morphological element or a root that takes the TAM, phi, and voice markers, etc., is a verb in Santali. This supports the central point of this dissertation by eliminating a typical N-V distinction pre-syntactically. I present some examples from Santali below to show what kind of morphological elements can act as verbs. Some of the examples are repeated from Chapter 1. In (47), repeated from (32) of chapter 1, both ‘sen’ (go) and ‘raajaa’ (king) take the identical verbal morphology by eliminating the N-V distinction.

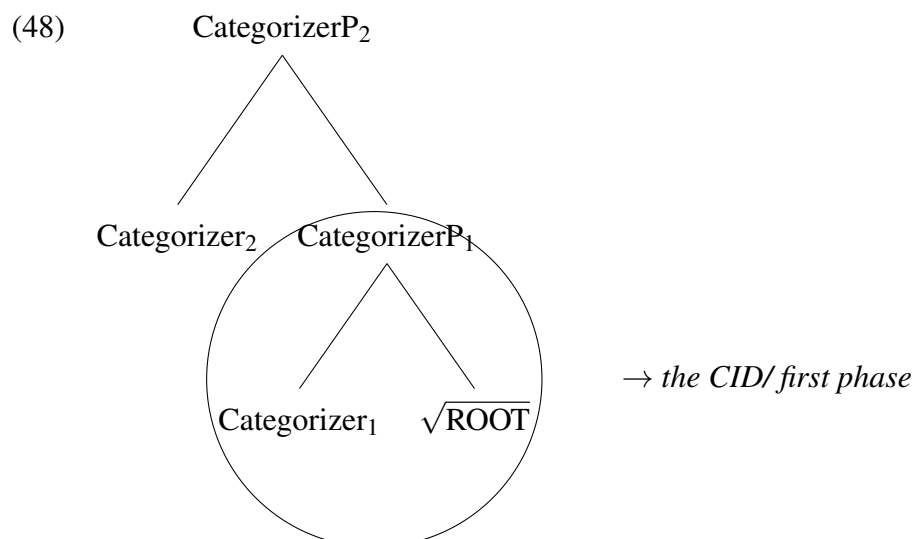
- (47) a. *raajaa sen-en-a-e*
King go-PST.MID-FIN-3SG
‘The king left.’
- b. *uni raajaa-en-a-e*
He king-PST.MID-FIN-3SG
‘He kinged.’ (He became a king.)

One of the major claims I make in this chapter is that Santali has both root-derived and stem-derived verbalization. Root-derived verbalization involves direct attachment of verbalizers to acategorial roots, typically resulting in idiosyncratic meanings. Stem-derived verbalization, by contrast, applies verbalizers to already categorized stems, maintaining compositional semantic relationships with the base category (Marantz, 1997; Arad, 2003). Even if on the surface there is no difference between the verbs in (47a) and (47b), I claim in this chapter that they differ depending on the height of the verbalizer in the derivation. This implies that there is a difference between the root-derived and the stem-derived verbalization in Santali. Opposing the claim for English that verbalization is always root-derived, (Harley and Haugen, 2007; Rimell, 2012; Alexiadou, 2020; a.o.), I show in the next few sections that the presence of stem-derived verbals is a productive system in Santali, along with the root-derived nominals. In the next subsections, I discuss these subtypes of verbals based on their formation in the

derivation.

3.4.1 Root-Derived Verbalization

This type of verbalization is the process to form prototypical verbs in any language according to underspecificational accounts, like DM. In Santali, the prototypical verbs are formed by merging a *v* with an acategorial ROOT to provide idiosyncratic verbal meaning to the derived structure. As argued by Marantz (2000, 2013), Arad (2003), Panagiotidis (2015), among others, when the first category-determining head merges with a root, it acts as the first phase and creates a closed interpretation domain (CID). The first phase/categorizer gets the interpretation of the root locally, and any idiosyncratic interpretation (special meaning or irregular form) is only possible inside the CID. In a structure like (48), repeated from (86) of Chapter 2, the first categorizing phase head (Categorizer₁) closes off the domain for root-specific interpretation. According to Marantz (*ibid*), followed by Arad, any categorizer that merges outside the CID only produces forms and meanings that are regular and computed compositionally from the already fixed interpretation of the categorized stem. This means in the derivation in (48), the Categorizer₂ (the second phase) doesn't access any idiosyncratic information from the root, that is not local to it.

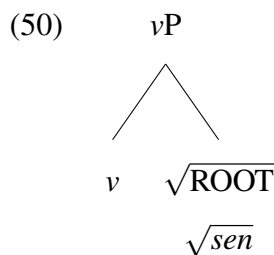


When we take this account forward to analyze Santali verbalization, the verbs like the ones in (49) are formed in the first merge position of a categorizer, as structurally represented in (50).

- (49) a. *binit sen-en-a-e*
 Vineet go-PST.MID-FIN-3SG

‘Vineet left.’

- b. *baahaa baŋ hej-en-a-e*
Baha not come-PST.MID-FIN-3SG
‘Baha did not come.’
- c. *uni goe-en-a-e*
He die-PST.MID-FIN-3SG
‘He died.’
- d. *aale eneh-ke-a-le*
we play-PST.ACT-FIN-1PL
‘We played.’
- e. *jon mili-ke puthi em-ked-e-a-e*
John Mili-DAT book give-PST.ACT-OBJ-FIN-3SG
‘John gave a book to Mili.’



The verbs in (49), represented in (50) are idiosyncratic in nature and do not carry the interpretation of any already categorized elements like a noun or an adjective. These idiosyncratic verbs are derived from roots, not nominal or adjectival stems. These verbs can participate in further derivation, i.e., they can be nominalized with a morphologically null nominalizer, producing ZN in Santali, as seen in Chapter 2.

However, verbs like *raajaa* ‘king’ in (47b) are not formed using the same strategy. Unlike English-type languages, they are not idiosyncratic and always carry the compositional meanings of another base category, like a nominal ‘king_N’ in (47b). In the subsequent subsection, I discuss the productive stem-derived verbalization strategy in Santali.

3.4.2 Stem-Derived Verbalization

Santali has a productive system like (47b) in which any prototypically non-verbal element can appear as a verb in the syntax. Constructions like (51) repeated from (34) of chapter 1, demonstrate that adjectives (51a), demonstratives (51b), kinship terms (51c), animal names

(51d), and even the most restricted kind of noun, like proper names (51e) can take the TAM, phi and voice markers to act as a verb.

(51) Santali

- a. *daare maaraang-en-a*
Tree big-PST.MID-FIN
'The tree bigged.' (The tree became big)
- b. *hana noa-en-a*
That this-PST.MID-FIN
'that this-ed.' (That thing became this thing)
- c. *uni ini bohya-en-a-e*
She my sister-PST.MID-FIN-3SG
'She my sister-ed.' (She became my sister)
- d. *uni seta-en-a-e*
He dog-PST.MID-FIN-3SG
'He dogg-ed.' (He became a dog)
- e. *uni binit-en-a-e*
He Vineet-PST.MID-FIN-3SG
'He Vineet-ed.' (He became Vineet)

The constructions in (51) have a change of state semantics where a theme goes through some kind of change to become the verbalized element, which is the state. These are unaccusative constructions showing some kind of a become/inchoative semantics. The constructions in (51) have their transitive counterparts in Santali, where the verbalized element takes an active voice marker (Bodding, 1929; Anderson, 2008; Ghosh, 2008, Rau, 2013 a.o.). These transitive constructions (52) show the same kind of change of state where an agent changes the theme to the state of the verbalized element, aligning with a causative/inchoative alternating structure (Harley, 2011b).

(52) Santali

- a. *jon daare-ke maaraang-ked-e-a-e*
John tree-OBJ big-PST.ACT-OBJ-FIN-3SG.SUBJ
'John bigged the tree.' (John turned the tree into big.)
- b. *uni hana-ke noa-ked-e-a-e*
He that-OBJ this-PST.ACT-OBJ-FIN-3SG.SUBJ
'He this-ed that.' (He turned that thing into this thing.)

- c. *jon uni-ke ini bohya-ked-e-a-e*
 John she-OBJ my sister-PST.ACT-OBJ-FIN-3SG.SUBJ
 ‘John my sister-ed her.’ (John turned her into my sister.)
- d. *jon uni-ke seta-ked-e-a-e*
 John he-OBJ dog-PST.ACT-OBJ-FIN-3SG.SUBJ
 ‘John dogg-ed him.’ (John turned him into/ made him a dog.)
- e. *jon uni-ke binit-ked-e-a-e*
 John he-OBJ Vineet-PST.ACT-OBJ-FIN-3SG.SUBJ
 ‘John Vineet-ed him.’ (John turned him into Vineet.)

Unlike (50), the verbal constructions in (51) and (52) are stem-derived because the verbs retain properties of their base categories after verbalization. If these constructions are really verbalizations (which I prove using certain diagnostics), they highlight two important points empirically. First, the generalization that verbalization is always idiosyncratic, therefore root-derived (Harley and Haugen, 2007; Rimell, 2012; Alexiadou, 2020), has to be cross-linguistically tested. The empirical evidence from Santali shows that verbs in (51) and (52) are not root-derived. They are compositionally formed from an already categorized element, showing more intermediate structures between the acategorical root and the verbalizer (*v*).

Secondly, if the constructions in question are indeed verbalizations, they are not as derivationally simple as the stem-derived structures (42) proposed by Arad (2003), among others. They show similarities with the SC-kind of analysis (37) where the base category of the verb corresponds to the result of the change-of-state construction. These Santali stem-derived verbs correspond to causative/ inchoative structures (Harley, 2011b). However, after strengthening my claim that the stem-derived constructions in question are instances of verbalization using diagnostic tests, I extend the SC-kind of analysis of the verbs to further decompose the base categories to align with the DM account.

Throughout this dissertation, I refer to the derivation of verbs in (51) and (52) as verbalization. However, there is another strategy to use non-verbal categories in the predicate position in Santali. That is by the help of a copula, like many other languages in the world. In (53), Ghosh (2008) shows that a genitive DP (53a) and a *wh*-morpheme (53b) take the copula *kan* to occupy the predicative position.

(53) Santali

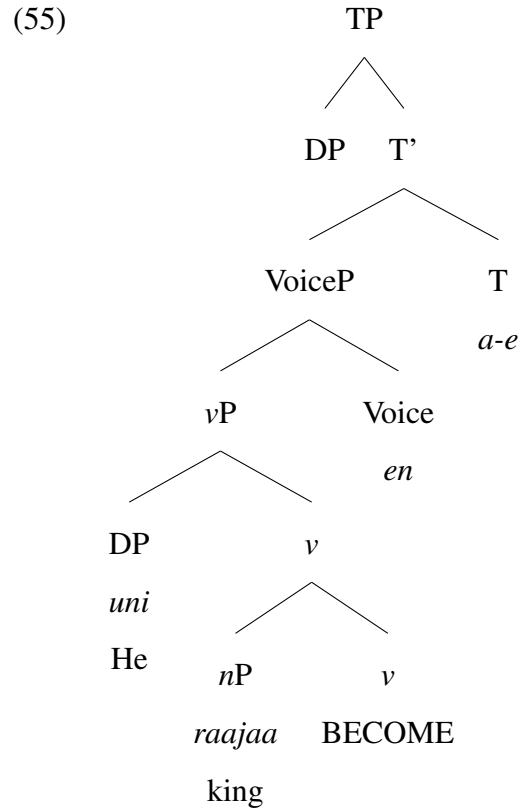
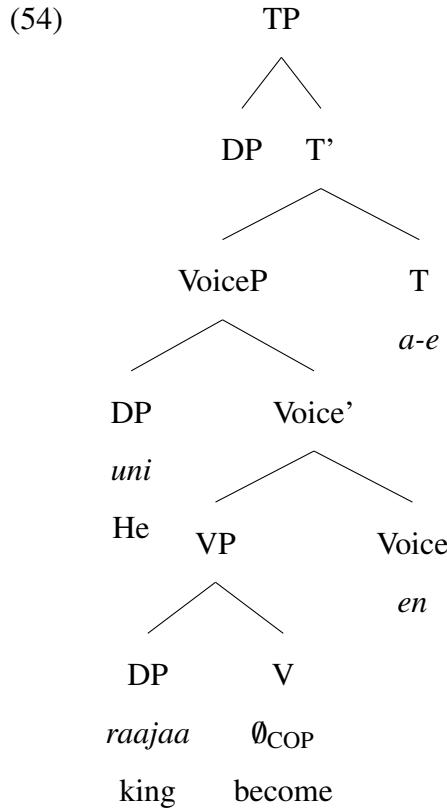
- a. *uni da am-ak-kan-a-e*
He TOP 2SG-GEN-COP-FIN-3SG.SUBJ
'He is yours.'
- b. *akae-kan-a-m*
who-COP-FIN-2SG.SUBJ
'Who are you?'

(Ghosh, 2008, p. 54)

There are some major differences between structures like (51) or (52) and (53). Firstly, and most importantly, the structures in (51) and (52) show verbalization, where non-verbal categories, like nouns or adjectives, take the verbal inflectional forms in syntax and change to verbs. The structures in (53) do not show verbalization of non-verbal categories. They instead show some non-verbal categories in the predicative position by keeping their categorial status intact. The constructions in (53) are examples of nominal predicates. Secondly, in structures like (51) and (52), there is a regular and productive change-of-state meaning. The verbalized element acts as the result of the change-of-state operation, where the theme argument (subject in (51) and objects in (52)) change to the verb. The constructions in (51) show inchoative structures, whereas the constructions in (52) have a causer involved. This is not the case with the structures in (53). Before moving into further analysis of the stem-derived verbs in Santali, we deal with this 'become' (change of state) meaning. There could be two possibilities of how the structures in (51) carry the 'become' semantics. One is that 'become' is a copula (Pushtet, 2003; Hengeveld, 1992, p. 36) with a null form in Santali. The second possibility is that 'become' is a feature on the verbalizer head.

These two analyses can be represented structurally as follows. Under the null copula hypothesis, the verbal markers attach to a covert copula, giving rise to a structure like (54) for the construction in (47b). On the other hand, under the verbalization hypothesis, they attach directly to the nominal/adjectival stem, projecting a structure like (55)³.

³This is a simplified structure of our analysis. We'll come back to such structures in Section 3.4.3.2 to explain in detail how such an approach can explain the Santali verbalization with a change-of-state interpretation



I propose that a structure like (54) showing a null copula will not work for the Santali structures under study. I argue against the null-copula hypothesis to claim that ‘become’ is a feature on the verbalizer head, following (Embick, 2004). To eliminate the null copula assumption in Santali, I propose the diagnostics ⁴ in the next subsection and compare Santali structures with similar constructions in Hindi/Urdu, a dominant language spoken in the Santali-speaking regions and the most common lingua franca. Hindi/Urdu shares a number of common characteristics with Santali, even if they belong to different language families. They are both SOV languages with free word order, allowing scrambling. Hindi/Urdu has an overt copula *ban*, which provides the ‘become’ semantics to the structures in the language (56)

(56) Hindi/Urdu

paudha peR ban-aa
 Plant tree COP.Become-PST.3SG

‘The plant became a tree.’

⁴I thank the participants of FASAL 14 for their suggestion to that led me to come up with the diagnostics to eliminate the null copular assumption.

If Santali has a similar copula as (56), it would behave like Hindi/Urdu in the following tests, proving that there is no verbalization of non-verbal categories in Santali. We perform three tests to see if the verbal clitics attach to the non-verbal entities like N or A, resulting in denominal/ deadjectival verbalization, or the verbal markers attach to the null copulas purporting to be verbalization in a structure like (57), repeated from (47b).

- (57) *uni raajaa-en-a-e*
 He king-PST.MID-FIN-3SG
 ‘He kinged.’ (He became a king.)

3.4.2.1 NP Scrambling

Let us assume the possibility of the presence of a null copula that takes the verbal markers in the predicate in (57).

Under this assumption, the structure would be analyzed as follows: *uni* ‘he’ functions as NP1 (the subject), *raajaa* ‘king’ functions as NP2 (the predicate nominal), and there is a phonologically null copula after NP2 that hosts the verbal morphology. This analysis treats *raajaa* as a simple NP rather than as a denominal verb. If this null copula assumption were true, we would expect Santali to pattern with any free word order language that expresses change-of-state semantics through an overt ‘become’ copula. In such languages, constructions like (57) would typically be copular sentences with two distinct nominal arguments (NP1 and NP2) and a copular verb. The copula (whether overt or null) then would carry the ‘become’ semantics and link the two NPs in a predication relationship.

Also, if the null copula hypothesis is correct, considering the free word order status of Santali, we should be able to scramble both the NPs of a structure like (57) and get the same meaning, making the rule in (58) true.

- (58) NP1 NP2 COP/ $\emptyset$ _{COP}-TAM.phi = NP2 NP1 COP/ $\emptyset$ _{COP}-TAM.phi

However, when we apply the NP scrambling test to Santali, changing the order of the NPs in (59), it doesn’t give the identical semantics as (57), contradicting the rule in (58).

- (59) **raajaa uni-en-a-e*
 King he-PST.MID-FIN-3SG
 ‘He became a king.’⁵

⁵The sentence itself is not ungrammatical. When we scramble the NPs leaving the verbal clitics *in situ*,

On the other hand, we can see that in Hindi, a free word order language that shows change of state semantics using a become copula, changing the order of NP1 and NP2 (60b) doesn't result in ungrammaticality, which supports the rule in (58).

(60) Hindi/Urdu

- a. *vah raajaa ban-a*
He king become-3SG.PST
'He became a king.'
- b. *raajaa vah ban-a*
King he become-3SG.PST
'He became a king.'

Based on this comparison with a similar construction in Hindi/Urdu, I conclude that the null copula assumption is not valid according to the NP scrambling test.

3.4.2.2 Displacement of Verbal Clitics

To further test whether constructions like (57) involve a null copula as opposed to verbalization, we can again study Santali's relatively free word order nature. If verbal clitics are hosted by a null copula rather than attached directly to NP2, we should be able to displace these clitics along with their null copula host, independently of the NPs. This prediction is formalized in rule (61).

(61) $COP/\emptyset_{COP-TAM,phi} NP1 NP2 = COP/\emptyset_{COP-TAM,phi} NP1 NP2$

In other words, if a null copula exists as an independent syntactic head, it should be able to move to different positions in the clause while carrying the verbal clitics with it. If the verbal clitics along with their null copula hosts can be successfully displaced from NP2 while maintaining grammaticality, this would support the null copula analysis. However, if displacement results in ungrammaticality, this would suggest that the verbal clitics are not hosted by an independent null copula but are instead directly attached to NP2 through verbalization. As shown in (62), when we attempt to front the verbal clitics in Santali, it produces an ungrammatical structure, rejecting the rule in (61).

assuming there is a copula hosting it, the meaning of the sentence changes. The meaning of the sentence changes to 'The king became him', which is different from the expected meaning, 'He became a/the king'.

- (62) **-en-a-e* *uni raajaa*
 -PST.MID-FIN-3SG he king
 ‘He became a king.’

In contrast, such a displacement (63) is allowed in Hindi/Urdu. This infers that Hindi/Urdu supports the rule in (61), unlike Santali.

- (63) Hindi/Urdu
- a. *vah raajaa ban-a*
 He king become-3SG.PST
 ‘He became a king.’
- b. *ban-a* *vah raajaa*
 become-3SG.PST he king
 ‘He became a king.’

The grammaticality of (63b) shows that Hindi/Urdu verbal elements can move independently along with the copula (overt), consistent with a copular verb structure. The contrast between Santali (62) and Hindi/Urdu (63) demonstrates that Santali lacks the ability to displace verbal markers along with the copula host (covert) in the manner expected if a null copula were hosting them. This provides further evidence that the ‘become’ meaning in the Santali constructions is not a result of a null copula, and the structures in the current study are really instances of verbalization.

3.4.2.3 Insertion of an Intervener

In the third and final test for verbalization, we can check if the TAM and phi markers are attached to the NP or a null copula by inserting an intervener between the NP2 and the verbal markers in (57). We can have a rule like (64) for the test.

- (64) NP1 NP2 COP/ $\emptyset$ _{COP} TAM.phi = NP1 NP2 [intervener] COP/ $\emptyset$ _{COP} TAM.phi

If the resulting construction is acceptable, supporting the rule in (64), it would mean that the null-copula assumption is correct. As we can see in (65), when we insert a focus marker between the NP2 and assumed null copula hosting the TAM phi markers, the structure (65) is not allowed in Santali. This again rejects the rule in (64) for the verbalization test.

- (65) **uni raajaa da-en-a-e*
 he king FOC-PST.MID-FIN-3SG

‘He became only a king.’

In contrast, the Hindi/Urdu structure in (66) shows that when we insert an intervener, like a focus marker in (66a), resulting in (66b), both the structures are acceptable, supporting a rule like (64).

(66) Hindi/Urdu

- a. *vah raajaa ban-a*
He king become-3SG.PST
‘He became a king.’
- b. *vah raajaa hi ban-a*
He king FOC become-3SG.PST
‘He became only a king.’

If a null copula is the host to the verbal clitics, inserting any marker after the NP2 shouldn’t have created any issue, considering we are not inserting anything between the hypothetical null copula and the verbal clitics. Still, the ungrammaticality in (65) shows that when we insert the focus marker *da*, it breaks the verbal structure of the denominal verbal and results in ungrammaticality.

These three tests show that the NP2 or any other category in a structure before verbalization can take verbal markers because of verbalization, where any category, like N or A, turns into verbs in Santali.

The three tests demonstrate that NP2, or any other category within the pre-verbal structure, is capable of receiving verbal inflectional markers as a consequence of the verbalization process. This phenomenon illustrates that in Santali, lexical categories such as nouns and adjectives undergo categorical conversion to verbs through verbalization processes.

After understanding the prototypical root-derived verbalization and proving through diagnostic tests that Santali also has stem-derived verbs, we now turn to the analysis of these stem-derived compositional verbs in Santali. In the subsequent subsection, we try to analyze the Santali change-of-state stem-derived verbs through two potential theories in the literature.

3.4.3 Stem-derived Verbalization in the Derivation

After understanding that verbs in Santali not only access the idiosyncratic information from acategorical roots, but they also can have semantics of an already categorized element (like a

noun or an adjective), it is clear that these structures call for an intermediate categorial structure before verbalization. That would mean in a compositional verbal construction, the verbalizer merges at the Categorizer₂ position in a structure like (48), following Marantz (2000, 2013), Arad (2003). In this subsection, I discuss two theoretical solutions that have been proposed to explain the verbalization of categorized stems with a change of state semantics in the literature.

3.4.3.1 The Small Clause Analysis

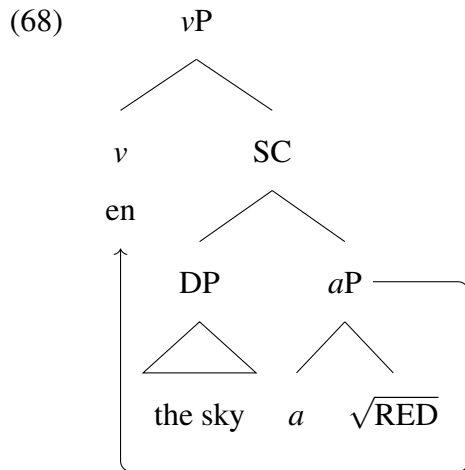
As we have already discussed, English deadjectival verbals are formed by the movement of an adjective from an SC to a verb position. According to Marantz (2013), small clauses usually consist of a DP/NP subject and a XP predicate (any categorial phrase), where the subject DP undergoes changes to become the predicate to be in the state of the predicate. This implies that small clauses give a change of state semantics. In terms of the relationship with the verb, a small clause gets the same interpretation as any complement DP of the verb. Building on this SC framework, Harley (2011b) shows that in languages like English and Hiaki (a Uto-Aztecan language of Sonora, Mexico), only adjectival predicates can get verbalized by moving from an SC. Harley notes that the verbs in Table (3.1) in both English and Hiaki show a causative/inchoative alternating construction. Harley shows a modified version of Hale

English		Hiaki	
Verb	Adj	Verb	Adj
to redden	red	<i>siki-si</i>	<i>siki</i>
to fatten	fat	<i>awi-a</i>	<i>awi</i>
to soften	soft	<i>bwalko-te</i>	<i>bwalko</i>
to sharpen	sharp	<i>bwawi-te</i>	<i>bwawi</i>
to warm	warm	<i>suka-we</i>	<i>suka</i>

Table 3.1: Deadjectival verbs in (Harley, 2011)

and Keyser’s incorporation approach to explain deadjectival inchoative verbs in Table (3.1). The V head, which was taken to be a lexical verb position, is taken as a verbalizing head (*v* in our analysis) by Harley to align with underspecificational accounts. A sentence like (67) instantiating the group of causative/inchoative verbs in (3.1) can have a representation like (68), according to Harley.

(67) The sky reddened.



Harley points out that the verbalizer in a change of state construction like (68) shows semantically inchoative properties. There is no specifier or agent theta role available on the verb (or the higher functional correspondence, Voice), and the theme DP acts as the surface subject. Also, the DP, subject of the SC, gets the change of state interpretation to the adjective (*aP*) by virtue of being in the SC.

However, as we have already seen in (51-52), Santali allows stems from various categories—not only adjectival stems but also nominal and other categorial stems—to be verbalized with a change of state interpretation. This shows extreme fluidity in verbalization. Theoretically, nothing in syntax should stop an SC from having predicates other than adjectives, as Marantz (2013) states that an SC can consist of an NP/DP subject and a predicate of different categories (XP). The restriction in English that limits SC predicates to adjectives reveals a fundamental limitation. The SC-based approach cannot account for Santali’s productive pattern of causative/inchoative verbalization, where stems of any category can undergo this process. Now, I list various limitations of the SC-based approach.

First, nouns (*nP*) cannot be used as predicates of causative/inchoative SCs. The verbs with a similar causative/inchoative semantics in (69) cannot form SCs with a change of state (cf. 70).

- (69)
- a. *John expected Kim his friend.
 - b. *John converted the cash gold.
 - c. *I changed him a good man.

- (70)
- a. John expected Kim to be his friend.

- b. John converted the cash to be gold.
- c. I changed him to be a good man.

Secondly, not all SCs show a change of state semantics in English. Even if SCs can have nominal predicates like (71), they do not show the causative/inchoative property. If SCs are claimed to deal with the change of state semantics, they should uniformly provide the semantics between the subject of the SC and the predicate of the same.

- (71)
- a. John considers Kim his friend.
 - b. I believe John to be a hero.
 - c. He proved himself a hero.

The constructions in (71), show that not all SCs provide causative/inchoative semantics, and syntax fails to explain the change from the subject of an SC to the predicate state.

Lastly, the verbalization of adjectives by head movement from an SC to the V is idiosyncratic. Not all adjectival predicates of SC can be verbalized (73). In constructions like (72-73), the movement of the SC predicate to form a verb, following the incorporation account, has to be lexically listed, or encoded at the root-derivation level. Even if ‘happy’ is an adjectival predicate in (73), it cannot be verbalized.

- (72)
- a. The news made John sad.
 - b. The news saddened John.
- (73)
- a. The news made John happy.
 - b. *The news happied John.

All these limitations put together tells us that the SC account cannot syntactically address the causative/inchoative verbalization with a productive change of state interpretation.

To summarize, in this subsection, I have evaluated the SC analysis as a potential solution for the regular stem-derived verbalization constructions in Santali. While Hale and Keyser’s (1993) incorporation approach, followed by Harley (2011b) successfully explains English deadjectival verbs through the movement of the predicate of SC to V, this analysis faces three critical limitations when extended to Santali. It fails to accommodate nominal predicates in causative/inchoative SCs, restricts some SCs from exhibiting change of state semantics, and

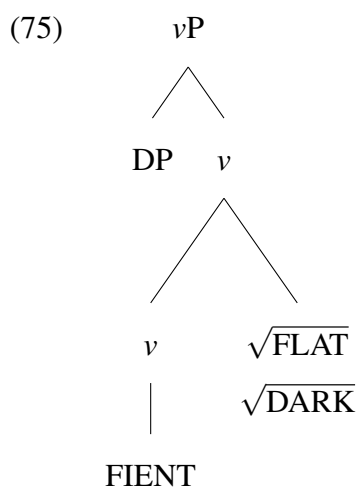
shows deadjectival verbalization as an idiosyncratic and lexically restricted operation. These limitations suggest that while SC-based movement may partially explain some verbalization patterns, Santali’s extreme fluidity in forming productive stem-derived verbs requires an alternative theoretical framework. In the following subsection, I examine Embick’s (2004) feature-based approach as a potentially more adequate account for these patterns.

3.4.3.2 The Feature-based Account

Another pertinent approach to explain the verbalization of English adjectival stems is associated with Embick (2004). Embick, followed by Borer (2012) and Alexiadou (2017), takes a feature based account to explain the deadjectival verbalization with a causative/ inchoative semantics. Building on Hale and Keyser (1993), Embick proposes a [FIENT] feature on the verbalizer head that explains the change of state in a deadjectival verbal structure in English. According to Embick, the verbalizer has different flavors, and one of them is the v [FIENT]. According to Embick, the [FIENT] feature is the become operator that denotes the transition event. In the process of verbalization, [FIENT] on the verbalizer (v), assigns the ‘became flat’ meaning to ‘flat(en)’ or ‘dark(en)’ in (74), structurally demonstrated as (75).

(74) Embick (2004, pp. 365, 381)

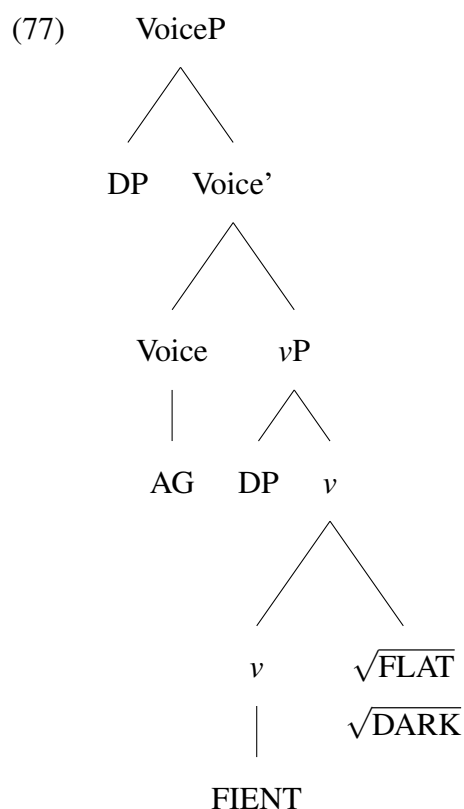
- a. The metal flattened.
- b. The hair darkened.



The v [FIENT] not only explains inchoative structures in English, but it also explains the change of state of a theme DP to a state in transitive structures with a causative semantics.

In a construction like (76), represented as (77), the ν [FIENT] merges with a root, then the verbalizer head merges with the theme DP. For the transitivity in the structure, the Voice head (the higher ν head, according to Embick), introduces the Agent in its specifier position. The Agent is introduced based on the [AG] feature on the Voice head.

- (76) a. The smith flattened the metal.
b. John darkened his hair.



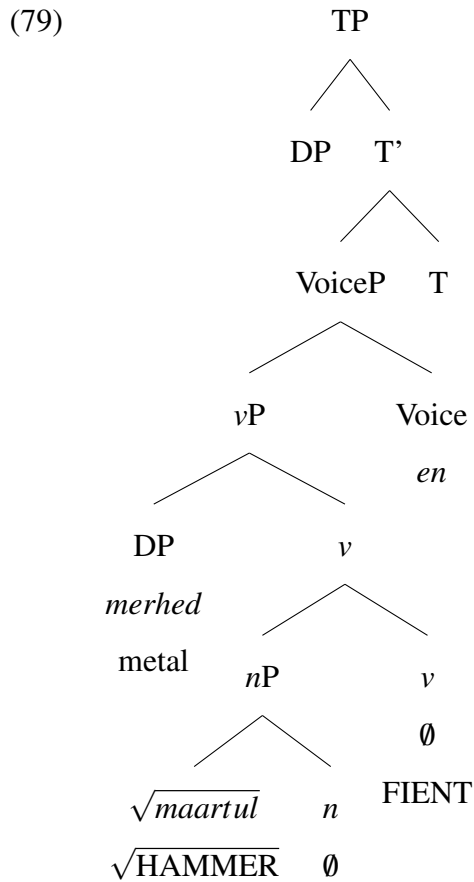
In Embick's approach, the labor of the change of state semantics has shifted from the presence of the subject and the predicate in the SC to the presence of a [FIENT] feature on the verbalizer. This account is not completely different from an SC-like approach, but the presence of the become semantics as a feature on the ν uniformly accounts for the change of state constructions to any category that merges with the ν [FIENT].

I extend this approach to account for the productive process of stem-derived verbalization seen in Santali (78). In a Santali construction like (78), where (78a) is an unaccusative structure with inchoative semantics, and (78b) is a transitive structure with causative semantics, there is always a change of state interpretation inside the SC. (78b) doesn't give a 'hit with a hammer-like object or manner', like English-type languages. The Santali structures, instead, provide

a ‘nail changing into a hammer (N)’ interpretation. This shows that, unlike the restricted adjectival states in English SC constructions (seen in the previous subsection), Santali can have any NP/DP as a result state. By the presence of the [FIENT] feature, a nominal (*n*P) can directly merge with a *v*[FIENT].

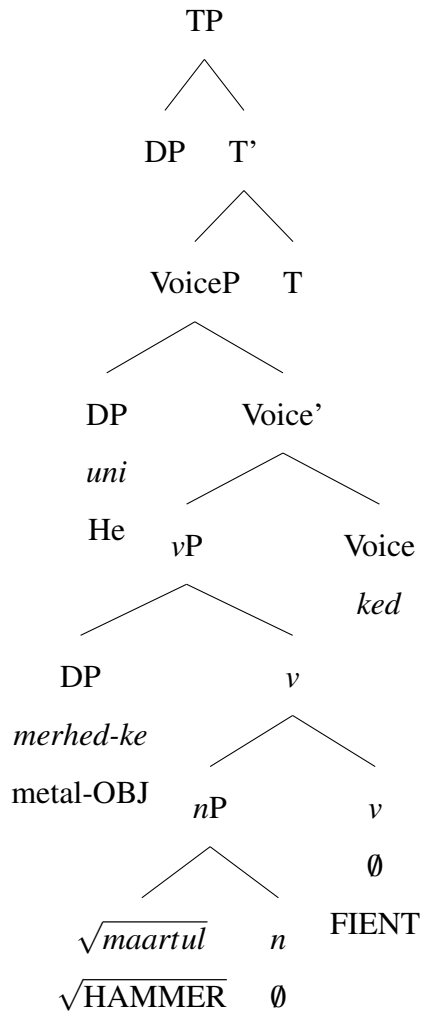
- (78) a. *merhed maartul-en-a*
 metal hammer-PST.MID-FIN
 ‘The metal hammered’. (The metal became hammer.)
- b. *uni merhed-ke maartul-ked-a-e*
 He metal-OBJ hammer-PST.ACT-FIN-3SG.SUBJ
 ‘He hammered the metal.’ (He turned the metal into a hammer.)

We can have a structure like (79) to explain (78a), following Embick’s account. In (79), the verbalizer merges with an already nominalized element, not a root. Following Marantz (2000) and Arad (2003), among others, I posit that since the verbalizer is merged with the root in the 2nd merge position (*n* being the intervening categorizer), the *v* fails to get the idiosyncratic interpretation of the root (cf. 48). It accesses the nominal semantics from the *n* and the nominalized element ‘hammer_N’ works as the result state of the transition event because of the *v*[FIENT].



The structure in (79) provides a change of state interpretation and explains the verbalization of an already categorized root. This kind of stem-derived structure is not idiosyncratic in nature and is seen productively in Santali, unlike languages like English. Even if the English CUC verbs according to Kiparsky (1997), followed by Arad (2003), show stem-derived verbalization, they do not show a change of state semantics like that of Santali. As seen in (78b), such Santali constructions also show transitive structures. As structurally demonstrated in (80), the transitive stem-derived change of state verbs differ from their intransitive counterparts (79), based on the [AG] feature on the Voice head, morphologically marked by *ked*, which is an active/transitive marker that licenses the EA in its spec position. The semantics of the EA is a causative semantics because of the transition event in the *vP* level (Embick, 2004).

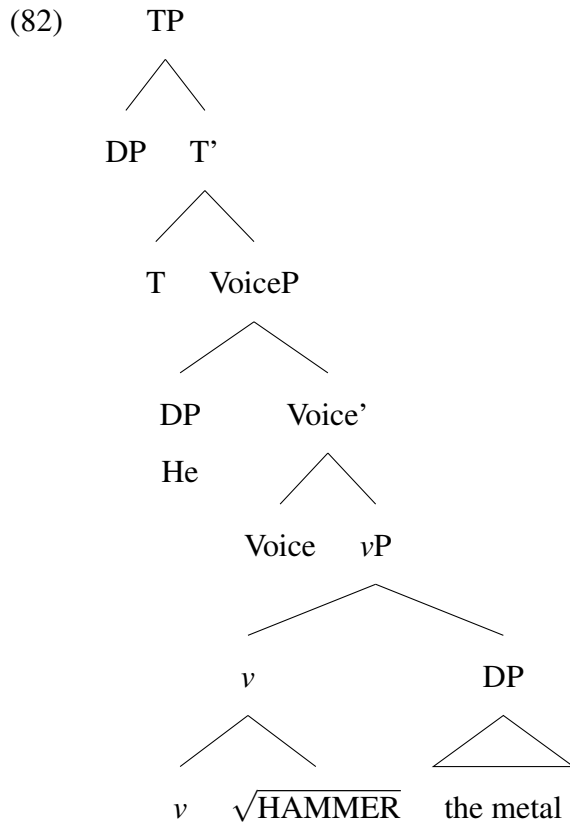
(80)



In Santali, the verb ‘hammer’ means ‘become a hammer’ in its intransitive form (78a), and ‘turn X into a hammer’ in its transitive form (78b). In both cases, the verbs have interpretive properties of a noun ‘hammer’, therefore analyzed as stem-derived.

This productive stem-derived verbalization in Santali contrasts with similar verbs in English. A verb like ‘hammer’ in English has an idiosyncratic interpretation of ‘hit’ or ‘pound’ (Arad, 2003; Panagiotidis, 2015). These hammer-type verbs in English are not formed from a nominal base, and do not show any structural or semantic interpretation of a nominal. Therefore, unlike Santali, ‘hammer_v’ in English is root-derived and doesn’t carry change of state semantics. Following Marantz (2013), I show that in a construction like (81), schematically represented as (82), there is no SC-like structure, and nominal stem-derivation to form verbs.

(81) He hammered the metal.



In (82), there is no *v* with a [FIENT] feature (Embick, 2004), and therefore, no change of state interpretation.

To summarize, in this subsection, we saw that the productive stem-derived verbs in Santali can be explained using the [FIENT]-feature-based account proposed by Embick (2004). The feature-based account works better than the incorporation-based SC analysis because it can account for all types of stem-derived verbalization, unlike the SC approach. Secondly, when we adopt Embick's analysis, we align with the underspecificational account like DM, assuming both nouns and verbs are further divisible and are constructed in the syntactic workspace. We also solve the issue of overgeneration by eliminating the SC analysis, because not all the subjects change to become the predicate state in an SC. And most importantly, we follow the locality constraint of interpretation of roots (LCIR) associated with Marantz (2000) and Arad (2003) to posit that categorization is a cyclic process. That means categorizers behave like phases in the process of categorization. A categorizer can access the interpretation of the root/stem local to it by allowing verbalizers to merge with both a root and a stem to allow idiosyncratic, as well as compositional verbalization in Santali.

3.5 A Note on Argument Alternating Operations

In the previous section, we established that the voice markers *en* and *ked* play an important role in indicating the argument structure of the verb. There is an inchoative semantics in the unaccusative structure when the past tense middle voice marker *en* is attached to the verb. Similarly, there is a causative semantics in the transitive structure when the past tense active voice marker *ked* or the present tense active voice marker *e* is attached to the verb.

This section examines how Santali realizes argument structure alternations, particularly focusing on the morphological realization of Voice heads on non-agentive and agentive constructions. According to Chandra (2014), languages are reported in the literature to vary depending on the middle-passive overlap they have. According to her, middles are marked either with active or non-active morphology cross-linguistically. English-type languages show middle constructions with an active voice morphology, whereas Greek-type languages carry non-active morphology for middle constructions. Since we have already seen in this chapter that Santali carries a morphological middle voice marker, I will demonstrate that the non-active voice marker is the middle marker *en* for middle constructions in Santali. I also demonstrate that Santali patterns with Greek-type languages (Alexiadou, 2014a) in exhibiting morphological syncretism across anticausatives, middles, and passives through middle voice morphology. Additionally, I take the discussion forward by showing that Santali Voice head productively interacts with both root-derived and stem-derived verbalizations, extending Alexiadou's analysis beyond the domain of root-derived verbs.

I first present the structural properties of non-agentive constructions. I then compare these with agentive constructions to show how the active/middle distinction systematically marks the agentive/non-agentive alternation in Santali.

3.5.1 Non-agentive Constructions in Santali

The middle voice morpheme is attached to the Santali verb to form constructions like anticausatives (83), middles (84), and passives (85). All these are unaccusative constructions because they show an inchoative semantics, like many other languages.

- (83) a. *arsi rapud-en-a*
mirror break-PST.MID-FIN

‘The mirror broke.’

- b. *duar khulo-en-a*
door open-PST.MID-FIN
‘The door opened.’
- c. *baraf chher-en-a*
ice melt-PST.MID-FIN
‘The ice melted.’
- d. *jom/Daakaa ranga-en-a*
food burn-PST.MID-FIN
‘The food burned.’

(Anticausatives)

- (84)
- a. *lugDi logonge sabodo-ge-a*
cloth quickly wash-PRS.MID-FIN
‘Cloth washes quickly.’
 - b. *noa kahni puthi sahajte paDhao-ge-a*
this story book easily read-PRS.MID-FIN
‘This story book reads easily.’
 - c. *noa haku sahajte gedo-ge-a*
this fish easily cut-PRS.MID-FIN
‘This fish cuts easily.’

(Middles)

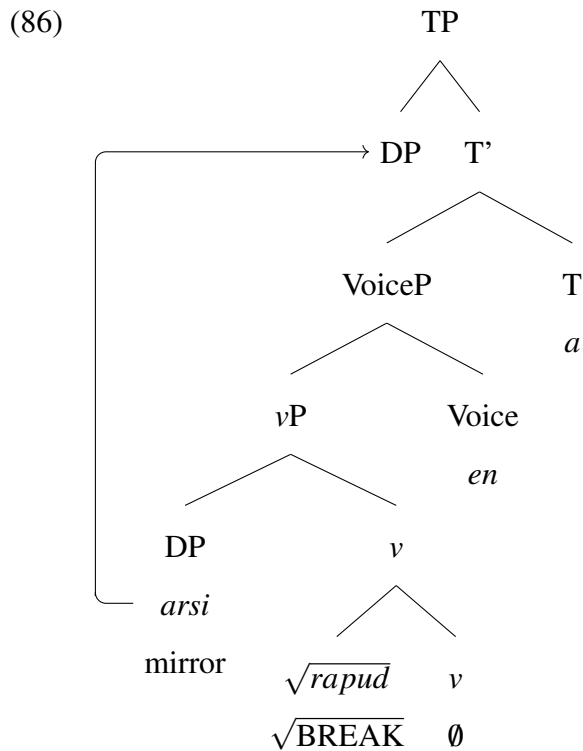
- (85)
- a. *puthi (jon-te) padhao-ocho-en-a*
book (John-by) read-PASS-PST.MID-FIN
‘The book was read (by John).’
 - b. *tarup (jon-te) goe-ocho-en-a-e*
tiger (John-by) kill/die-PASS-PST.MID-FIN-3SG.SUBJ
‘The tiger was killed (by John).’
 - c. *jon (mili-te) Daal-ocho-en-a-e*
John (Mili-by) hit-PASS-PST.MID-FIN-3SG.SUBJ
‘John was hit by Mili.’

(Passives)

In (83-85), one of the crucial morphological commonalities is the presence of the middle voice marker *en* and *ge* in past and present forms, respectively. Santali voice system shows similarity to the voice system reported in Greek by Alexiadou (2014a). Alexiadou proposes three Voice heads, such as active, middle, and passive, among which, Greek has only active and middle. All the transitive constructions in Greek have active voice marking, and intransitives, specifically, unaccusative (non-agentive) constructions, carry middle voice morphology. I extend Alexiadou’s proposal for Greek to Santali to claim that, like Greek, Santali also projects two

Voice heads, i.e., active and middle. Active Voice marks transitive constructions with *ked/e* and middle Voice marks intransitive unaccusatives with *en/ge* in Santali. Unlike active Voice, which introduces an external argument in its specifier, middle Voice generates unaccusative structures with no projected external argument.

To illustrate the structural properties of the middle Voice marker in Santali, we can see the anticausative construction in (83), structurally represented in (86). In (86), the middle Voice head *en* projects a VoiceP without projecting a specifier position. The theme argument *arsī* ‘mirror’ is generated at the IA position and moved to the subject (Spec, TP) position. The middle Voice head assigns inchoative semantics to the construction, yielding the interpretation ‘the mirror broke’, without an agent.



Also, as seen in (85), the passive structures in Santali are formed by the morphological middle markers. The difference between Santali middle constructions (84) and passives (85) lies in the tolerance of by-phrases. Santali passives allow by-phrases (85), while middles do not (87). The construction in (87) is ungrammatical with the by phrase.

- (87) **lugDi jon-te logonge sabodo-ge-a*
 cloth John-by quickly wash-PRS.MID-FIN
 ‘Cloth washes quickly by John.’ (Santali)

This contrasts with Greek, since Greek middle constructions can tolerate by-phrases (88), like passives (Tsimpili, 1989; Alexiadou, 2014a).

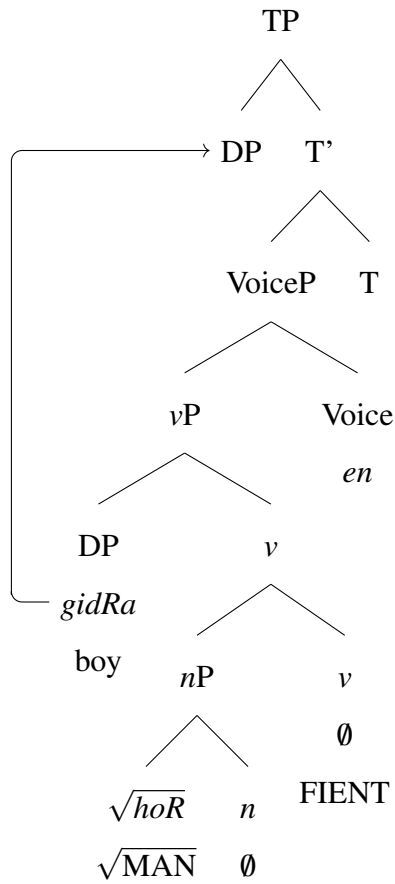
- (88) *Afto to vivlio diavazete efxarista (apo opiondipote)*
 this the book read.MID.IMPERF.3SG with pleasure by anyone
 ‘This book reads with pleasure by anyone.’ (Greek: Alexiadou, 2014, p. 25)

All Santali non-agentive constructions have the middle Voice head. Unlike an active Voice head, the middle Voice head generates unaccusative structures and introduces no external argument in its specifier. It triggers non-active morphology, like *ge* and *en*. However, the middle Voice head is not restricted only to root-derived verbs. As seen in the previous section, the middle Voice head in Santali is compatible with change-of-state semantics via [FIENT]. In (89), the morphological middle marker is seen in an anticausative construction, which is a stem-derived non-agentive construction.

- (89) *gidRa hoR-en-a-e*
 boy man-PST.MID-FIN-3SG
 ‘The boy manned (became a man).’ (Anticausative)

The structural representation of the stem-derived anticausative in (89) can be seen in (90). As we have already seen in (79), the verbalizer merges with an already categorized *nP*, not an acategorial ROOT, to form the stem-derived verb. This follows the analysis in Section 3.4.3.2, where we established that *v*[FIENT] can select for nominal stems to yield change-of-state semantics. The middle Voice head *en* then selects this stem-derived *vP*. This shows that a verb can combine with verbal functional projections regardless of whether the verb is root-derived or stem-derived. This productive pattern distinguishes Santali from the languages discussed in Alexiadou (2014a), where the focus is primarily on root-derived verbs.

(90)



To extend the discussion further, the constructions in (91) show that middle Voice head combines productively with stem-derived verbs created via [FIENT]. This is theoretically significant because it shows that Voice heads in Santali can select for already-categorized stems, not just acategorial roots, as seen in section 3.4.3.2.

- (91) a. *noa daare logonge maarang-ge-a*
 this tree quickly big-PRS.MID-FIN
 ‘This tree bigs (becomes big) quickly.’ (Middle)
- b. *gidRa (jon-te) hoR-ochο-en-a-e*
 boy (John-by) man-PASS-PST.MID-FIN-3SG
 ‘The boy was manned (made a man) by John.’ (Passive)

The productivity of this pattern distinguishes Santali from Greek, because Alexiadou’s discussion focuses primarily on root-derived verbs.

3.5.2 Agentive Constructions

As we have already seen in this chapter, both root-derived and stem-derived verbs can be in agentive construction with the presence of an active Voice head like *ked/ke* (92-93).

- (92) a. *uni bhit-ke Daal-ked-a-e*
 He wall-OBJ hit-PST.ACT-FIN-3SG.SUBJ
 ‘He hit the wall.’
 b. *uni jon-ke Daal-ked-e-a-e*
 He John-OBJ hit-PST.ACT-FIN-3SG.SUBJ
 ‘He hit John.’ (Root-derived)
- (93) a. *uni bhit-ke haryaal-ked-a-e*
 He wall-OBJ green-PST.ACT-FIN-3SG.SUBJ
 ‘He greened the wall.’
 b. *uni jon-ke dhiri-ked-e-a-e*
 He John-OBJ stone-PST.ACT-FIN-3SG.SUBJ
 ‘He stoned John (turned John into a stone).’ (Stem-derived)

Unergative root-derived verbs, which are also agentive constructions, take the Active voice marker like *ke* in Santali (94).

- (94) a. *uni japid-ke-a-e*
 He sleep-PST.ACT-FIN-3SG
 ‘He slept.’
 b. *uni daD-ke-a-e*
 He run-PST.ACT-FIN-3SG
 ‘He ran.’

In (94), the subjects of the constructions are the agents. Stem-derived unergative counterparts are impossible to form in Santali because the subject of a stem-derived verb must be a theme undergoing change of state, yielding an inchoative/unaccusative structure (cf. Section 3.4.2). This constraint follows from the presence of the [FIENT] feature on *v* in stem-derived verbals, as seen in (Section 3.4.2), which requires a result state interpretation.

All agentive constructions in Santali can have their argument structure increased by getting causativized using a causative morpheme, *ocho*. The root-derived verbs in (95) and the stem-derived verbs in (96) are in causative constructions due to the presence of the morphological causative marker *ocho*.

- (95) a. *uni lili-te bhit-ke Daal-ocho-ked-a-e*
 He Lily-by wall-OBJ hit-CAUS-PST.ACT-FIN-3SG.SUBJ
 ‘He made Lily hit the wall.’
 b. *uni lili-te jon-ke Daal-ked-e-a-e*
 He Lily-by John-OBJ hit-CAUS-PST.ACT-FIN-3SG.SUBJ
 ‘He made Lily hit John.’ (Root-derived)
- (96) a. *uni lili-te bhit-ke haryaal-ocho-ked-a-e*
 He Lily-by wall-OBJ green-CAUS-PST.ACT-FIN-3SG.SUBJ
 ‘He made Lily green the wall.’
 b. *uni lili-te jon-ke dhiri-ocho-ked-e-a-e*
 He Lily-by John-OBJ stone-CAUS-PST.ACT-FIN-3SG.SUBJ
 ‘He made Lily stone John (turned John into a stone).’ (Stem-derived)

The Santali constructions in this section showed three significant theoretical points. First, Santali’s Voice system exhibits morphological syncretism patterns similar to Greek-type languages (Alexiadou, 2014a), with systematic active/middle distinctions marking agentive/non-agentive constructions. Secondly, Santali extends this system beyond root-derived verbs to productively include stem-derived verbalizations. This distinction is not addressed in Alexiadou’s account. Thirdly, the Voice head plays an important role in argument-increasing and argument-decreasing operations. This role remains consistent in both root-derived and stem-derived verbalization by supporting the claim that Santali Voice heads show uniform structural behaviour regardless of the derivational history of the verb they merge with.

3.6 Summary and Conclusion

In this chapter, I have shown, focusing on Santali, that verbs are both root-derived and stem-derived. This contributes to the generative studies on verbalization that state that verbs are always idiosyncratic and derived directly from a root by showing a contrary trend. I have used diagnostic tests, such as NP scrambling, displacement of verbal clitics, and insertion of an intervener, to prove that Santali indeed shows stem-derived verbalization. Further, using two potential theoretical accounts, such as the incorporation account, associated with Hale and Keyser (1993) and Harley (2011), and the feature-based account, associated with Embick (2004), Alexiadou (2017), among others, I show that the incorporation account best suits for Santali stem-derived verbalization, as it uniformly accounts for verbalization across all lexical

categories without stipulating category-specific restrictions. Further, we also understood that verbs can be configurationally formed from roots, or have intermediate categories like nouns or adjectives, and the idiosyncratic or compositional interpretation of roots depends on the merge position of the categorizer.

Chapter 4

Structure and Derivation of Adjectives

4.1 Introduction

The category of adjective has long been puzzling in the study of categorization. Starting from the traditional typological literature to the recent generative literature, the categorial status of adjectives have been an unsolved issue. Researchers argue that adjectives constitute a marked LC in comparison to other LCs like nouns or verbs, since adjectives are not present in the same way in all natural languages (Croft, 1991; Beck, 1999b; Dixon, 2004; Alexiadou, 2014b). Arguments like Adjectives constitute a closed class in some languages, and open class in others, some languages in the world do not have adjectives as a composite category, etc., makes adjectives known in both the typological and the generative literature as a problematic category. Despite all the markedness adjectives carry as a category, Baker (2003) has termed adjectives as one of the core lexical categories, along with nouns and verbs.

The goal of this chapter is to understand the distribution and formation of adjectives in Santali. As we have seen in Chapters (2) and (3), nouns and verbs show categorial fluidity in Santali as they can be formed from acategorial roots to show lexical idiosyncrasies and already categorial stems to demonstrate regularity and compositionality. In this chapter, I ask being the most marked category and being much more restricted than nouns and verbs, cross-linguistically, if adjectives still follow the derivational structure similar to that of nouns and verbs. The second major question I deal with in this chapter is if adjectivization can happen at different hierarchical stages in the derivation to involve compositional interpretation

of categories like nouns and verbs and their extended functional projections.

The chapter is structured as follows. I begin from defining adjectives in Section 4.2. I discuss the categorial status and the puzzles involved with adjectives, followed by an overview of Aikhenvald's cross-linguistic survey of adjectives and their properties. Section 4.3 talks about the theoretical core of the chapter, examining the process of adjectivization. Then I move to adjectivization in Santali to first prove that adjectives exist as a composite category in Santali unlike what is stated in the previous typological literature. Further, I critically evaluate both root-derived and stem-derived adjectival structures in Section 4.4. After discussing the differences between nominal and verbal inputs to adjectivization in terms of adjectivization height in this section, I summarize and conclude the chapter in Section 4.5.

4.2 What is an Adjective?

4.2.1 Adjectives as a Category

Adjectives are defined as linguistic elements that can clarify, adjust or alter the meaning of nouns (Huddleston and Pullum, 2002, p. 526; Kennedy, 2012). Adjectives are semantically defined as expressions that denote properties and syntactically as modifiers of nouns.

Because of their lack of cross-linguistic presence, or presence in some languages as a closed class, adjectives have remained in the literature a controversial category (Beck, 1999a). However, before going to the categorial status of adjectives and the problems associated with it, we will understand what adjectives are in both traditional and generative literature. In the traditional grammar, the core function associated with adjectives is their modificational properties. Adjectives modify nouns in both attributive (1a) and predicative (1b) positions. However, Alexiadou (2014b) shows that there are adjectives that can be used only as predicative modifiers (2a), not attributive modifiers (2b).

- (1) a. The happy man.
b. The man is happy.
- (2) a. The dog is asleep.
b. *The asleep dog.

Similarly, there are adjectives, according to Kennedy (2012), which can only be used in attributive position. In English, the adjectives in (3) fall into such a class.

- (3) a. This is our former/principal/main objective.
- b. *This objective is former/principal/main.

Adjectives can also have their own modifiers. As can be seen in (4), they can be modified by degree modifiers (4a) and measurement modifiers (4b). Adjectives carry phi features of the nouns they modify in some languages (5). In the Hindi/Urdu example in (5), the adjective *chhoTi* ‘little’ agrees with the noun *bacchi* ‘girl’ in number and gender.

- (4) a. The man is very/too/little/quite happy.
- b. The water below is 20 meters deep.

- (5) Hindi/Urdu

<i>chhoTi</i>	<i>bacchi jaa rahi</i>	<i>hai</i>
little.SG.FEM girl	go stay.PROG.3SG.FEM	COP.PRS.3SG

‘The little girl is going.’

Studies on adjectives gained generative attention after Chomsky (1970), where Chomsky defines adjective as a category with [+N], [+V] features. Stowell (1981) supports Chomsky’s claim showing that nouns and adjectives form a natural class since they share [+N] feature. [+N] categories like nouns and adjectives can have ‘of’ insertion in English (6).

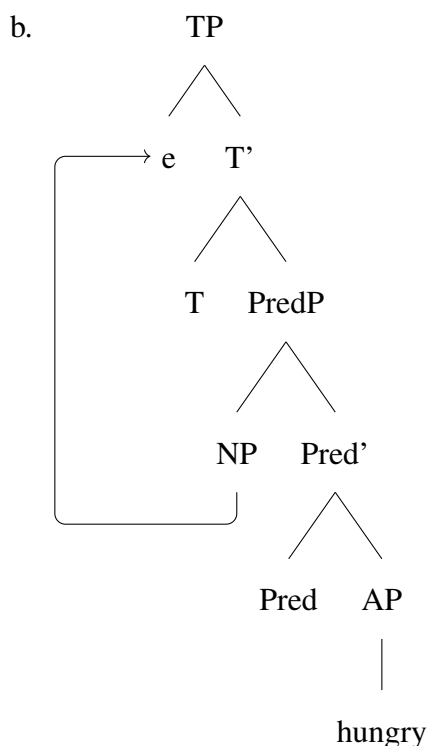
- (6) a. The enemy’s destruction of the city.
- b. John’s fear of heights. (Repeated from (15) of Chapter 1)

This kind of view, however, has been argued against by Baker (2003). In Chomsky’s binary feature system, the [+N +V] feature would incur difficulty for the CI interface to encode. The CI interface cannot interpret the same morpheme as both noun and verb simultaneously. Baker rejects such an approach using the Reference-Predication Constraint, stating that referential identity and predicative function contradict each other. According to the constraint, no category can simultaneously behave as a noun and a verb. This means no category can bear [+N +V] feature. Mitrović and Panagiotidis (2020) further strengthen Baker’s position by arguing

that Nouns have sortal characteristics and verbs extend into time. Putting both nominal and verbal characteristics together as [+N +V] would be paradoxical.

Baker puts adjectives in elsewhere category arguing that adjectives lack both nominal and verbal properties. According to Baker, adjectives are the default lexical category. They do not have a criterion of identity, like a noun or predicative function to license a subject. The defining characteristic of adjectives is that they lack the defining properties of nouns and verbs. They are property-denoting entities that act as modifiers of nouns. Like nominal predicates, they can occur predicatively as well, when they are supported by a functional head Pred. Baker (2003, p. 35) proposes a structure like (7b) for an adjectival predicate like (7a).

(7) a. Chris is hungry.



Chomsky's description of adjectives as having both nominal and verbal properties, and Baker's characterization of adjectives as an elsewhere category make adjectives a problematic category in the generative tradition. Chomskyan notion of [+N +V] faces empirical challenges as no category can be nominal and verbal at the same time. It also violates Baker's Reference-Predication Constraint. Baker's elsewhere condition becomes problematic when we consider adjectives cross-linguistically, that partially behave like/ formed from nouns or verbs (not both at the same time). Adjectives exhibit significant variation in their distribution, size as a word

class, and morpho-syntactic behavior across languages. In the subsequent sub-section, we can understand how adjectives are probably the most controversial of the major lexical categories.

4.2.2 Adjectives as a Puzzling Category

It is evident from the above discussion that adjectives are a puzzling category in the study of categorization. According to Dixon (2004, p. 9), adjectives constitute a distinguishable class in almost all languages. However, identifying adjectives as a distinguishable class is not as straightforward as that of noun and verb. Unlike noun or verb, adjectives do not constitute an open class in all languages (Hyslop, 2004). Even though many languages have adjectives as an open class, it is much smaller than noun class or verb class. In terms of adjectives as a closed class, there are languages with the smallest closed class containing three to five adjectives, and other languages containing several hundred adjectives, according to Dixon (2004). Both the types of languages, however, have a closed class of adjectives, meaning no new adjectives may be added to them. Segerer (2008) lists a number of African languages that have less than thirty adjectives. This blurs the lexical categorial status of adjectives in some languages, because usually, open classes have lexical categorial status.

Another well-known issue discussed in the literature is that adjectives do not take a fixed position in a DP in some languages. In attributive modification, adjectives yield interpretation differences depending on their pre-nominal or post-nominal position. Following Bolinger (1967), and Cinque (2010), Alexiadou (2014b) discusses how the interpretation varies in a language like English depending on the adjective position. According to this view, pre-nominal modification by adjectives represents an enduring, permanent, and characteristic property of the noun they modify (8). On the other hand, post-nominal adjectives address temporary and non-typical properties of the noun (9).

- (8) a. the visible stars
- b. the explorable rivers
- c. the stolen jewels
- d. the present cats

- (9) a. the stars visible

- b. the rivers explorable
- c. the jewels stolen
- d. the cats present

Alexiadou points that in (8), the prenominal modification can include temporary properties, along with the intrinsic properties they carry, making the interpretation ambiguous.

The other puzzle adjectives show is the embedded nominal or verbal nature across various languages. They often display morphological and syntactic behaviors that overlap with both nouns and verbs. According to Dixon (2004) and Hyslop (2004), a language like Ambae has adjectives as a subclass of verb (10).

- (10)
- a. *huhue* ‘to paddle’
 - b. *aka huhue* ‘paddling canoe’
 - c. *vavatu* ‘to weave’
 - d. *gamali vavatu* ‘weaving house’

Similarly, According to Baker (2003) an adjective can have properties of a noun or a verb when they are converted to an adjective through derivation (11) to modify another noun. In (11), the noun ‘wealth’ or the verb ‘shine’ cannot modify another noun before they turn into an adjective.

- (11)
- a. a wealthy man
 - b. a shiny coin

According to the WALS data by Stassen (2013), cross-linguistically, adjectives behave like nouns (34%), verbs (39%), have a mix behaviour of both noun and verb (27%). Figure (4.1) shows such a polyfunctional behaviour of adjectives across languages.

Adjectives in general have a higher number of derived forms than nouns and verbs (Dixon, 2004). This entails that our stem-derived categorization in Chapter 2 and 3 can be applied to adjectives also. Adjectives or prototypical adjectives can be assumed as root-derived and the adjectives derived from nouns and verbs can be considered stem-derived. This can be supported by the cross-linguistic survey by Aikhenvald (2011) to discuss the properties of derived (stem-derived) adjectives. In the subsequent subsection we explore the properties associated with the derived adjectives in different languages in the world based on Alexiadou’s survey.

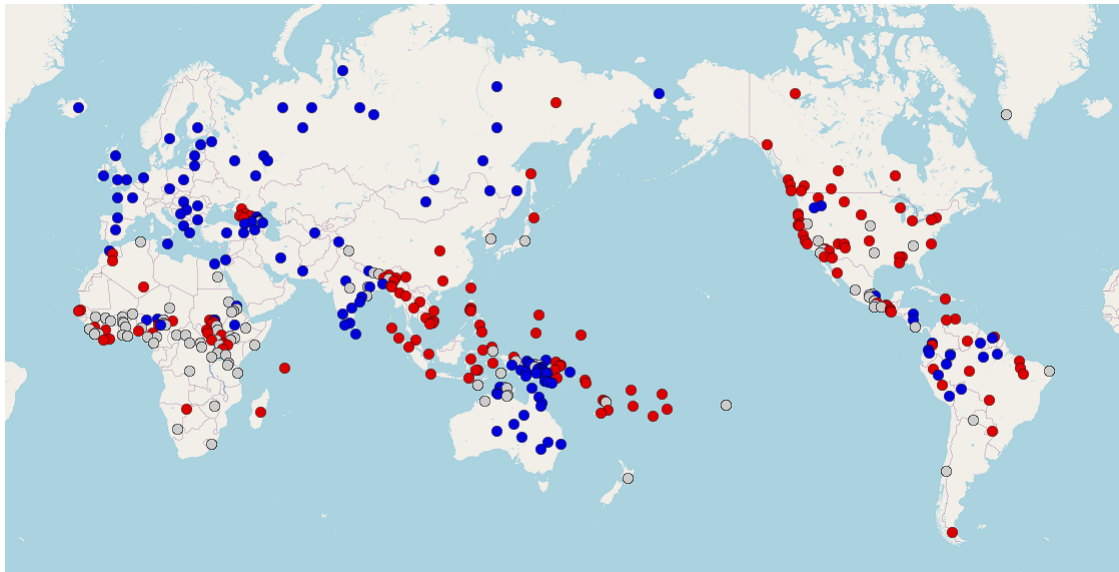


Figure 4.1: Cross-linguistic distribution of three-way categorial features of adjectives. Adjectives behave like **nouns** (blue), **verbs** (red), or **both** (grey) from *WALS* data (Stassen, 2013)

4.2.3 Adjectives Cross-linguistically

Aikhenvald (2011) shows that adjectives can carry properties of other LCs like that of nouns and verbs across languages. The classification of such adjectives is semantic according to Aikhenvald. Adjectives derived from verbs can have various semantic properties related to the activity or the result of the verb and the core argument structure of the verb. Adjectives in (12-14) show the properties associated with the action and the adjectives in (15-16) exhibit the property of the result of their verbal counterparts.

(12) Hungarian (Kenesei et al., 1998, p. 367)

- a. *viszket* ‘itch’
- b. *visket-ős* ‘itchy’

(13) Turkish (Kornfilt, 1997, p. 458)

- a. *ak* ‘flow’
- b. *ak-ici* ‘fluent, fluid’

(14) Tamambo (Jauncey, 1997, p. 11)

- a. *mana* ‘laugh’
- b. *manamana* ‘friendly’

- c. *sale* ‘float’
- d. *salesale* ‘light (in weight)’

(15) Amharic (Amberber, 1996, p. 10)

- a. *saffa* ‘become wide’
- b. *saffi* ‘wide’

(16) Awa Pit (Curnow, 1997, p. 11)

- a. *ii* ‘die’
- b. *iita* ‘dead’

There are also deverbal adjectives present cross-linguistically that show the argument structure properties or the object argument of the verbal base. Aikhenvald shows examples of such adjectives from various languages (17-18).

(17) Hungarian (Kenesei et al., 1998, p. 366)

- a. *olvas* ‘read’
- b. *olvas-ható* ‘readable’

(18) Basque (Saltarelli, 1988, p. 260)

- a. *farre* ‘laugh’
- b. *farre-garri* ‘laughable’
- c. *jan* ‘eat’
- d. *jan-garri* ‘edible’

According to Aikhenvald, English ‘-able’ suffix has deverbal adjectival properties (in read-able, breath-able, walk-able, etc.) that carries the internal argument information of the corresponding verb. In the construction, ‘a read-able book’ the adjective ‘read-able’ has the internal argument ‘book’ that it modifies.

The other type of derived adjective seen across languages is the denominal adjectives. According to Aikhenvald, denominal adjectives across languages can have semantic properties such as, ‘made of N’ (19), ‘be like N’ (20) ‘be characterized by’ (21), propriative or ‘having N’ (22), and privative or ‘absence of N’ (23).

- (19) Evenki (Nedjalkov, 1997, p. 305)
 - a. *altan* ‘gold’
 - b. *alta-ma* ‘golden’
- (20) Turkish (Kornfilt, 1997, p. 457)
 - a. *masa* ‘table’,
 - b. *masa-msI* ‘table-like’
- (21) Jacalteco (Day, 1973, p. 49)
 - a. *pojoj* ‘dust’
 - b. *pojoj-taj* ‘dusty’
- (22) Hungarian (Kenesei et al., 1998, pp. 362-363)
 - a. *kazetta* ‘cassette’
 - b. *kazettá-s* ‘having cassettes’
- (23) Bengali (Onishi, 1997)
 - a. *sim (a)* ‘boundary’
 - b. *a-sim* ‘boundless’

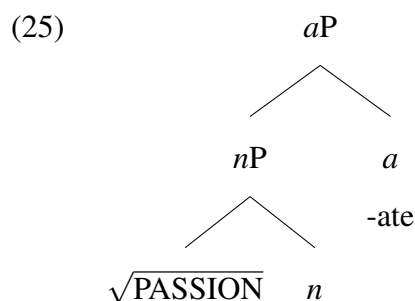
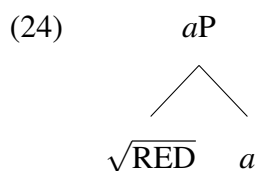
Such denominal adjectives (19-23) are also seen in English (Aikhenvald, 2011, p. 274). The systematic patterns observed across languages, where adjectives derived from verbs retain aspectual or argument information, and adjectives derived from nouns express relational meanings like ‘made of,’ ‘like,’ or ‘having’, etc., suggest that adjectivization involves regular structural operations. These cross-linguistic regularities point towards a unified syntactic mechanism underlying adjective formation. The subsequent section deals with this mechanism, known as adjectivization.

4.3 Adjectivization

Adjectivization can refer to the structural mechanism of forming adjectives from non-adjectival categories like a noun or a verb or an acategorical root. According to Marantz (2007, 2013),

an adjectivizer (*a*) merges with an acategorial root or an already categorized stem to form adjectives within the process of adjectivization. This mechanism is parallel to nominalization through *n* and verbalization through *v* as discussed in Chapters 2 and 3, respectively. In Chapters 2 and 3, we have discussed that nouns and verbs are syntactically formed, not only by the merge of the respective categorizers with acategorial roots to show idiosyncrasies, but also with already categorized stems to generate compositional interpretations.

Within the DM framework, the adjectivizer head ‘*a*’ serves as the categorizing head that determines both the syntactic and semantic interpretation of the derived form. Crucially, whether *a* merges with a root, a verbalized stem, or a nominalized stem predicts the different semantic types we observed in the cross-linguistic data. Deverbal adjectives retain regular verbal properties, where as denominal adjectives express relational meanings, such as possession, similarity, etc. A prototypical adjective like ‘red’ in English is formed as (24) and a noun-derived adjective like ‘passionate’ (Aikhenvald, 2011) is formed as (25). Due to the difference in the merge positions in the derivation, ‘red’ has idiosyncratic interpretation, where as ‘passionate’ has a nominal compositional interpretation from an already nominalized structure ‘passion_N.’



Building on this, adjectivization can be understood as a syntactic operation whose outcomes vary depending on the structural environment in which *a* is merged. When *a* attaches directly to a root, the resulting adjective often displays non-compositional or idiosyncratic meaning, as seen in (24). When it merges with a nominal or verbal stem, however, the output is semantically transparent and predictable, reflecting the structure that *a* categorizes.

However, recent literature also demonstrates contrary evidence to this view. According to Takehisa (2017), some adjectives that appear to be stem-derived show root-sensitive idiosyncrasies and have to be formed as (24). According to such works, adjectives like ‘crooked’, ‘dogged’, ‘ragged’, etc., do not show any compositional interpretation of another categorized

element. As we will see in the next section, Santali shows more regularity in the derivation of adjectives and the interpretation of the resulting adjectives than what we see in English. We have both idiosyncratic and compositional adjectives that follow regular syntactic rules in the adjectivization process.

A different point of view in adjectivization by Panagiotidis (2020) argues that there is no adjectivizer cross-linguistically because of the lack of interpretative perspective for adjectivizers. Baker puts adjectives in the elsewhere position claiming that any LC, that is not a noun or a verb, can be termed as an adjective. Therefore, following such a claim, Panagiotidis argues that, like *n* has ‘sortality’ and *v* has ‘extend into time’, there is no interpretative feature for *a*. The second reason for arguing against an adjectivizer for the process of adjectivization is the polycategorial nature of adjectives. Panagiotidis states that adjectives in some languages, like Korean, demonstrate verbal properties and in some other languages, like Japanese, show nominal characteristics. That is why adjectives do not have their own independent interpretive perspective cross-linguistically.

I argue against such a claim in this chapter proposing that the interpretive perspective of adjectives are ‘properties’ and ‘modification.’ Even if a few nouns also can denote properties, all adjectives are property denoting categories. Along with that the core function adjectives play in both attributive and predicative positions are that of modification of nouns. I follow Marantz (1997, 2000, 2007, 2013), Nevins and Myler (2014), Takehisa (2017), among others, to support that adjectivizers exist and they play an important role in the formation of adjectives. They have property denotation and modification as their interpretative characteristics. To solve the second issue that Panagiotidis points about the polycategorial nature of adjectives, I argue that the position of attachment of an adjectivizer in the derivational spine decides their root-interpretation and nominal or verbal interpretation. In other words, the low attachment point of an *a* shows idiosyncrasy and corresponds to root-adjectivization. On the other hand, higher attachment with an already categorial element like a noun or a verb determines the nominal or verbal compositional interpretation. We will also see in Santali that, the attachment position of an *a* can go as high as the DP level to show extended nominal meaning in adjectives.

Now, we can move to Santali to see how the process of adjectivization affects Santali categorization to form adjectives.

4.4 Adjectivization in Santali

Due to the problematic nature of adjectives as a grammatical category cross-linguistically, studying adjectives in Santali similarly presents many challenges. Despite being a major word class, this category has received almost no attention in the limited typological literature available on Santali. In one of the earliest available works, Bodding (1929) claims that there is no restriction on what can behave as an adjective in Santali. According to him, when two words appear side-by-side and one is a noun, the other word functions as an adjective. As a rule in Santali, the first word in such a combination acts as an adjectives, making adjectives pre-nominal.

I argue that this characterization of adjectives is misleading and shows problems of over-generation. In a DP, not every word co-occurring with a noun gives an adjective-noun combination. Nouns in Santali can be preceded by various elements including demonstratives (26a-26b), numerals (26c-26d), possessors (26e-26f), etc. While these elements have modificational properties, they should not be categorized as adjectives on distributional and categorial grounds.

- | | |
|---|--|
| (26) a. <i>uni hoR</i>
that man
‘that man’ | d. <i>baareyaa daare</i>
two tree
‘two trees’ |
| b. <i>noa lugDi</i>
this dress
‘this dress’ | e. <i>uni-a oDaa</i>
he-GEN house
‘his house’ |
| c. <i>peyaa hoR</i>
three man
‘three men’ | f. <i>jon-rena lugDi</i>
John-GEN dress
‘John’s dress’ |

However, Bodding’s position is correct in two aspects. First, Santali adjectives can occur pre-nominally in the attributive position. Second, any category, such as noun or verb can behave as an adjective. This doesn’t mean that any morphological element in a pre-nominal position is an attributive adjective, as Bodding incorrectly predicts. Also, Bodding doesn’t explain much on the prototypical adjectives, that are not derived from any other category. As we will see in Section 4.4.1, these prototypical adjectives are the root-derived adjectives in Santali, which carry the idiosyncratic modificational characteristics and cannot be composed from any other

categorial stem, like a noun or a verb. These adjectives are also true property concepts that can be modified by degree elements like ‘too’, ‘very’, etc. Santali also exhibits predicative adjectives that Bodding does not talk about.

Providing an exactly contrary view, Ghosh (2008) argues that there is no adjective at all in Santali. According to him, apart from a handful of borrowed words behaving as adjectives, Santali doesn’t have adjectives as a category. However, I show in this section that adjectives constitute a category in Santali like many other languages. I use a number of diagnostics to prove that Santali recognizes adjectives as a separate LC. These diagnostics are the most common tests in various languages in the world that correspond to the defining and the descriptive properties of adjectives.

There are typologically attested diagnostics to distinguish the functions of adjectives across languages. Beck (1999b), Baker (2003), Kennedy (2012), among others, have used different tests to check the adjectival status of a morpheme in a language. Compiling them together, Venkatesan (2025) shows four empirical diagnostics to argue for the presence of adjectives in Tamil. I argue that these diagnostics, although are not the defining properties of adjectives, are interpretative properties of adjectives in many languages. I apply these diagnostics to Santali to argue the presence of adjectives in Santali.

- i. **Modification of nouns:** Both attributive and predicative modification is considered to be the most fundamental characteristic of adjectives (Dixon, 2004, p. 10). Natural languages have a particular type of words that act as adjectives carrying the modificational properties in both attributive and predicative positions. Like many other languages, Santali has a fixed position for the occurrence of adjectives in a DP, which is the pre-nominal position. In Santali, words like *bes* are used as adjectives in both attributive (27a) and predicative (27b) positions.

- (27) a. *bes hoR*
 good man
- b. *uni hoR bes-ge-a*
 that man good-FOC-FIN
 ‘That man is good.’

- ii. **Co-occurrence with other modifiers by qualifying N in parallel:** An important trait adjectives show is that they can have other modifiers, like numerals, quantifiers, and

other adjectives co-occur with them modifying the same noun (head noun) in a DP. Since, Santali has a distinct class of such words, it can be inferred that the language recognizes adjectives as an independent grammatical category. In (28), the adjective *bes* co-occurs with numerals (28a), quantifiers (28b), and other adjectives (28c), modifying the head noun in parallel.

- (28) a. *peyaa bes hoR-ko*
 three good man-PL
 ‘three good men’
- b. *aaemaa bes hoR-ko*
 many good man-PL
 ‘many good men’
- c. *naawaa bes lugDi*
 new good dress
 ‘The new good dress’

iii. **Modification by degree heads:** Another key defining property of adjectives across languages is that only they can be modified by a degree elements or intensifiers (Beck, 1999b; Baker, 2003, p. 212; Kennedy, 2012, p. 329). According to Baker, degree words always modify adjectives. In the predicative position, they refine the meaning of the adjectival predicate by indicating to what extent the property applies to the subject. However, degree heads can modify adjectives in the attributive position, as well. The constructions in (29) show the presence of prototypical adjectives in Santali, since they can be modified by degree elements in both predicative (29a-29b) and attributive (29c) positions.

- (29) a. *noaa hor aDi jhal-ge-a*
 this road very long-FOC-FIN
 ‘This road is very/too long/ the longest.’
- b. *daah kaTi reaaD-ge-a*
 water little cold-FOC-FIN
 ‘The water is a little cold.’
- c. *aDi maaraang oDaa*
 very big house
 ‘A very big house’

Baker (2003, p. 213) notes some exceptions in this well-known defining property of adjectives. Not all adjectives can be modified by degree heads. According to Baker, adjectives, like ‘prime’, ‘pregnant’, and ‘three-legged’ in (30) exhibit absolute, ungradable properties, and cannot be modified by degree words.

- (30) a. *Seven is as prime as two is.
 b. #Mary is too pregnant to go on the trip.
 c. *How three-legged is that stool?

We can infer from the ungrammaticality of (30) that not all adjectives are modified by degree heads, where as all degree heads modify only adjectives.

- iv. **Comparative properties:** Adjectives cross-linguistically exhibit comparative properties. This diagnostic shows the gradable character of adjectives like their ability to be modified by degree words. Adjectives can be expressed in comparative and superlative terms. In Santali too, modifiers can have post-position, like *khon* to show comparative and superlative characteristics (31).

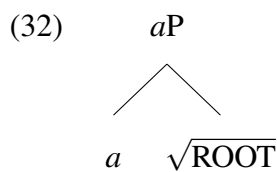
- (31) *noaa daare onaa daare-khon maaraang-ge-a*
 this tree that tree-ABL big-FOC-FIN
 ‘This tree is bigger than that tree.’

Based on these empirical diagnostics, we can argue that adjectives constitute a word class of its own in Santali. After explaining that adjectives exist in Santali, We can go into the derivational differences between two types of adjectives. In the subsequent subsections, I claim that like nouns and verbs in Santali, adjectives also are formed structurally yielding in a division between root-derived and stem-derived adjectives. When Boddington (1929) claims that any word can behave as an adjectives, it becomes partially correct in our theory as stem-derived adjectives are formed from categories like nouns and verbs.

4.4.1 Root-derived Adjectivization

In this section I discuss how the prototypical adjectives, are formed in syntax by root-derivation. Following Halle and Marantz (1993), Marantz (1997, 2007, 2013), Embick (2004), Siddiqi

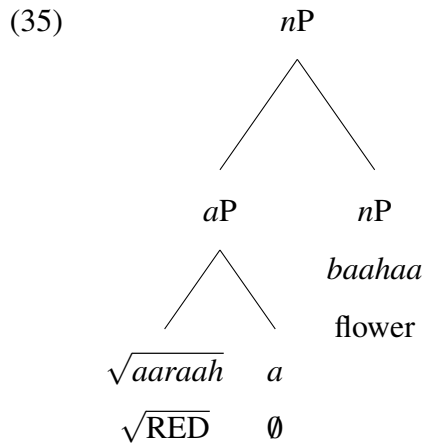
(2010), among others, I propose that prototypical adjectives are formed by merging an adjec-tivizer with an acategorial root, like prototypical nouns and verbs as seen in Chapter 2 and 3, respectively. Since according to DM, interpretive properties of adjectives are dependent on the structural height at which an *a* merges with a root, a structure like (32) can explain the idiosyncratic interpretation of prototypical adjectives. The acategorial root that lacks any in-herent semantic content, gets interpreted as an adjective at the first merge position. Also there is no intervening categorizer to create the first phase (Marantz, 2000) or the CID (Arad, 2003) to make the adjec-tivizer get compositional semantics.



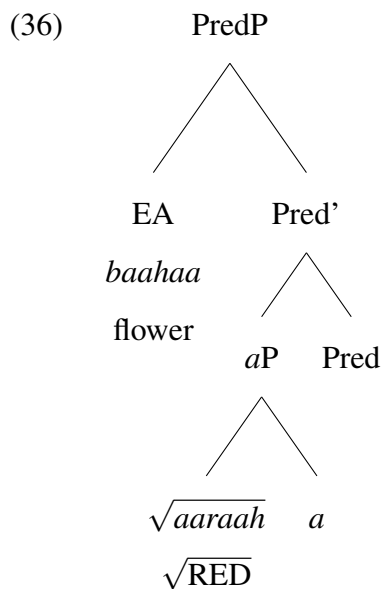
This structure in (32) can explain the formation of prototypical adjectives which can acquire the attributive modifier (33) and the predicative (34) position in Santali.

- | | |
|--|---|
| <p>(33) a. <i>aaraah baahaa</i>
red flower
'A red flower'</p> <p>b. <i>huDing gidRa</i>
small child
'A small child'</p> <p>c. <i>keTe dhiri</i>
hard stone
'A hard stone'</p> <p>d. <i>heDem uul</i>
sweet mango
'A sweet mango'</p> <p>e. <i>reaaD daah</i>
cold water
'Cold water'</p> | <p>(34) a. <i>onaa baahaa aaraah ge-a</i>
That flower red FOC-FIN
'That flower is red.'</p> <p>b. <i>gidRa huDing ge-a-e</i>
child small FOC-FIN-3SG
'The child is small.'</p> <p>c. <i>dhiri keTe ge-a</i>
stone hard FOC-FIN
'The stone is hard.'</p> <p>d. <i>noaa uul heDem ge-a</i>
this mango sweet FOC-FIN
'This mango is sweet.'</p> <p>e. <i>daah reaaD ge-a</i>
water cold FOC-FIN
'Water is cold.'</p> |
|--|---|

For attributive modification, these prototypical adjectives merge with the nouns they modify by XP adjunction. They are formed in a different syntactic workspace before merging with nouns as adjuncts (Nevins and Myler, 2014; Morita, 2021). Adjectives like (33) can be formed like (35) to modify nouns in a DP.



On the other hand, the same adjectives in the predicative position do not co-occur with nouns in a pre-nominal position. As Baker (2003) points out, predicative adjectives, like nominal predicates need a functional head ‘Pred’ to license the external arguments. ‘Pred’ is the functional head equivalent to the verbal functional head ‘Voice’. Drawing from this, we can have a structure like (36) to explain the prototypical predicative adjectives in Santali (34).



In both (35) and (36) the adjectives are formed using the inner morphology through root-derivation to correspond to the idiosyncratic interpretation of the adjectives. However, Santali also exhibits adjectives with transparent nominal and verbal interpretation in adjectivization. In the next sub-section, we can see the stem-derived adjectivization process that includes nominals or verbals in the derivational structure and semantic interpretation.

4.4.2 Stem-derived Adjectivization

In this section we address one of the major questions of this chapter: whether Santali exhibits stem-derivation for the formation of adjectives in the same way it demonstrates stem-derivation in nominalization and verbalization. We will see that it does indeed exhibit stem-derived adjectivization. Stem-derived adjectivization is not unique to Santali. As we have seen in Section 4.2.3, many languages cross-linguistically show adjectives derived from other categories. However, the current finding is novel because Santali, which has been described as a language where any prenominal morpheme is an adjective (Bodding, 1929), or as a language lacking adjectives as a composite category (Ghosh, 2008), not only shows root-derived adjectives as a distinct subclass but also exhibits adjectives formed through derivation, specifically stem-derivation.

A further point of inquiry concerns the extent to which Santali can adjectivize non-adjectival categories such as nouns or verbs. We will demonstrate in this section that Santali not only adjectivizes nouns or verbs through re-categorization in a higher merge position (Marantz, 2000, 2007; Arad, 2003), but also adjectivizes nominal and verbal extended structures by allowing different adjectivization heights in the derivation.

Constructions in (37) and (38) demonstrate that Santali has denominal and deverbal adjectivization, respectively.

- (37) a. *paandu uup-ih hoR*
white hair-ADJZ man
'The white haired man'

- b. *daaDhi-ihaa hoR*
beard-ADJZ man
'bearded man'

- (38) a. *japid-ih setaa*
sleep-ADJZ dog
'sleepy dog'

- b. *laTu-ih gidRa*
grow-ADJZ boy
'grown boy'

- c. *sereng-ihaa koDaa*
sing-ADJZ boy
'the singing boy'

- d. *eneh-ihaa setaa*
 dance-ADJZ dog
 ‘dancing dog’

The adjectives in (37) and (38) are stem-derived adjectives unlike the adjectives in (33). There are two empirical pieces of evidence that argues for the adjectives in (37-38) not being root-derived. First is their compositional nature. These adjectives show interpretations related to the nominal or verbal counterparts they are derived from, not idiosyncrasies we expect from root-derivation. We will see in next two subsections that these adjectivizations also accommodate the nominal and verbal extended structures, which proves that they have both structural and semantic properties of their categorial bases.

Secondly, like root-derived adjectives, stem-derived adjectives in Santali do not show gradability, so they do not take degree modifiers (39). The stem-derived adjectives in (39) are unacceptable when they are modified by intensifiers/ degree modifiers, whereas root-derived adjectives (40) can be modified by degree heads.

- (39) a. **aDi daaDhi-ihaa hoR*
 very/too beard-ADJZ man
 ‘very bearded man’
 b. **aDi laTu-ih gidRa*
 very/too grow-ADJZ boy
 ‘very grown boy’
 c. **aDi sereng-ihaa koDaa*
 very/too sing-ADJZ boy
 ‘very singing boy’

- (40) a. *aDi bes hoR*
 very/too good man
 ‘very good man’
 b. *aDi heDem uul*
 very/too sweet mango
 ‘very sweet mango’
 c. *aDi reaaD daah*
 very/too cold water
 ‘very cold water’

The stem-derived adjectives can appear predicatively as well, in Santali. The constructions in (41) and (42) demonstrate the noun-derived and verb-derived adjectives in predicative positions. Santali adjectives take a focus copula *ge* when they appear in the predicative position.

- (41) a. *uni hoR paandu uup-ih ge-a-e*
 that man white hair-ADJZ FOC-FIN-3SG
 ‘That man is white haired.’
 b. *uni hoR daDhi-ihaa ge-a-e*
 that man beard-ADJZ FOC-FIN-3SG

‘That man is bearded.’

- (42) a. *uni setaa japid-ih ge-a-e*
 that dog sleep-ADJZ FOC-FIN-3SG
 ‘That dog is sleepy.’
- b. *uni gidRa laTu-ih ge-a-e*
 that boy grow-ADJZ FOC-FIN-3SG
 ‘That boy is grown.’

After understanding the distributional patterns of stem-derived adjectives in Santali, we can now move on to a detailed understanding of both noun-derived and verb-derived adjectives in the subsequent subsections. In the next two subsections, I explain how the extended structures of nouns and verbs can work as bases for adjectivization in the derivational spine.

4.4.2.1 Derivation from Nominal Bases

Santali uses two strategies to derive modifiers from nominal bases. First, similar to what we have seen in nominalization and verbalization, it uses a null adjectivizer to change nominal stems to modifiers, producing a noun-noun compound kind of structure. Secondly, it uses the overt adjectivization morpheme *-ih(aa)*¹ to adjectivize already nominalized constructions or extended nominal structures. In the first strategy, the nominal modifier of the compound doesn’t show adjectival property. However, it acts as an attributive modifier in an endocentric compound structure (43). The constructions in (43) demonstrate that the modifiers do not have the adjectival marker. As we can see in (44), they cannot take the predicative position like other adjectives in Santali.

- (43) a. *kaaTh chiyaar*
 wood chair
 ‘wooden chair’
- b. *sonaa paagraa*
 gold earring
 ‘golden earring’

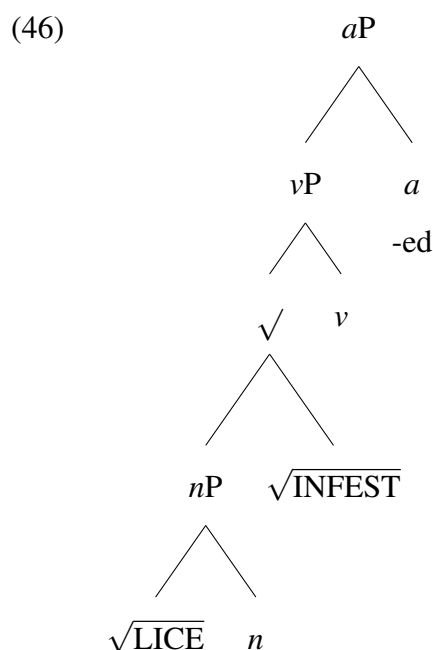
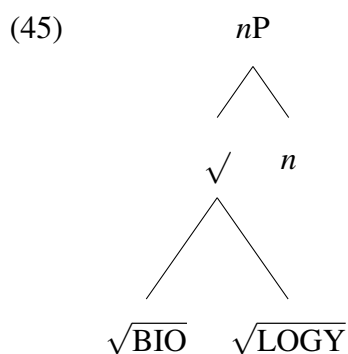
¹Here, the *aa* marker, co-occurring with the adjectivizer *ih* could be a borrowed adjectivizer from Odia. *-iaa* or *aa* is a productive adjectivizer in Odia, an IA language, spoken in the same geographical region where Santali is spoken (1).

- (1) a. *paaNi* ‘water’ *paaNia* ‘watery’
 b. *daaDhi* ‘beard’ *daaDhia* ‘bearded’

- (44) a. **noa chiyaar kaaTh ge-a*
 this chair wood FOC-FIN
 'This chair is wooden'
- b. **onaa paagraa sonaa ge-a*
 that earring gold FOC-FIN
 'That earring is golden.'

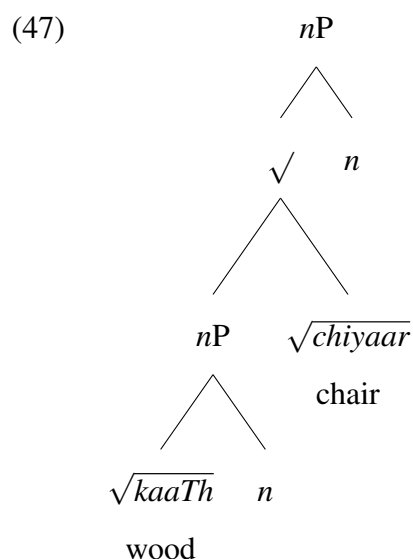
We cannot argue that these modifiers are formed through the adjectivization process. Even if they have modificational properties, they behave like a prototypical noun-noun compound. Such compound-like modifiers are quite rare Santali.

Siddiqi (2006), Harley (2011a), Okubo (2013), Nevins and Myler (2014), among others, have proposed both a root-merge and a stem-merge approach to explain compounding. According to their analysis, compounds can be formed in syntax by merging two acategorial roots, then a categorizer (45), or merging a root with an already categorized element (46). According to Okubo (2013), Booij (2012), constructions like 'bio-logy', 'geo-graphy', etc. are formed by using a root-derived compounding structure (45). On the other hand, Siddiqi (2006, p. 91), and Okubo (2013) propose that compounds like 'lice-infested' are formed following a structure like (46).



Following such an approach, I argue that the noun-noun construction in (43) is formed by merging a nominalized element with a root before categorization. In other words, in a structure

like (47), the modifier root gets nominalized, not adjectivized to modify the head noun in the attributive position.



The second type of noun-derived adjectivization includes adjectives like (48, repeated from 37b). I propose that not only they get adjectivized by the merge of an *a* with a nominalized element, they also have different structural positions in the derivation for adjectivization, which I refer to adjectivization height. In the generative literature that concerns with categorization, adjectivization height is not a very well-studied phenomenon. However, as we have already seen in Chapter 2, nominalization can have different structural *n*-attachment points in the derivation demonstrating the extent to which something (a ‘verb’ or a ‘phrase’) can be nominalized. I assume such an extended structure as the complement of *a* to propose adjectivization height. Even if denominal adjectivization is more restricted than deverbal adjectivization, cross-linguistically, and such adjectivization even more restricted in Santali, Santali shows a regular pattern in showing the adjectivization of extended nominal structures. Not only just prototypical nouns (48), but NPs with adjectival modification (49), NumPs (50), and DPs (51) can be turned into adjectives in Santali.

- (48) *daaDhi-ihaa hoR*
 beard-ADJZ man
 ‘bearded man’

- (49) a. *uni hende daaDhi-ihaa hoR*
 that black beard-ADJZ man

‘that black bearded man’

- b. *uni hoR hande daaDhi-ihaa ge-a-e*
that man black beard-ADJZ FOC-FIN-3SG
‘That man is black bearded.’

- (50) a. *punia jahngaa-ih hoR*
four leg-ADJZ man
‘four-legged man’

- b. *uni hoR punia jaangaa-ih ge-a-e*
that man four leg-ADJZ FOC-FIN-3SG
‘That man is four-legged.’

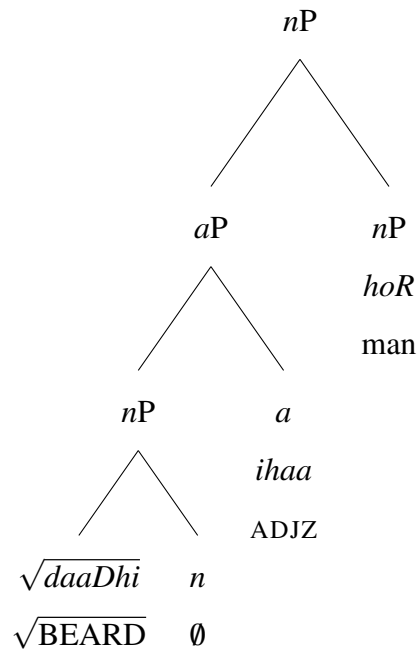
- (51) a. *uni [han teren-ih] koRaa*
that.ANIM [that.INANIM side-ADJZ] boy
‘that [that sided] boy’

(Context: When someone is at the other side of the house/village)

- b. *uni koRaa [han teren-ih] ge-a-e*
that.ANIM boy [that.INANIM side-ADJZ] FOC-FIN-3SG
‘That boy is that sided.’

A prototypically denominal adjective like (48), can be formed at a higher merge position. The root first merges with a nominalizer, closing the interpretation domain of the root. In other words, after the *n* merges, by the virtue of creating a phase, the root gets spelt out leaving the nominal interpretation at the *nP* level. The adjectivizer merges in the derivation to interact with the *nP*, following the LCIR. Therefore, the denominal adjective in (48), can be structurally formed as (52).

(52)

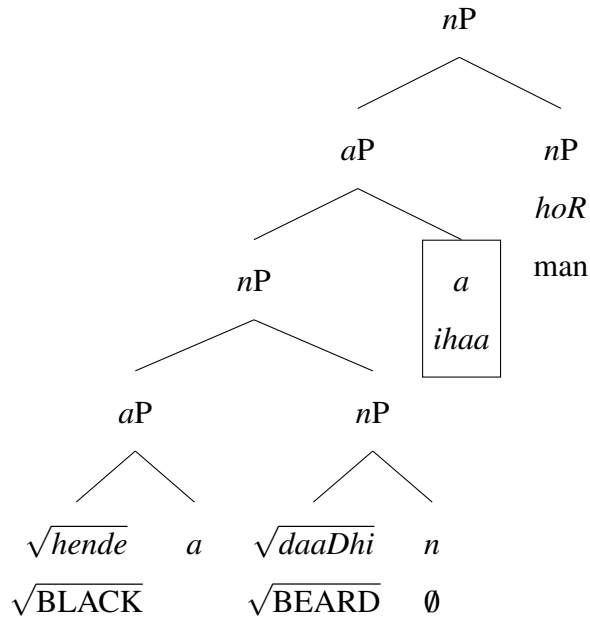


Similarly, the construction in (49) involves not just a noun, but a noun with its own adjectival modifier as an input for adjectivization. Such an adjective-noun combination constitutes an NP with the adjective as an XP adjunct at the specifier position. This means that the nominal input for adjectivization involves extended nominal elements higher than just the nominal level. I show in (53) that the adjectivizer in Santali can merge at phrasal level like NP, as well.

Here (53), the root $\sqrt{daaDhi}$ ‘beard’ merges with a null nominalizer to form a root-derived noun. It then takes an adjectival modifier in its specifier position through XP adjunction. This adjective $\sqrt{hende}$ ‘black’ is formed through root derivation in its own CID (Marantz, 2000; Arad, 2003; see also 35). These two then form an *nP* which is equivalent to the lexical NP. This NP subsequently gets adjectivized by the overt marker $-ih(aa)^2$ to modify the noun *hoR* ‘man’.

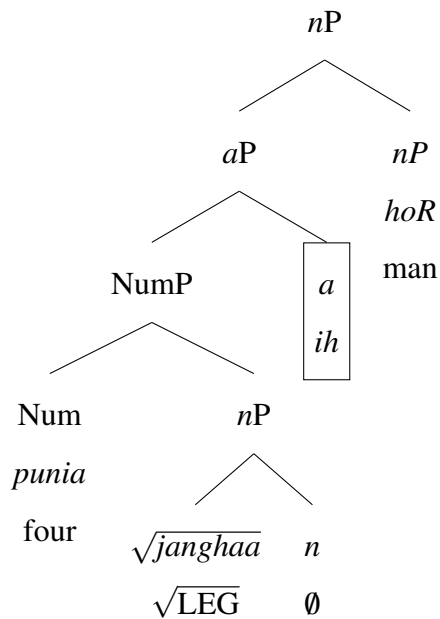
²As we have seen in this chapter, overt adjectivizers are used only in the formation of stem-derived adjectives. Root-derived adjectives are formed with a null categorizer, similar to nominalization and verbalization, as discussed in Chapters 2 and 3, respectively.

(53)



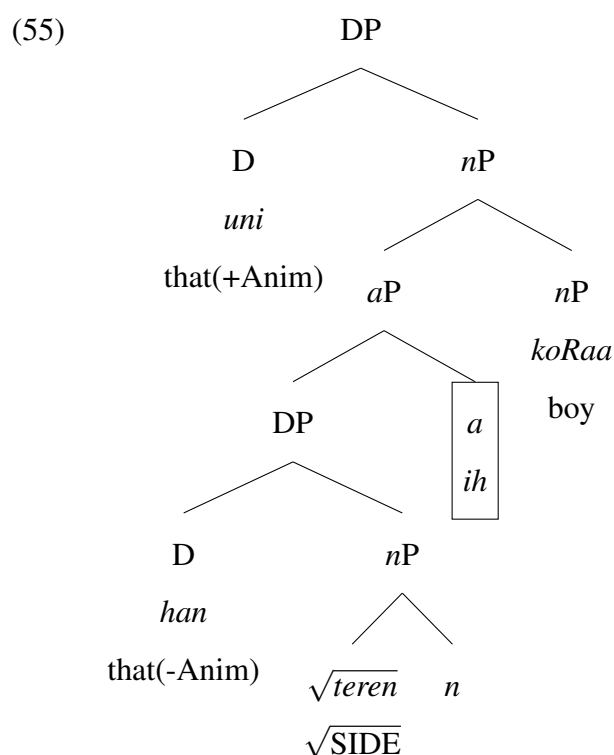
Moving a step further, the construction in (50) demonstrates that the numeral-noun complex, which constitutes a NumP, acts as the base for adjectivization. In (54), we can see even a higher attachment point for the ‘*a*.’ In this structure, the root *janghaa* ‘leg’ first merges with a null nominalizer to form an *nP*. This *nP* then combines with the numeral *punia* ‘four’ in the Num head position, projecting a NumP. Crucially, it is the entire NumP that serves as the input for adjectivization. The adjectivizer *-ih* merges with the NumP, taking the whole numeral-noun complex within its scope and creating an adjective (*aP*). This *aP* in turn modifies the noun *hoR* ‘man’, yielding the structure ‘four-legged man’.

(54)



Along the same lines, the adjectivizer can merge as high as the DP level to interact with the maximal nominal structure. The construction in (51), structurally represented as (55), shows that an entire DP can act as the input for adjectivization in Santali. In (55) the root *teren* ‘side’ first merges with a null nominalizer to form an *nP*. This *nP* then merges with an inanimate determiner *han* ‘that’ to form a complete DP ‘that side’. Crucially, the entire DP undergoes adjectivization when the adjectivizer *-ih* merges with the DP. This derived adjective further modifies the noun *koRaa* ‘boy’, which later merges with an animate determiner to form a DP.

The structure in (55) demonstrates the highest point of attachment for the adjectivizer to form denominal adjectives in Santali, showing that adjectivization can target not just LCs but also functional projections up to the DP level, encompassing the extended structure of nominals including determiners.



After understanding the formation of stem-derived from nominal bases, we can move to the other type that deals with the adjectival derivation from verbal bases. Like in the current section we saw different structural positions for adjectivization, we aim to understand the adjectivizer height in the verbal domain, as well. In the subsequent subsection, we analyze how the structural position of adjectivization has direct consequences for the predictability and transparency of the derived adjective’s meaning.

4.4.2.2 Derivation from Verbal Bases

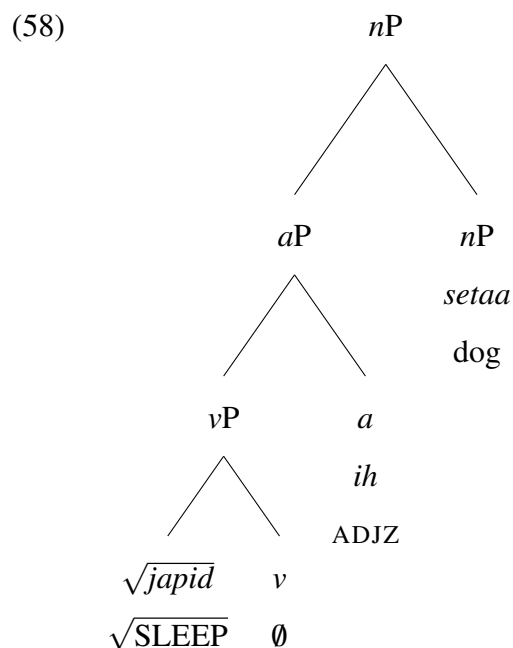
In this subsection, we deal with the adjectival constructions, in which the *a* takes an already verbalized element as the input. This kind of verb derived adjectives are more frequently used than noun-derived ones in Santali. These adjectives have compositional verbal interpretation, unlike root-derived adjectives. As demonstrated in (56) and (56), the attributive and predicative adjectives, respectively have verbal structural and semantic interpretation.

- (56) a. *japid-ih setaa*
 sleep-ADJZ dog
 ‘sleepy dog’
- b. *koko-ih hoR*
 beg-ADJZ man
 ‘begging man’ (beggar)
- c. *sereng-ihaa koDaa*
 sing-ADJZ boy
 ‘singing boy’
- d. *eneh-ihaa setaa*
 dance-ADJZ dog
 dancing dog
- (57) a. *uni setaa japid-ih ge-a-e*
 that dog sleep-ADJZ FOC-FIN-3SG
 ‘that dog is sleepy’
- b. *uni hoR koko-ih ge-a-e*
 that man beg-ADJZ FOC-FIN-3SG
 ‘that man is a beggar’

Not all deverbal attributive adjectives can be predicativized in Santali. As we have seen in (3), the distributional restriction of adjectives is not limited to Santali. It is an idiosyncratic property of adjectives as a category. Even in languages like English, adjectives like ‘former’, ‘main’, etc. do not have a predicative equivalent.

Coming back to the Santali deverbal adjectives, to derive the compositional verbal interpretation observed in Santali deverbal adjectives (56), I propose that the adjectivizer merges at a higher (the second categorizer here) categorizer position in the derivational structure. This position allows the adjectivizer to retain its verbal semantics from *v*, the first categorizer, following the LCIR. As we can see in (58), the adjectivizer *a* merges with *v*P, where the first

category has already been formed through the merge of *v* with the acategorial root.

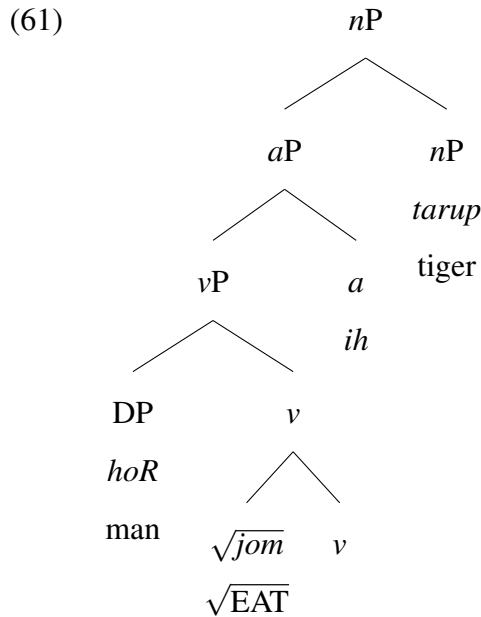


The structure in (58) works as any deverbally derived adjective structure we saw cross-linguistically in Section (4.2.3). The claim that Santali has verb-derived adjectivization can be further attested by the presence of argument structure in adjectives. In Santali, some deverbally derived adjectives retain the argument structure information from their verbal bases. Since adjectives are known as one place predicates (Baker, 2003), we can expect internal argument of the verb after adjectivization if Santali indeed shows deverbally derived adjectives. As we can see in (59), the constructions show the retention of the internal arguments of the verbal bases after adjectivization. Like denominal adjectives, they take *-ih* marker for adjectivization and can occur at predicative position (60), although the predicative use is not very common in natural speech.

- (59)
- a. *hoR jom-ih tarup*
man eat-ADJZ tiger
'man eating tiger'
 - b. *daare geged-ih buDiaa*
tree cut-ADJZ axe
'tree cutting axe'
 - c. *daare geged-ih cheNe*
tree cut-ADJZ bird
'wood cutting bird (woodpecker)'

- (60) a. *uni tarup hoR-jom-ih ge-a-e*
 that tiger man-eat-ADJZ COP-FIN-3SG
 ‘That tiger is man-eating.’
- b. *noa buDiaa daare-geged-ih ge-a*
 this axe tree-cut-ADJZ COP-FIN
 ‘This axe is tree-cutting³.’
- c. *noa cheNe daare-geged-ih ge-a-e*
 this bird tree-cut-ADJZ COP-FIN-3SG
 ‘This bird is tree-cutting.’

In underspecificational approaches for categorization, like DM, the verbal categorizer *v* introduces the internal argument (Bešlin, 2025). The Voice head introduces the external argument and assigns accusative case to the internal argument (Kratzer, 1996; Bešlin, 2025). As can be seen in (61), the acategorial root first merges with the verbalizer to form *jom_v* ‘eat’, which subsequently merges with the IA *hoR* ‘man’. The *vP* marks the phase boundary, following the LCIR. The *a* later merges with the *vP* to get the compositional verbal interpretation.



Now the question is whether Santali allows different structural positions for adjectivization above the *vP* level, like we saw in the case of denominal adjectivization. In other words, if we can have adjectivization of different verbal extended structures like AspectP, TP, etc. I will show that the answer to this question is not straightforward. I argue that *vP* is the maximum

³The lack of phi agreement in this example is due to the -animate feature of the subject. As we have discussed in section 3.2.3 of Chapter 3, only animate subjects trigger phi agreement with the verb

point of adjectivization in the verbal domain. Even if the extended verbal layers like AspectPs and TPs act as modifiers of nouns, they are not adjectives in Santali. The constructions in (62) and (63) may look like adjectives because of their attributive position to modify nouns. But deeper analysis of these data points will show that these constructions are not adjectival constructions.

- (62) a. *rapud-aaka Dekchi*
 break-PERF pot
 ‘the broken pot’
 b. *isin-aaka daakaa*
 cook-PERF rice
 ‘the cooked rice’
 c. *nyur-aaka daare*
 fall-PERF tree
 ‘the fallen tree’
- (63) a. *baDaae-en-a kaathaa*
 know-PST.MID-FIN fact/topic
 ‘known fact/topic’
 b. *chaabaa-en-a kami*
 complete-PST.MID-FIN work
 ‘completed work’

I use diagnostics like (i) Verbal interpretation in predicative position, (ii) Restriction on adjectival co-ordinate structure, and (iii) Absence of morphological adjectivizer, to prove that the constructions in question are verbal participle structures, not adjectivization of extended verbal layers.

- i. **Verbal interpretation in predicative position:** True adjectives (both root-derived and stem-derived) have similar adjectival interpretation in both attributive and predicative domains. As we have seen in the denominal adjectivizations and the deverbal adjectivizations, the predicative constructions also show adjectival modificational properties. However the constructions in the questions (62-63) have canonical verbal interpretations (64-65), even if the verbal modifiers can acquire a predicative position.

- (64) a. *noaa Dekchi rapud-aakan-a*
 this pot break-PERF.COP-FIN
 ‘This pot has broken.’

- b. *daakaa isin-aakan-a*
rice cook-PERF.COP-FIN
'The rice has cooked.'
 - c. *noaa daare nyur-aakan-a*
this tree fall-PERF.COP-FIN
'This tree has fallen.'
- (65)
- a. *kaathaa baDaae-en-a*
topic know-PST.MID-FIN
'The topic became known.'
 - b. *kami chaabaa-en-a*
work complete-PST.MID-FIN
'The work became completed.'

This diagnostic suggests that the attributive constructions in (62-63) are not instances of true adjectives.

- ii. **Restriction on adjectival co-ordinate structure:** According to one of the diagnostics⁴ proposed by McIntyre et al. (2013), true adjectives can form co-ordinate structures with other adjectivals. When we use this test to the verbal modifiers in the current question (62-63), we realize that they produce unacceptable structures (66-67).

- (66)
- a. **rapud-aaka aar bes Dekchi*
break-PERF and beautiful pot
'the broken and beautiful pot'
 - b. **isin-aaka aar heDem daakaa*
cook-PERF and sweet rice
'the cooked and sweet rice'
 - c. **nyur-aaka aar haraa daare*
fall-PERF and green tree
'the fallen and green tree'
- (67)
- a. **baDaae-en-a aar nawaa kaathaa*
know-PST.MID-FIN and new topic
'known and new topic'

⁴The other diagnostics they propose to distinguish true adjectives are English centric. They include diagnostics like, 'un-' prefixation, selection by verbs like 'seem', etc. Even if their diagnostics include modification by 'very', we do not use it because as we have already discussed, not all adjectives are modified by degree heads/ intensifiers.

- b. **chaabaa-en-a* *aar bes kami*
 complete-PST.MID-FIN and good work
 ‘completed and good work’

However, true stem (verb)-derived adjectives can accept co-occurrence with other adjectival coordinate structures (68).

- (68) *japid-ih* *aar laTu-ih* *setaa*
 sleep-ADJZ and grow-ADJZ dog
 ‘The sleepy and grown dog’

According to this diagnostic also, the modifiers in (62-63) are not true adjectives and behave like verbal participles.

- iii. **Absence of morphological adjectivizer:** The third diagnostic is a morphological one. As we can see when we compare the true deverbal adjectives (56-59) with the verbal participle constructions (62-63), we can realize that there is a difference in morphological marking on the modifiers. The *-ih* is a stem-derived adjectivizer and can be used for adjectivization of both nouns and verbs. On the other hand, this marker is absent and the canonical verbal morphology, like the aspect marker and the finite marker, are present in the verbal participial constructions. This also supports our claim that the constructions in (62-63) are not adjectives.

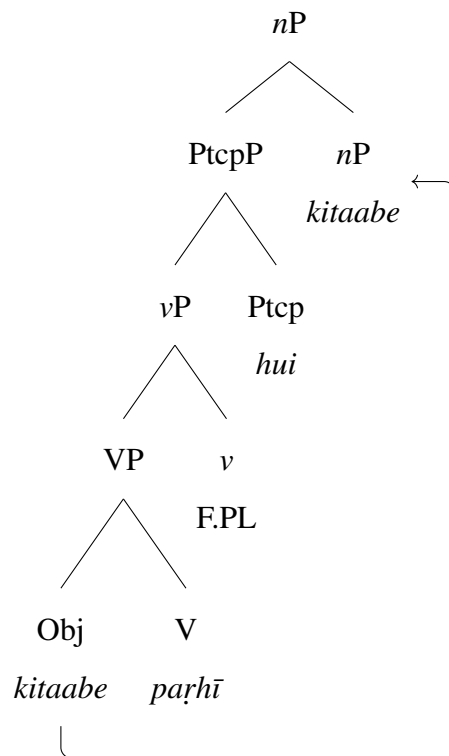
Based on the above three diagnostics, we can conclude that phrasal structures above vP level do not go through adjectivization to modify nouns.

Another difference between the true verbal adjectives and the verbal participial modifiers in Santali lies in the type of the verbs. As we can see, the verbs in the participial constructions (62-63) are unaccusative ones. In other words, the head noun (the noun that is modified in the NP) is base generated in the internal argument position of the verbal modifiers. In the recent literature, Bhattacharya (2021) has studied such participial modifications in Hindi-Urdu. Unaccusative participial modifiers carry the head nouns as the objects in the derivation, which, according to Bhattacharya, later moves to merge with the structure as the head noun. We can see such a participial construction in Hindi-Urdu and the proposed structure in (69).

(69) a. Hindi-Urdu

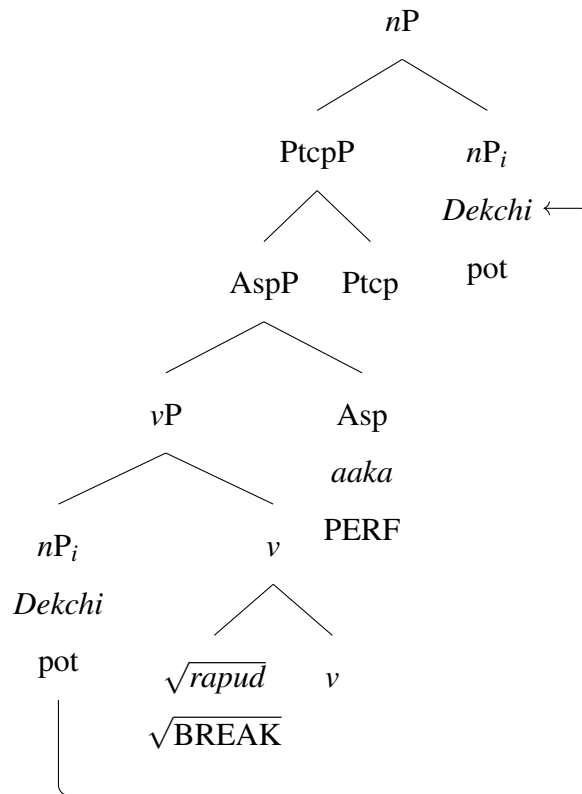
parḥī hui kitaabe
 read.PFV.F PTCP.F book.F.PL
 ‘books that have been read’ (read books)

b.



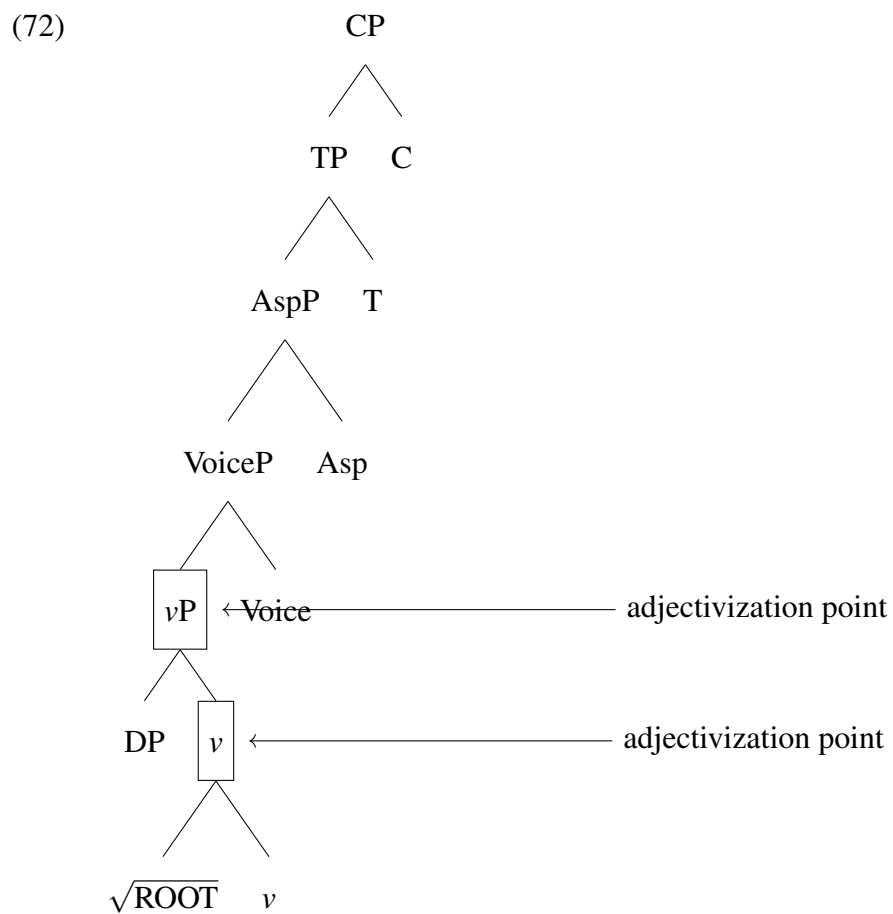
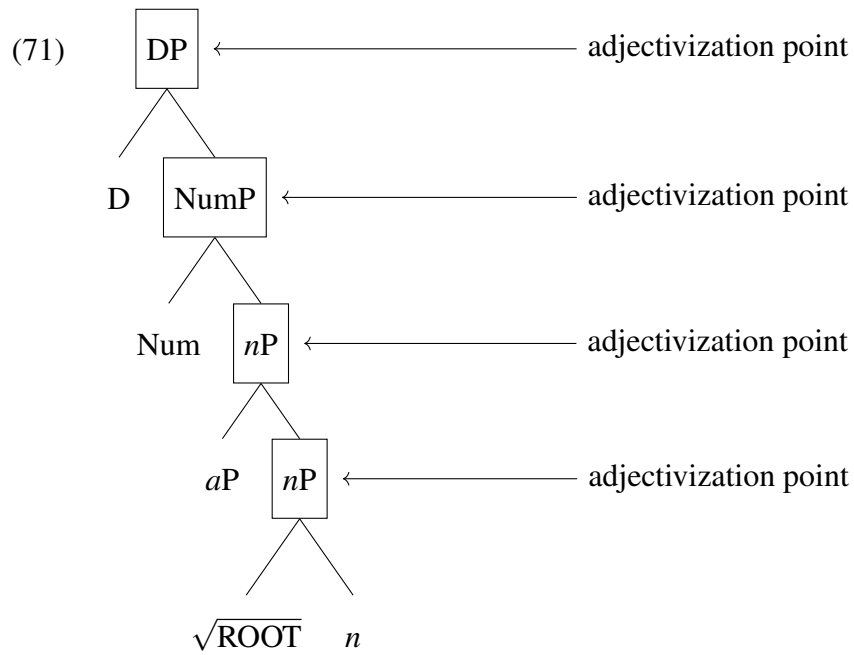
Extending such an approach to Santali verbal participles, we can claim that even though the position for the PtcpP and the *a*P are the same, i.e., the XP adjunct to the head noun, they are different modifiers. Santali has attributive adjectival modification till the *v*P level (58) in deverbal adjectivization, where as it shows participial modification above AspectP level (70).

(70)



This analysis shows a clear structural distinction between true deverbal adjectives and participial constructions in Santali. Even if both can occupy attributive positions and superficially appear to modify nouns in similar ways, their underlying syntactic structures and categorical identities differ. True deverbal adjectives, marked by the adjectivizer *-ih*, are derived at the *vP* level and retain compositional verbal semantics while occupying both attributive and predicative positions. On the other hand, constructions of AspectP level and TP level, despite their surface similarity, behave as verbal participles rather than adjectives. This has been clear from the three diagnostic tests we saw in this subsection.

Collating all these points together, we can have some generalizations for the adjectivization height in Santali. Even if both nominal and verbal extended structures can be inputs to adjectivization, they differ in terms of the structural positions of the input. In the nominal domain, the *a*-attachment point can be as high as the DP level (71), where as the verbal domain stops being the base for adjectivization at the *vP* level that carries the IA information of the verb (72).



4.5 Summary and Conclusion

In this chapter I have shown, arguing against the previous typological accounts, that Santali has adjectives as a composite category. Like nominals and verbals, adjectives are also formed syntactically by merging an acategorial root with an adjectivizer. This adjectivizer not only forms an adjective out of a root, but it also recategorizes already categorial stems like nouns and verbs. Crucially, the attachment site of the adjectivizer determines both the syntactic and semantic interpretation of the resulting structure. Extending this insight further, I have proposed the adjectivizer height as different attachment points in the derivation. With this I claim that, similar to what has been proposed for nominalization, which allows roots or verbal structures to be nominalized at different structural heights, Santali can show adjectivization of extended nominal and verbal stems at different levels. I have shown that nominal stems can be adjectivized up to the DP level with different *a*-attachment points. On the other hand verbal stems can be turned into adjectives only till the *v*P level with the internal argument information of the verb retained after adjectivization.

Taken together, the findings demonstrate that Santali not only exhibits a composite adjectival category, but also allows adjectivization of both nominal and verbal structures at multiple structural heights. This contributes to a broader theoretical picture in which adjectivization, like nominalization, is sensitive to the structural hierarchy of the derivation. Santali therefore provides empirical support for an underspecificational approach to category formation.

Chapter 5

Conclusion

5.1 Summary of the Thesis

This dissertation investigates how lexical categories, like nouns, verbs, and adjectives are formed in the syntactic workspace using Santali as an empirical testing ground. Theoretically, by studying the structures and formation of categories, this dissertation tests the core arguments of an underspecification approach like DM. As we have already seen in Chapter 1, typological literature (Bodding, 1929; Rau, 2013) tags Santali as a language with no categorial distinction among nouns, verbs and adjectives because they do not show any distinction until they are used in a sentence. Such claims also lead to overgeneralization that some Austroasiatic languages lack categories completely (Peterson, 2003, 2011). However, using a non-lexicalist model like DM, we see in this dissertation that Santali, along with any other language, shows a clear distinction between categories when we consider their formation in the syntactic derivation.

By closely examining nominalization, verbalization, and adjectivization in Santali, the dissertation demonstrates that category formation is uniformly syntactic and configurationally defined at various points in the derivational spine.

In this dissertation, **Chapter 2** attempts to understand the formation of three different types of nominals, namely prototypical nominals, affixal nominals, and zero nominals (ZN) in Santali. Both prototypical nominals and affixal nominals are formed by root-nominalization in Santali to show idiosyncratic interpretation, whereas ZNs show interesting patterns in their

structure and formation. Arguing against what has been claimed for the formation of ZNs in the generative literature, this chapter has presented four diagnostics to claim that ZNs in Santali have larger verbal structures, behave like GNs, and should be reanalyzed as stem (verb)-derived with a higher *n*-attachment point. These tests show how much nominal or verbal characteristics the Santali ZNs carry. First, Santali ZNs possess argument structures identical to their verbal counterparts. Second, they co-occur with adverbial modifiers, which conventionally appear with verbs, establishing their verbal characteristics. Third, Santali ZNs allow aspectual modification, confirming their verbal nature. Finally, the high degree of productivity in forming Santali ZNs out of unaccusative, unergative, transitive, and ditransitive verbs, retaining the compositional verbal interpretation, shows the presence of verbal structure in these nominals. These tests, which are important indicators of verbal nature of a nominal, suggest that Santali ZNs are not directly formed from acategorial roots. They rather have a verbal intervener in the nominalization process, where the roots become verbs before becoming nominals.

This contributes to the discussion on nominalization in the generative literature by showing how morphologically simplex nominals have syntactic complexity and can be formed at different stem-derivational points, such as *vP*, *VoiceP*, and *AspectP*.

After understanding how nominals are formed with a null categorizer and carry verbal structural and semantic properties, the dissertation moves on to the internal structures of verbs in Santali. **Chapter 3** studies how verbs are formed from both category less roots and already categorized stems. Santali shows root-derived verbs such as *sen* ‘go’, *japid* ‘sleep’, etc., like any other language along with stem-derived verbals by carrying information of already categorized elements (noun or adjective) in the verbal internal structure. These stem-derived verbals carry structural and semantic information of the base categories they are derived from with a systematic and regular ‘CoS/ become’ semantics. This stem-derived verbalization not only contradicts the claim that verbalization is always idiosyncratic and root-derived, it also goes a step further to show how the feature on the verbalizer can explain the systematic compositionality in Santali verbs. By using a feature-based account, this chapter demonstrates how any non-verbal base category can be verbalized by retaining the change of state interpretation of the base category itself by the help of a [FIENT] feature.

This chapter contributes to the generative literature by showing that both idiosyncratic and compositional interpretation can be accommodated in verbalization when we adopt a config-

rational approach to study categorization. The empirical contribution shows that Stem-derived verbalization is productive in Santali, demonstrating that *v* can merge with categorized stems to provide compositional interpretations. Further, it confirms that verbs can be configurationally formed from roots, or have intermediate categories like nouns or adjectives, and the idiosyncratic or compositional interpretation of roots depends on the merge position of the categorizer.

Chapter 4 takes the discussion forward by studying the structure and formation of adjectives in Santali. The limited typological literature presents two opposing views on the categorial status of adjectives in Santali. Boddling (1929) claims any morpheme can act as an adjective when it appears pre-nominally. Conversely, Ghosh (2008) argues that Santali lacks adjectives entirely. Using certain diagnostics, this chapter first establishes the presence of adjectives as a composite category in Santali which is a result of the adjectivization process in derivation, taking a root or a stem (a noun or a verb) as input. Santali root-derived adjectives are idiosyncratic, formed low in the derivation to lack compositionality, and lacks morphological adjectivizers. Stem-derived adjectives, in contrast, are regular and compositional to the input category, and have a overt morphological adjectivizer *-ih(aa)*. This chapter also extends the study of structural height of categorization to adjectives to analyze the structural asymmetry in stem-derived adjectivization in Santali. In this vein, we see that Santali adjectivization is input sensitive which makes the *a*-attachment points vary depending in case on nominal and verbal bases. In the nominal input domain, *a* can merge with the noun (*nP*) and its phrasal level, NumP, and DP, leveling only these *a*-attachment points to accommodate the compositional structure of the functional layers of the nouns. On the other hand, it shows asymmetry by not adjectivizing the whole verbal extended functional layer and stopping at the *vP* level with the internal argument information only.

This chapter contributes to the studies on categorization by confirming the universality of adjectives as a composite category. Adjectives are not an elsewhere category like it has been claimed in the literature. Santali shows us that adjectives also follow the same configurational rules in the derivational spine when we adopt the non-lexicalist approaches of word formation. Also, like what has been seen for nominalization, adjectivization shows various categorization points in the derivational spine by showing an asymmetry between nominal and verbal inputs.

5.2 Key Takeaways

This dissertation leads to various broader theoretical points. It challenges fundamental assumptions about word formation by examining category fluidity in Santali. It argues against the assumption that categories are primitive elements, which are lexically listed to be used as inputs for structure building. Instead, it supports the underspecification approaches to word formation by arguing that words/ categories are results of syntactic operations and they show hierarchical structures similar to phrase building. Also, we have seen in this dissertation that stem-derived nominals and verbals show zero morphology even when they are configurationally formed. LCs, such as nouns, verbs, and adjectives which were taken as smallest linguistic units also have their complex derivational structures. This also means that elements with simple/ no morphological complexity also can have complex syntactic structures. In other words, this means that zero morphology does not mean lack of derivational complexity in languages. Secondly, a major takeaway of the current research can be that structural height plays an important role in understanding the semantic interpretation in Santali categorization. In other words, the merge position of a categorizer determines the idiosyncratic or compositional nature of LCs in Santali. Thirdly, this research makes some important points on the input sensitivity of category formation in Santali. Since categorizers in Santali merge with different structural components like a root, an already categorized stem or higher functional structures of those categorized stems, the takeaway for us is that even languages with extreme categorial fluidity show systematic derivational constraints.

In this dissertation we have also seen that nouns, verbs, and adjectives in Santali are uniformly formed using the operation merge, a syntactic operation that deals with phrase/ sentence building. Through this, the current research provides strong support for non-lexicalist accounts, aligning with DM. This confirms that a single derivational engine for both word (category) and sentence building is not only theoretically economical, it is also empirically crucial since it explains the fluid categorial system of languages like Santali.

Broadly, this study also contributes supports from an understudied Austroasiatic language from South Asia, like Santali to the ongoing debates on categorization. This not only helps us in understanding cross-linguistic variation, but this also helps us refine and develop the theories in the generative literature.

5.3 Way Forward

5.3.1 Categorizers and Functional Heads

In this dissertation, we have seen that categorization has been dependent on the merge of categorial roots with categorizing heads/ categorizers. This raises some questions on the nature of the categorizers and their relation with functional heads. As seen throughout Chapters 2-4, these categorizers are distinct from lexical heads and purely functional heads, like Complementizer, Tense, and Determiner, etc. In this subsection, we try to address the theoretical status of these categorizers and how they interact with functional heads.

Functional heads are known to play a crucial role in Syntax. According to (Chomsky, 1995, p. 232), functional heads carry strong features and are responsible for the movement of constituents. Functional categories (FC) play a central role in Agree operation by their ability of feature checking. Functional categories carry -interpretable [-i] features and probes for the +interpretable [+i] features of the lexical heads to satisfy the Agree operation. Studies say that FCs unlike LCs cannot assign theta roles (Abney, 1987; Radford, 1997). FCs are also considered to be the locus of parameters (Borer, 1983; Pollock, 1989; Smith and Tsimpli, 1995). According to the Borer-Chomsky conjecture, Parametric variations are attributed to the functional heads in the lexicon (Chomsky, 1981; Borer, 1984). Muysken (2008) discusses the universality of FCs by stating how FCs are crucial even when they are not lexically or morphologically realized. Different languages have different elements, like clitics, affixes, compounds, and morphologically null entities, etc. that belong to the FC. In other words, they do not receive universal lexical representations in all languages. Muysken also specifies how FCs act as the heads of syntactic projections in the generative tradition.

However, functional heads are defined differently in DM, since there is no distinction between lexical or functional category in the lexicon. Withing the DM account, Harley and Noyer (1999) propose a difference between the f-morpheme and the l-morpheme to correspond to the traditional division between closed class and open class words, or between FC and LC. F-morphemes are elements whose vocabulary insertion is deterministic, meaning the syntactic and semantic feature of an f-morpheme determine a unique phonological content. They express purely grammatical properties or have interpretation determined by Universal Grammar.

Tense markers, case morphemes, determiners, etc. are the examples of f-morpheme, according to Harley and Noyer.

F-morphemes contrast with l-morphemes as l-morphemes can be filled with Vocabulary Items that contain language specific concept. L-morphemes are nouns, verbs, and adjectives, which are formed from acategorial roots. This distinction is crucial for our discussion on the relation of categorizers with the FCs. By connecting Harley and Noyer's distinction with the core idea of this dissertation, we understand that LCs or if we assume a lexically listed l-morpheme doesn't have an universal significance. They, instead of being lexically listed, are formed configurationally with the help of a local relation with category-defining f-morphemes. This means that categorizers can be taken as a subset of functional heads. Similar to some higher functional heads, categorizers also act as phase heads by making the structures they dominate opaque for further derivations.

However, this classification of categorizers as a subgroup of FCs is not as straightforward as it seems. Categorizers play an important role in the derivation that other functional heads do not, i.e., they are responsible for assigning categorial status to roots (l-morphemes) to make them interpretable at the CI interface. This follows the categorization assumption (Embick and Marantz, 2008, p. 6) because roots need to be categorized and cannot directly merge with functional heads like D, T, C, etc. Categorizers work as a bridge between acategorial roots and true functional heads.

Categorizers in Santali play a dual role in the derivation. First, as we discussed, they assign categorial status to the roots. Second, they select various extended functional projections. One of the important nature of the categorizer is that they select and merge not only with roots, but with extended functional projections as well. As we have seen in Chapters 2-4, categorizers can merge with already roots, already categorized stems, and functional projections of those categorized stems. When they merge directly with roots, they yield idiosyncratic interpretations and when they merge with already-categorized stems and their extended projections, they yield compositional interpretations. This flexibility is not seen in other functional heads, which typically have fixed merge position in the derivational spine.

Another way categorizers differ from functional heads depends on the interpretability of features. Functional heads are understood to carry uninterpretable features that must be valued through Agree operation (Chomsky, 2001). According to Panagiotidis (2015), on the other

hand, categorizers carry interpretable categorial features. Panagiotidis proposes that categorizers like *n* or *v* carry LF-interpretable privative features like [N] or [V] which assigns the interpretive perspective to the roots they categorize. This contrasts with the [-i] features the higher functional heads carry which get deleted after being checked and are not interpretable at the LF.

An important finding of this dissertation also includes that categorizers exhibit selectional restrictions that vary cross-linguistically and across different derivational contexts within a language. The categorizers in Santali actively select their complements and their behaviour vary in nominalization, verbalization, or adjectivization. As seen across Chapters 2-4, categorizers can select acategorial roots yielding idiosyncratic semantics and already categorized elements, showing systematic and compositional interpretation. Higher in the structure, above the categorized stems, categorizers do not have the freedom to merge at any point in the functional spine. Connecting this to what we saw in Chapter 2, even if a nominalizer can merge as high as a CP, there are language specific restrictions which stops it at the AspectP-level in Santali. The verbalizer can merge with stems like *nP* and *aP*, as well as phrases like DP. Also, we saw in Chapter 4 that the adjectivizer shows asymmetry, merging with nominal projections up to DP but stopping at *vP* in the verbal domain. Apart from this, the presence of [FIENT] feature on the verbalizer head in Chapter 3 shows how the feature on a categorizer can interact with the complements of the categorizers. The presence of this feature is also dependent on the selectional domain of the categorizer. These selectional properties distinguish categorizers from more rigid functional heads like Tense or Complementizer, which have fixed positions in the clause structure and more predictable selectional requirements.

Our understanding of categorizers from this dissertation supports certain theoretical points. Firstly, categorizers do not show purely functional or a purely lexical characteristics. Even if they assign the categorial status to roots, they are not defined as LCs like nouns, verbs, or adjectives. Similarly, although they carry features, are abstract morphemes and receive phonological interpretation through late insertion, like functional heads, they do not trigger Agree, not being categorized as purely functional. Secondly, the idiosyncratic and compositional interpretation categorizers trigger depending on the categorization height provides insights for the claim that semantic interpretation is determined configurationally in the derivation.

After understanding that the position of categorizers in the division of functional and lex-

ical category is fuzzy, we can have some unanswered question that future works should take up. This dissertation has shown that categorizers may vary cross-linguistically in terms of their merge position they have in the derivation. An important question here would be what determines the maximal height at which a categorizer can merge in a language. Some other questions would include why nominalization stops at AspectP in Santali but can extend to CP in other languages. It should also be inquired if there are universal hierarchies constraining categorizer attachment.

We have also seen that the categorizer is null in case of nominalization and verbalization. On the other hand, it is partially overt in adjectivization. At this point we can ask what role morphological overtiness plays in categorization and if categorization varies across languages depending on whether the categorizer is covert or overt.

5.3.2 Empirical Goals

The subsequent future study we can have is the cross-linguistic extension of our analysis on categorial fluidity. In this dissertation, we have studied the categorial fluidity of Santali, a major Munda language of the Austroasiatic language family. Many other Austroasiatic languages, such as Mundari and Kharia, as well as Indo-Aryan languages like Odia, Hindi-Urdu, and Bangla, are spoken in the same geographical regions as Santali. Two separate studies involving languages within Austroasiatic family and languages outside Austroasiatic family can be done to understand the categorial fluidity of the neighboring languages.

Within the Munda subgroup of the Austroasiatic family, both Mundari (Evans and Osada, 2005, p. 373), and Kharia (Peterson, 2003, 2010, 2011) have been reported to have categorial fluidity identical to Santali. As we can see in (1), *buru* ‘mountain’ can occur both as a noun and a verb in Mundari.

- (1) a. *buru-ko* *bai-ke-d-a*
 mountain-3PL.SUBJ make-COMP-TR-IND
 ‘They made the mountain.’
- b. *saan-ko* *buru-ke-d-a*
 firewood-3PL.SUBJ mountain-COMP-TR-IND
 ‘They mountained the firewood.’ (They heaped up the firewood)

Similarly, Peterson (2003, 2010, 2011, 2008) has shown that Kharia uses the same morpho-

logical element both as an argument and a predicate (2).

- (2) a. *lebu Del-ki*
 Man come-MID.PST
 ‘The/a man came.’
- b. *bhagwan lebu-ki ro Del-ki*
 God man-MID.PST and come-MID.PST
 ‘God man-ed and came.’ (God became a man and came.)

Apart from this, Peterson shows that pronominals (3) and adverbs like ‘today’ and ‘yesterday’ (4) can be verbalized in Kharia.

- (3) *naaTak-te in-ga ho-kaR-na-in ro am-ga in-na-m*
 play-OBL 1SG-FOC He-MID.IRR-1SG and 2SG-FOC I-MID.IRR-2SG
 In the play, I will be him and you will be me.
- (4) a. *musa ida?-ki*
 today yesterday-MID.PST
 ‘Today yesterday-ed. (Today became yesterday.)’
- b. *ponomosor musa-te ida?-yo?*
 God today-OBL yesterday-ACT.PST
 ‘God yesterday-ed today’ (God changed today into yesterday.)

However, during my field visit in the Kharia spoken regions of Jharkhand and Odisha, it was noticed that Kharia is losing its categorial fluidity. Young Kharia speakers do not accept constructions like (2) - (4) to have a compositional CoS interpretation. Young Kharia speakers produce constructions like (5) to get an interpretation of ‘God became a man’.

- (5) *bhagwan lebu bone-ki*
 God human become-PST.MID
 ‘God became human.’

On the surface, it seems from (5) that Kharia is losing the stem-derived verbalization. It would be interesting to study in which categorial domains Kharia has lost categorial fluidity and what is the reason behind it.

The study within different language families can involve Odia, a major Indo-Aryan language spoken in the Santali spoken regions and Tamil, a Dravidian language that shows patterns similar to Santali nominalization.

also saw that the merge position of the categorizers like *n*, *v*, and *a* can vary cross-linguistically. In case of nominalization, we saw that as Ross (1973) has proposed, nominals occur as a continuum universally and Santali stops at the AspectP-level. ZNs, which are known in the literature as root-derived and idiosyncratic elements, can have extended verbal functional structures inside them. Taking this forward to verbalization, we saw that not only stem-derived verbs in Santali are formed by the merge of the verbalizer with a categorized stem, but the feature present on the verbalizer head also plays an important role in assigning the ‘CoS/ become’ semantics to the category at the complement position. In case of adjectivization, we saw that Santali not only possesses adjectives as a composite category, it is also formed at different derivational heights varying depending on the input categories.

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Dutta) Western Conference of Linguistics California State University, Fresno, November 2024

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